Labelling where models disagree: label-efficient comparison of classifiers on group fairness

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Abstract

Comparing candidate classifiers on group fairness requires true labels, and under simple random sampling such comparisons need far more labels than comparisons of accuracy, because group rates have small denominators and the contrasts of interest are differences of group gaps. We study how to spend a fixed labelling budget on an unlabelled pool when several trained models are to be compared on group-specific true positive rates, false positive rates and error. Two structural facts are exploited: differences between models arise only on instances where their predictions differ, which are few, and group membership is known before labelling. Our design stratifies the pool by group and by the number of models predicting the positive class, allocates labels by a weighted generalised Neyman rule that uses the models' predicted probabilities as proxies for the unknown labels, and estimates pairwise contrasts by stratified expansion and ratio estimators with linearisation confidence intervals. Agreement strata are always sampled, because group-prevalence denominators involve every instance. Disagreement-based sampling and discordant-result designs are not new; our contribution is their adaptation to ratio-type fairness contrasts among several models, with theory and empirical evaluation. Across eleven data settings with nine models, the median efficiency relative to simple random sampling is 3.7 at 250 labels and 4.4 at 1000 labels (in the eight settings large enough for that budget), whereas stratification by group alone gives about 1.0. The proxy allocation performs almost as well as one computed from the true labels, and the gain is larger when models disagree less.

Keywords: algorithmic fairness; model comparison; stratified sampling; Neyman allocation; label-efficient evaluation; equalised odds

1 Introduction

A model that is to be deployed in lending, hiring, health care or public administration is rarely judged on accuracy alone. Auditors and regulators also ask whether its errors fall evenly on the groups it affects, and the most common way of putting that question into numbers is to compare group-specific true and false positive rates, as in the equal opportunity and equalised-odds criteria of Hardt et al. (2016). These criteria are not free to evaluate. Unlike demographic parity, they depend on the true outcome, so every instance that enters the calculation must be labelled, and labels for the instances on which fairness questions arise are often expensive: they may require a clinician to review a record, an investigator to resolve a case, or a domain

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expert to adjudicate a disputed decision. In practice the question is seldom about one model. A team typically has several candidate classifiers, trained with different methods or settings and of similar accuracy, and wants to know which of them is fairer, by how much, and whether the difference can be told apart from sampling noise. This paper is about answering that question with as few labels as possible.

Comparisons of fairness are label-hungry for two reasons. The first is that group rates are ratios whose denominators are small: a true positive rate in a minority group is estimated from the positives of that group only, which may be a few per cent of a test sample. The second is that the quantity of interest is a difference of differences, namely how the gap between groups for one model compares with the gap for another, so the variance of four group rates enters every contrast. Statistical treatments of fairness metrics have drawn attention to the resulting instability, particularly for small groups (Lum et al., 2022; Ferrara et al., 2025). Our own earlier analysis of nine models on twelve data settings, using held-out pools as a reference population, gave two concrete illustrations. With 500 randomly drawn test labels, the model with the best equal-weight composite of error, calibration, robustness and equalised-odds gap coincided with the reference winner in a median of 36% of test samples (range 12–96% across settings). And when the test sample size needed to certify, on a single criterion, that a chosen model is within 0.01 of the best was extrapolated from pilot certificates, the median requirement was about 1,100 random labels for error but about 135,000 for the equalised-odds gap. A comparison that is cheap for accuracy is two orders of magnitude more expensive for fairness under simple random sampling. Labelling budgets of that size are rarely available, and the conclusion we draw is not that fairness comparisons should be abandoned but that the labels should be spent where they are informative.

Two structural facts make this possible. The first is that competing models disagree on few instances. For a pair of models m and j, the difference in true positive rate within group g is a sum over the positives of that group of $\hat{y}_{mi} - \hat{y}_{ji}$, divided by the number of positives P_g; the summand is zero wherever the two models make the same prediction. The same holds for false positive rates and for error. Among the three most accurate of our nine models, the average pairwise disagreement rate is between 3.0% and 9.5% of the pool in ten of our eleven settings and essentially zero in the eleventh. This is the familiar observation that near-equally accurate models can still disagree on individual predictions (Marx et al., 2020), read from the other side: the disagreements are numerous enough to matter for fairness but rare enough to be targeted. The second fact is that group membership is known before labelling. Unlike the outcome, the protected attribute is usually recorded with the instance, so the pool can be divided by group in advance and a small group can be sampled more heavily than its share of the population. Neither fact helps under simple random sampling, which spends most of its budget on instances where all models agree and on the majority group.

The problem we address is therefore the following. We are given a pool of N unlabelled instances, M fixed classifiers with their binary predictions and predicted probabilities on every instance, and the group of every instance. A budget of L labels may be spent once, before any label is seen. We wish to choose which instances to label and how to estimate, for every pair of models, the difference between their group gaps in true positive rate, the difference between

their group gaps in false positive rate, and the difference between their error rates, together with confidence intervals that are valid under the sampling design. Inference is conditional on the trained models and the target is the value of each contrast on the pool, which is the natural reference when the pool is the population on which the models will be used or a large sample from it.

Our design, which we call PATTERN, stratifies the pool by group and by the number of models that predict the positive class. Instances on which all models predict the same label form the two agreement strata of each group, and the strata in between collect the instances on which the models split. Labels are allocated across strata by a weighted generalised Neyman rule that uses the models' averaged predicted probabilities as a proxy for the unknown labels, and are drawn by stratified simple random sampling without replacement. Each contrast is a smooth function of a small number of population totals, which are estimated without bias by the stratified expansion estimator; the contrasts are estimated by plug-in and their variances by linearisation, which gives Wald intervals. The agreement strata are never left empty. This matters because the denominators P_g and Q_g of the group rates involve every instance of the group, including those on which all models agree, so a design that labelled only disagreements would identify the numerators of the contrasts but not the rates themselves.

None of the ingredients is new on its own, and we state the contribution narrowly. Sampling where models disagree in order to compare their risks goes back at least to Sawade et al. (2012), who derived the power-optimal sampling distribution for the difference in risk of two models, noted that it favours instances of disagreement, and extended it heuristically to several models; related ideas appear in active model selection (Hara et al., 2024; Kay et al., 2025) and in discordant-pair evaluation (Musgrove et al., 2023a). In biostatistics, comparing two screening tests by verifying discordant or test-positive subjects is decades old and covers sensitivity and specificity directly (Schatzkin et al., 1987; Chock et al., 1997; Pepe and Alonzo, 2001), and optimal two-phase allocation for these quantities has been worked out (McNamee, 2002). Stratification on classifier scores with optimal allocation is established for estimating the performance of one classifier (Bennett and Carvalho, 2010a; Kumar and Raj, 2018a; Fisch et al., 2024), and label-efficient estimation of group true and false positive rates for one model, or for several models with a given labelled set, has been studied with Bayesian, semi-supervised and mixture-model methods (Ji et al., 2020; Gao and Gronsbell, 2025; Shanmugam et al., 2025). What we add is the fixed-budget design and allocation for ratio-type group-fairness contrasts between several models, its theory and empirical evidence on its performance. Specifically, we (i) define a stratification that crosses the protected group with the models' agreement pattern and an allocation that minimises, up to finite-population corrections and integer rounding, a weighted sum of the linearised variances of all pairwise contrasts at once, with weights that make the contrasts comparable; (ii) give the design-based theory for the plug-in estimators, including the way the linearised variance of a ratio contrast draws on disagreement strata through its numerator and on all strata of a group through its denominator, and conditions under which the linearisation intervals are asymptotically valid; and (iii) evaluate the design on eleven data settings with nine models, in simulations with a known truth, and against baselines and ablations that isolate where the gain comes from. We do not claim that using disagreement is new, that

proxy-based Neyman allocation is new, or that this is the first label-efficient method for fairness estimation or for comparing several models.

The empirical gain is substantial and its source is clear. We measure efficiency as the ratio of the mean squared error under simple random sampling to that under the design, which is the factor by which simple random sampling would need more labels. Taking the median over the 108 pairwise contrasts of nine models within each setting and then the median over settings, the efficiency of PATTERN relative to simple random sampling is 3.7 at a budget of $L = 250$ (eleven settings), 3.9 at $L = 500$ (ten settings), 4.4 at $L = 1000$ (eight settings) and 7.3 at $L = 2000$ (six settings; 5.4 if law school, where the leading models agree on almost every instance, is set aside). Stratifying by group alone, with the same allocation rule, gives a median efficiency between 0.99 and 1.04 at every budget, so the gain comes from the disagreement strata and not from oversampling small groups. The proxy allocation loses little against an allocation computed from the true labels (median 3.7 against 3.8 at $L = 250$), and the gain is larger when the models disagree less (Spearman correlation -0.72 between efficiency and the disagreement rate of the leading models across the eleven settings at $L = 250$). Average coverage of nominal 95% intervals is close to nominal (median across settings 0.944 at $L = 250$, against 0.931 under simple random sampling), although individual contrasts can undercover at the smallest budget, and the share of correctly signed estimates for contrasts larger than 0.02 rises from 0.86 to 0.92. Six of the eleven settings use arbitrary median splits as groups, which exercise the method but carry no fairness meaning, and the gain varies widely between settings, from 1.65 to 10.2 at $L = 250$.

The paper is organised as follows. Section 2 reviews related work and positions the contribution. Section 3 fixes the setting, the estimands and the notation. Section 4 describes the stratification, the allocation and the estimators, and Section 5 gives their design-based properties. Section 6 reports simulations with a known truth, and Section 7 the study on eleven real data settings. Section 8 compares the design with alternative methods from the literature, and Section 9 isolates the contribution of each component. Section 10 turns the results into practical guidance on when the design pays off and how to choose a budget, and Section 11 discusses limitations and open problems.

2 Related work

The problem sits where several literatures meet: the statistical comparison of classifiers, the label-efficient evaluation of machine learning models, the design of studies that compare diagnostic tests, the estimation of fairness metrics, prediction-assisted inference and classical survey sampling. We review each in turn, stating what the work does and how our setting differs, and close with a short positioning statement.

2.1 Statistical evaluation and comparison of classifiers

The comparison of two classifiers on a common test set is usually treated as a paired problem. McNemar's test (McNemar, 1947) uses only the discordant pairs of a two-by-two table of correct and incorrect classifications, and Dietterich (1998) studied its behaviour, together with resampled and cross-validated t tests, as a tool for comparing learning algorithms. For areas under correlated

ROC curves the nonparametric approach of DeLong et al. (1988) is standard. Demšar (2006) recommended rank-based tests for comparing several classifiers over several data sets. These methods share two assumptions that do not hold here. The test set is taken as fully labelled, so the question of which instances to label does not arise, and the metric is overall accuracy or a ranking measure, not a difference between models in group-specific rates. The use of discordant pairs in McNemar's test is, however, the simplest instance of the first structural fact on which our design rests: for a paired comparison of accuracy, concordant instances carry no information about the difference.

Group-fairness criteria that depend on the outcome, such as equal opportunity and equalised odds (Hardt et al., 2016), add group-specific denominators to every comparison, and it is known that several such criteria cannot in general be satisfied at once when base rates differ between groups (Chouldechova, 2017). We do not take a position on which criterion is appropriate; we take the true and false positive rate gaps and error as given and ask how to compare several models on them with few labels.

2.2 Active testing and active evaluation of one model

A large literature reduces the labelling cost of estimating the performance of a single trained model. Active testing (Kossen et al., 2021) selects test points sequentially with acquisition functions tailored to evaluation and corrects the resulting bias with an importance-weighted estimator; active surrogate estimators (Kossen et al., 2022) instead interpolate the loss with an actively trained surrogate, and the approach has been scaled to large language models with cheap in-context surrogates (Berrada et al., 2025). Prediction-powered active testing (Ashouritaklimi et al., 2026) combines the importance-weighted estimator with a prediction-powered control variate, derives variance-minimising acquisition rules and proves asymptotic normality. Bayesian assessment of black-box classifiers selects instances by Thompson-type rules (Ji et al., 2021). Importance sampling designs for F-measures and other ratio metrics were derived by Sawade et al. (2010) and, for rare categories and extreme class imbalance, by Poms et al. (2021) and Marchant and Rubinstein (2017); Yilmaz et al. (2021) compared importance and Poisson sampling designs for accuracy and F1 with confidence intervals.

Closer to our design are the stratified approaches. Bennett and Carvalho (2010a) stratified on classifier scores to estimate the precision of deployed classifiers with an online approximation to optimal allocation, Katariya et al. (2012) adaptively refined a stratification for accuracy estimation, and Kumar and Raj (2018a) studied stratification on classifier output for risk estimation and reported large variance reductions against simple random sampling. More recently, approximate Neyman allocation from surrogate signals has been used to stratify evaluation pools for generative models (Liu et al., 2026), and stratified importance sampling with one-shot batch acquisition has been proposed for monitoring deployed classifiers (Marsigli and Lopez de Haro, 2026). All of this work estimates the performance of one model, usually a mean or an overall ratio such as precision, and builds strata from that model's scores. Our strata come from the agreement pattern of several models crossed with the protected group, our targets are between-model contrasts in group-specific rates, and our allocation must serve many such contrasts at once. Our design is also non-sequential: the whole budget is allocated before any label is seen,

which suits labelling campaigns run as a single batch.

2.3 Label-efficient model selection and comparison

When several models are available, labels can be chosen to discriminate between them rather than to estimate each one well. Sawade et al. (2012) is the closest precedent for our use of disagreement. They compare the risks of two fixed models under a labelling budget, derive the instrumental distribution that maximises the power of a test of the risk difference, observe that it favours instances on which the models disagree, include a baseline that samples only where the models' predictions differ, and extend the approach to more than two models through a heuristic mixture of pairwise distributions, using the models' own predictive distributions as proxies for the unknown labels. Hara et al. (2024) develop an adaptive labelling strategy that minimises the variance of estimated differences in test loss for selecting the best of several candidates. Consensus-driven active model selection (Kay et al., 2025) uses agreement and disagreement among candidate models in a probabilistic model to decide which points to label and updates a posterior over the identity of the best model. Discordant pair analysis (Musgrove et al., 2023a) evaluates a new classifier against a baseline of known performance by adjudicating only the cases on which they differ, and Kuboyama (2026) shows that labelling every row on which at least two candidates disagree settles the accuracy ranking of several models exactly on a finite pool. Other work selects the best of several pretrained models from a stream or a pool with few labels (Karimi et al., 2021; Okanovic et al., 2025), chooses instances for annotation from clusters of output differences between two text generators (Ashury Tahan et al., 2024), or studies whether a fixed budget should be spent on more items or on repeated labels when identifying the more accurate of two classifiers (Dorner and Hardt, 2024). The only work we found that selects among several models on both accuracy and subgroup fairness under label costs is the Bayesian bandit formulation of Kher et al. (2026), in which an online policy decides which agent to query and the target is regret.

This literature establishes that disagreement is where comparative information lies, and we cite it as such. It differs from our setting in three respects. Its target is a risk or an accuracy ranking rather than a contrast of ratios with group-specific denominators; most of it is adaptive or Bayesian and aims at identifying a best model, not at estimating contrasts with design-based confidence intervals; and the designs that label disagreements only, such as Musgrove et al. (2023a) and Kuboyama (2026), rely either on known baseline performance or on exhaustive labelling of the disagreement set. For error rates the denominator is the pool size and disagreements suffice, but for true and false positive rates the group prevalences P_g and Q_g depend on the agreement strata too.

2.4 Verification designs for comparing diagnostic tests

Biostatistics has long faced the problem of comparing two screening tests when the reference standard is too costly or invasive to apply to everyone. Schatzkin et al. (1987) proposed applying the reference procedure only to people positive on at least one of the two tests and comparing the tests through ratios of sensitivities and of false positive rates; Chock et al. (1997) studied the related design in which individuals negative on both tests are not verified; and Pepe and

Alonzo (2001) showed that, under verification of screen positives only, absolute sensitivities and specificities are not identified but their ratios are, and gave a regression framework for relative accuracy. Correcting for verification that depends on the test result is the subject of a classic literature on verification bias (Begg and Greenes, 1983), and the practice of resolving only discordant results with the reference standard, so-called discrepant analysis, has been criticised because it assumes that concordant results are correct (Hadgu, 1996). Two-phase designs with a cheap first-phase classification and an optimally allocated second-phase verification were developed for the sensitivity, specificity and predictive value of one test by McNamee (2002), within the broader two-phase framework of Neyman (1938) and its epidemiological descendants (Breslow and Cain, 1988; Reilly and Pepe, 1995).

Our true and false positive rate contrasts between two classifiers are this design problem transplanted to machine learning, and the debt should be acknowledged. The differences are in scope and in what is identified. The diagnostic literature compares two tests, not several models; it does not have protected groups or contrasts of group gaps; its strata are the test results themselves, not results combined with predicted probabilities used to set the allocation; and several of its designs leave the doubly negative cell unverified on purpose, so that only relative measures are identified. Our design samples every stratum with known positive probability, so that differences in rates, not only their ratios, are identified, at the price of spending some labels where all models agree. How many such labels are worth spending is one of the allocation questions the design answers.

2.5 Fairness estimation and auditing with few labels

Several lines of work estimate fairness quantities with limited data. One line audits a model by querying it rather than by acquiring ground truth: active fairness auditing estimates demographic parity with few queries to the audited model (Yan and Zhang, 2022), and query-efficient auditing of language models gives intervals for metrics such as group differences in AUC with active selection of queries, reporting large savings over stratified sampling (Hartmann et al., 2026). Demographic parity needs no outcome labels, so the query-based setting is different from ours, although the second of these papers addresses an outcome-dependent metric for one model. A second line provides valid inference: Chugg et al. (2023) give sequential, anytime-valid tests of fairness that allow data to be collected by changing and group-dependent policies, Cherian and Candès (2023) construct simultaneous bounds on performance disparities over many subgroups of one fully labelled model, Ferrara et al. (2025) adapt hypothesis tests of fairness to group size, and Lum et al. (2022) show that common disparity summaries are biased estimators when groups are small and propose corrections. A third line uses unlabelled data to improve estimates of group-specific rates. Ji et al. (2020) calibrate scores per group in a hierarchical Bayesian model to infer group differences in accuracy, true positive rate and false positive rate of one model, and remark that stratified sampling by group could help but is not always possible. Gao and Gronsbell (2025) impute missing outcomes and augment for robustness to estimate group-specific true and false positive rates and calibration with semi-supervised efficiency gains, Gronsbell et al. (2022) evaluate one prediction rule semi-supervisedly when the labels come from a stratified sample, Miller et al. (2021) estimate subpopulation performance with model-based metrics, and Das et al.

(2025) audit fairness under partial feedback with a cost model for acquiring extra labels. Closest in setting is SSME (Shanmugam et al., 2025), which fits a semi-supervised mixture model to the labels and the scores of several classifiers and estimates any metric for each classifier, including within demographic subgroups.

These methods either take the labelled set as given, so there is no design to optimise, or concern one model, or both. Our contribution is complementary: a sampling design and design-based estimators for between-model fairness contrasts. Model-based and semi-supervised estimators could in principle be applied to the labels our design collects, with design weights.

2.6 Prediction-powered and model-assisted inference

Prediction-powered inference (Angelopoulos et al., 2023) combines a small labelled sample with machine learning predictions on a large unlabelled sample and gives confidence intervals that are valid whatever the quality of the predictions; PPI++ (Angelopoulos et al., 2026) tunes the weight on the predictions so that the interval is never wider than the classical one asymptotically. Stratified prediction-powered inference (Fisch et al., 2024) applies the idea within strata and allocates the label budget across strata by a Neyman-type rule. Active statistical inference (Zrnic and Candès, 2024) samples labels with probabilities driven by model uncertainty and combines the resulting design-unbiased estimator with prediction-powered corrections; robust sampling (Li et al., 2025) interpolates between uniform and active sampling so that a poor proxy does not make matters worse than uniform sampling, and fixed-size variants replace Poisson sampling by conditional Poisson sampling (Skalnes and Oberst, 2026). Prediction-powered ranking (Chatzi et al., 2024) gives simultaneous rank-sets for several language models from pairwise comparisons. In these methods predictions enter either as a control variate in the estimator or as a guide to inclusion probabilities, and the target is typically a single mean or a vector of means for one model. We use predictions in the second way only, to set the allocation, and keep the estimator purely design-based; the target is a vector of nonlinear contrasts between models, and the strata are defined by the models' agreement pattern and the group.

The same idea is much older in survey sampling, where auxiliary information is used to build the design and to sharpen the estimator. Model-assisted estimation, in which a working model improves efficiency while validity rests on the design (Särndal et al., 1992; Breidt and Opsomer, 2017), and calibration estimators (Deville and Särndal, 1992) are the classical forms, and prediction-powered inference can be read as a member of this family. A model-assisted version of our estimator, which would use the models' probabilities also in the estimation step, is a natural extension that we do not pursue here.

2.7 Survey-sampling foundations

The design is assembled from standard survey-sampling components. Stratified sampling with allocation proportional to stratum size times stratum standard deviation, which minimises the variance of an estimated total for a fixed sample size, is due to Neyman (1934). When several quantities are estimated from one sample, a single allocation must compromise between them; optimum allocation in multivariate surveys has a long history, including analytical solutions (Kokan and Khan, 1967), and the weighted generalised Neyman rule we use minimises a weighted

sum of variances, the simplest such compromise. Unbiased estimation of totals under unequal inclusion probabilities goes back to Horvitz and Thompson (1952). Group rates are ratios of totals, and the standard treatment of ratio estimators and of smooth functions of totals by Taylor linearisation is given by Woodruff (1971), Binder (1983), Särndal et al. (1992) and Wolter (2007), with a practical account in Lumley (2010). Two-phase sampling, in which a cheap first-phase measurement guides the selection of a second-phase sample, was introduced by Neyman (1938); our setting is a two-phase design in which the first phase, the models' predictions on the whole pool, costs nothing. What these foundations do not supply is the choice of strata and of the variance targets for comparing classifiers on fairness, which is the part we work out.

2.8 Positioning

Disagreement-based sampling for comparing models, discordant-result designs for comparing diagnostic tests, proxy-based optimal stratification for evaluating one classifier, and label-efficient fairness estimation for one model are all established, and we build on each. To our knowledge they have not been combined into a fixed-budget design for estimating fairness contrasts between several classifiers. Such a design must treat group-prevalence denominators that depend on all strata, allocate labels for many contrasts at once, and provide design-based confidence intervals. That combination, its design-based theory and an empirical assessment of when it pays off are the subject of this paper.

3 Problem setting

3.1 Pool, models and labels

We consider a pool $U = \{1, \dots, N\}$ of instances drawn from the deployment population and $M \geq 2$ binary classifiers that have already been trained elsewhere. For every instance $i \in U$ the following quantities are available before any label is collected: the predictions $\hat{y}_{mi} \in \{0, 1\}$ and predicted probabilities $\hat{p}_{mi} \in [0, 1]$ of every model $m = 1, \dots, M$, and the value $G_i \in \{0, 1\}$ of a binary group attribute. The label $Y_i \in \{0, 1\}$ is unknown. It can be obtained for any instance, at a cost of one unit, and the analyst can afford $L \ll N$ labels. The group attribute is assumed to be recorded for all instances, which is the usual situation when a protected attribute is part of the data used for auditing even when it is not used for prediction. The labels are treated as fixed; all randomness comes from the choice of which instances to label. Inference therefore refers to the finite pool, which is the natural target when the pool is the population on which the models will be used, and it extends to the superpopulation in the usual way when the pool is a random sample from it (Särndal et al., 1992, Chapter 13).

This setting differs from the classical evaluation protocol, in which a labelled test set of fixed size is given and the only question is how to analyse it. Here the question is also how to spend a labelling budget. It also differs from active learning: the models are fixed, and labels are collected only to compare them.

3.2 Totals and estimands

All quantities of interest are smooth functions of a small number of population totals, each of which is linear in the labels. For a group $g \in \{0, 1\}$ and model m define

$$P_g = \sum_{i:G_i=g} Y_i, \qquad Q_g = \sum_{i:G_i=g} (1 - Y_i), \tag{1}$$

$$TP_{mg} = \sum_{i:G_i=g} Y_i\, \hat{y}_{mi}, \qquad FP_{mg} = \sum_{i:G_i=g} (1 - Y_i)\, \hat{y}_{mi}, \qquad E_m = \sum_{i \in U} |\hat{y}_{mi} - Y_i|. \tag{2}$$

Each total has the form $T_c = \sum_{i \in U} t_{ic}(Y_i)$ with $t_{ic}(Y_i) = Y_i u_{ic} + (1 - Y_i) v_{ic}$, where u_{ic} and v_{ic} are known for every instance. For example, TP_{mg} has $u_{ic} = \mathbf{1}\{G_i = g\}\hat{y}_{mi}$ and $v_{ic} = 0$, and E_m has $u_{ic} = 1 - \hat{y}_{mi}$ and $v_{ic} = \hat{y}_{mi}$. We collect the totals in a vector T of length $C = 4M + 4 + M$.

The group-wise true and false positive rates are $\mathrm{TPR}_{mg} = TP_{mg}/P_g$ and $\mathrm{FPR}_{mg} = FP_{mg}/Q_g$. The between-group gaps of model m are

$$\Delta_m^{\mathrm{TPR}} = \mathrm{TPR}_{m1} - \mathrm{TPR}_{m0}, \qquad \Delta_m^{\mathrm{FPR}} = \mathrm{FPR}_{m1} - \mathrm{FPR}_{m0}, \tag{3}$$

and the equalised-odds gap is $\mathrm{EO}_m = \max(|\Delta_m^{\mathrm{TPR}}|, |\Delta_m^{\mathrm{FPR}}|)$ (Hardt et al., 2016). A model with small gaps treats the two groups more similarly in terms of error rates.

Our primary estimands are comparisons between models. For a pair (m, j) let

$$\theta_{mj}^{\mathrm{TPR}} = \Delta_m^{\mathrm{TPR}} - \Delta_j^{\mathrm{TPR}}, \qquad \theta_{mj}^{\mathrm{FPR}} = \Delta_m^{\mathrm{FPR}} - \Delta_j^{\mathrm{FPR}}, \qquad \theta_{mj}^{\mathrm{ERR}} = (E_m - E_j)/N. \tag{4}$$

A negative $\theta_{mj}^{\mathrm{TPR}}$ means, for instance, that model m has a smaller signed TPR gap than model j. We write $\theta = f(T)$ for the vector of all $3\binom{M}{2}$ comparisons, where f is a known smooth map on the region where $P_g, Q_g > 0$. The signed gaps rather than their absolute values are used as primary estimands because they are smooth functions of the totals; the non-smooth equalised-odds gap and the ranking of models by it are studied in Section 9.

3.3 Why comparisons on fairness are label-hungry

Under simple random sampling (SRS) of L labels, the TPR of a group is estimated from the positives of that group that happen to be sampled, about $L\,P_g/N$ of them. In the Adult data with sex as the group, female positives make up 3.6% of the pool, so a budget of 1,000 labels yields on average 36 relevant instances for TPR_{m1}. An earlier analysis of the same data settings (Section 1) found that certifying a regret of 0.01 in the equalised-odds gap required on the order of 10^5 randomly sampled labels, against about 10^3 for the error rate. Simply oversampling the minority group helps less than one might expect, because each group-wise rate is also a ratio whose denominator must be estimated, and because the comparison of two models depends on the difference of their predictions rather than on each prediction separately.

3.4 Two structural facts

Two features of the problem can be exploited by the sampling design.

Comparisons depend only on disagreements. Write $s_1 = +1$ and $s_0 = -1$. Then

$$\theta_{mj}^{\mathrm{TPR}} = \sum_{g \in \{0,1\}} \frac{s_g}{P_g} \sum_{i : G_i = g} Y_i \, (\hat{y}_{mi} - \hat{y}_{ji}), \tag{5}$$

and similarly for $\theta_{mj}^{\mathrm{FPR}}$ with $(1 - Y_i)$ and Q_g, while $N\theta_{mj}^{\mathrm{ERR}} = \sum_i (\hat{y}_{mi} - \hat{y}_{ji})(1 - 2Y_i)$. The numerators vanish on every instance where m and j predict the same label. Competing models that are all reasonably accurate agree on most instances; in our eleven data settings the three most accurate of nine models disagree, averaged over their three pairs, on between 0.0% and 9.5% of the pool (Table 6). Labels spent on instances where all models agree inform only the denominators P_g and Q_g.

Everything except the label is known in advance. The group of every instance and the full vector of predictions are known before labelling. A design can therefore condition on them freely, and on the predicted probabilities, without introducing any bias, because the inclusion probabilities are known.

The design proposed in Section 4 uses both facts: it stratifies the pool by group and by how many models predict the positive class, and it allocates labels across strata to minimise the variance of all comparisons at once.

4 Design and estimation

4.1 Stratification

Let $K_i = \sum_{m=1}^{M} \hat{y}_{mi} \in \{0, \ldots, M\}$ be the number of models that predict the positive class for instance i. The proposed design, which we call PATTERN, partitions the pool into strata indexed by (G_i, K_i). The two extreme strata of each group, $K_i = 0$ and $K_i = M$, contain the instances on which all models agree; the strata with $0 < K_i < M$ contain every instance on which at least one pair of models disagrees, and the value of K_i indicates how evenly the models are split. Strata with fewer than 50 instances are merged with the adjacent stratum (in K) of the same group, so that within-stratum variances can be estimated; this gives $H \leq 2(M + 1)$ strata, of size N_h.

The count K_i is a coarse summary of the prediction pattern $(\hat{y}_{1i}, \ldots, \hat{y}_{Mi})$, which has up to 2^M values. It is chosen because it is invariant to the ordering of models, it keeps the number of strata linear in M, and its extreme values isolate the agreement region exactly. Finer stratifiers are compared in Section 9.

4.2 Estimation

Within each stratum a simple random sample of n_h instances is drawn without replacement and labelled ($\sum_h n_h = L$). The total T_c is estimated by the stratified expansion estimator

$$\hat{T}_c = \sum_{h=1}^{H} N_h \, \bar{t}_{hc}, \qquad \bar{t}_{hc} = \frac{1}{n_h} \sum_{i \in s_h} t_{ic}(Y_i), \tag{6}$$

Algorithm 1 PATTERN design for comparing M classifiers on group-fairness contrasts

Require: pool U with predictions $\hat{y}_{mi}$, probabilities $\hat{p}_{mi}$, groups G_i; budget L

1: $K_i \leftarrow \sum_m \hat{y}_{mi}$; strata $h \leftarrow (G_i, K_i)$; merge strata with $N_h < 50$ within group
2: $\pi_i \leftarrow M^{-1} \sum_m \hat{p}_{mi}$; proxy totals $\tilde{T} \leftarrow \sum_i \{\pi_i u_i + (1-\pi_i) v_i\}$
3: $A_i \leftarrow \nabla f(\tilde{T})^\top u_i$, $B_i \leftarrow \nabla f(\tilde{T})^\top v_i$; proxy moments of z_i
4: S^2_{hk}, $S^2_k \leftarrow$ proxy within-stratum and overall variances
5: $n_h \leftarrow$ allocation (8) with $w_k = 1/S^2_k$, minimum 10, capped at N_h, $\sum_h n_h = L$
6: draw n_h instances without replacement in each stratum and query their labels
7: $\hat{T} \leftarrow (6)$; $\hat{\theta} \leftarrow f(\hat{T})$; $\hat{V} \leftarrow (7)$; Wald intervals
8: **return** $\hat{\theta}$, standard errors, intervals

which is unbiased for T_c (Cochran, 1977). The comparisons are estimated by the plug-in $\hat{\theta} = f(\hat{T})$, a ratio-type estimator. Its variance is estimated by Taylor linearisation (Särndal et al., 1992, Section 5.5): with the influence values $z_i = \nabla f(\hat{T})^\top t_i(Y_i)$,

$$\hat{V}(\hat{\theta}_k) = \sum_{h=1}^{H} N_h^2 \Big(1 - \frac{n_h}{N_h}\Big) \frac{s^2_{z_k,h}}{n_h}, \tag{7}$$

where $s^2_{z_k,h}$ is the sample variance of the kth influence values in stratum h. Wald intervals $\hat{\theta}_k \pm 1.96\, \hat{V}(\hat{\theta}_k)^{1/2}$ are reported. If a sampled denominator total is zero (no sampled positive in a group), the estimate is undefined; this happens with small probability under SRS at small budgets and never occurred under PATTERN in our experiments (Section 7).

4.3 Allocation

The allocation $(n_1, \ldots, n_H)$ determines the efficiency of the design. For a single estimand the variance of the linearised estimator is minimised by Neyman allocation, $n_h \propto N_h S_h$, where S_h is the within-stratum standard deviation of the influence values. With many estimands we minimise the weighted sum $\sum_k w_k V_k$ of their linearised variances, which leads (Section 5) to

$$n_h \;\propto\; N_h \Big(\sum_k w_k S^2_{hk} \Big)^{1/2}. \tag{8}$$

We use $w_k = 1/S^2_k$, where S^2_k is the population variance of the influence values of estimand k; this weights each estimand by the inverse of its variance under SRS, so that every comparison counts equally in relative terms.

The stratum variances depend on the unknown labels. We replace them by their expectations under a proxy model in which Y_i is Bernoulli with probability $\pi_i = M^{-1} \sum_m \hat{p}_{mi}$, the average predicted probability of the models. Because the influence values are linear in Y_i, $z_{ik} = Y_i A_{ik} + (1 - Y_i) B_{ik}$ with known A_{ik}, B_{ik} evaluated at the proxy totals, and the proxy moments $\mathrm{E}_\pi z_{ik} = \pi_i A_{ik} + (1-\pi_i) B_{ik}$ and $\mathrm{E}_\pi z^2_{ik} = \pi_i A^2_{ik} + (1-\pi_i) B^2_{ik}$ give closed-form proxy variances. No label is used. A poor proxy can only reduce efficiency; the estimator remains unbiased for the totals because the inclusion probabilities n_h/N_h are known. Each stratum receives at least ten labels (or all of its instances if it is smaller), and rounding to integers is done greedily so that the budget is met exactly.

4.4 Reference designs

We compare PATTERN with three designs that use the same estimators. SRS samples L instances without replacement from the whole pool. GROUP stratifies by the group attribute only and allocates with (8); it oversamples the minority group when that reduces variance, which is the natural first idea for fairness auditing. ORACLE-ALLOC uses the PATTERN strata with the allocation computed from the true labels instead of the proxy. It is not usable in practice and is not an upper bound on efficiency, since (8) minimises a weighted sum of first-order variances rather than the mean squared error of each estimand; it shows how much is lost by using a proxy. Further baselines from the active-testing and survey-sampling literatures are compared in Section 8.

4.5 Computational cost

The design needs one pass over the pool to compute K_i, π_i and the proxy moments, which costs $O(NMC)$ operations with $C = O(M)$ totals, and the estimation costs $O(LMC)$. For the pools used here ($N \leq 19{,}537$, $M = 9$) both steps take well under a second on a laptop.

5 Theory

This section collects the statistical properties of the design and of its two estimators, the plain Horvitz–Thompson plug-in and the model-assisted difference estimator. We work in the design-based framework of survey sampling (Cochran, 1977; Särndal et al., 1992): the pool U, the predictions $\hat{y}_{mi}$, the probabilities $\hat{p}_{mi}$, the groups G_i and the labels Y_i are fixed numbers, and the only randomness is the selection of the labelled sample. Two facts make this framework convenient. First, the strata, the allocation and the proxy are computed from $(\hat{y}_{mi}, \hat{p}_{mi}, G_i)_{i \in U}$ alone, so they are fixed before any label is seen. Second, no model for Y given the predictions is needed for validity; the proxy π_i enters the allocation and, for the difference estimator, the estimator itself, but only through label-free quantities. Both estimators are unbiased for the totals for every proxy (Section 5.2); a wrong proxy costs efficiency, not validity (Sections 5.4 and 5.7). Section 5.2 gives unbiasedness, consistency and asymptotic normality; Section 5.3 the structure of the influence values that drives the gains; Section 5.4 the comparison of the difference and the plain estimator, including tuning of the difference coefficient and zero-variance strata; Section 5.5 the allocation; Section 5.6 the design effect as a function of the disagreement rate. Proofs are in Appendix A.

5.1 Set-up

Write $t_i(Y_i) = Y_i u_i + (1 - Y_i) v_i \in \mathbb{R}^C$ for the vector of contributions of instance i to the totals, $T = \sum_{i \in U} t_i(Y_i)$, and $\tau = T/N$. All entries of u_i, v_i lie in $\{0, 1\}$, so $\|t_i\|_\infty \leq 1$. Every comparison estimand is a smooth function of the totals, $\theta = f(T)$, and is scale free in the sense that $f(T) = \phi(T/N)$: the group rates are ratios of totals and $\theta^{ERR}_{mj} = (E_m - E_j)/N$. The function ϕ is infinitely differentiable on the set where the denominators P_g/N and Q_g/N are

positive. The *influence value* of instance i for a scalar estimand $\theta = f(T)$ is

$$z_i = \nabla f(T)^\top t_i(Y_i) = N^{-1} \nabla \phi(\tau)^\top t_i(Y_i). \tag{9}$$

Strata $h = 1, \ldots, H$ partition U into sets U_h of sizes N_h, with $W_h = N_h/N$; n_h labels are drawn from U_h by simple random sampling without replacement, independently across strata (STSI). For any variable x we write $\bar{x}_h = N_h^{-1} \sum_{U_h} x_i$, $S_{x,h}^2 = (N_h - 1)^{-1} \sum_{U_h} (x_i - \bar{x}_h)^2$ and $f_h = n_h/N_h$, and $\bar{x}_{s_h}$, $s_{x,h}^2$ for the corresponding sample mean and sample variance, and $c_h = N_h^2(1 - f_h)/n_h$. The plain estimator and its variance estimator are

$$\hat{T} = \sum_{h=1}^{H} N_h \bar{t}_{s_h}, \qquad \hat{\theta} = f(\hat{T}), \qquad \hat{V} = \sum_{h=1}^{H} N_h^2 \, \frac{1 - f_h}{n_h} \, s_{\hat{z},h}^2, \quad \hat{z}_i = \nabla f(\hat{T})^\top t_i(Y_i). \tag{10}$$

$\hat{T}$ is the Horvitz–Thompson estimator (Horvitz and Thompson, 1952) for the inclusion probabilities n_h/N_h, and $\hat{V}$ is the usual linearisation (Taylor) variance estimator (Särndal et al., 1992, Ch. 5). Its population counterpart is the linearised variance

$$V(\theta; n) = \sum_{h=1}^{H} N_h^2 \, \frac{1 - f_h}{n_h} \, S_{z,h}^2 = \sum_h c_h S_{z,h}^2. \tag{11}$$

Proxy means and the difference estimator. Let $\pi_i \in [0, 1]$ be any proxy for $P(Y_i = 1)$ that is a function of $(\hat{y}_{mi}, \hat{p}_{mi}, G_i)_{i \in U}$ only (the design uses $\pi_i = M^{-1} \sum_m \hat{p}_{mi}$). Its *proxy mean* is the expected contribution under $Y_i \sim \text{Bernoulli}(\pi_i)$,

$$\psi_i = \pi_i u_i + (1 - \pi_i) v_i, \qquad \xi_i = t_i(Y_i) - \psi_i = (Y_i - \pi_i)(u_i - v_i), \tag{12}$$

with $\|\psi_i\|_\infty \leq 1$ and $\|\xi_i\|_\infty \leq 1$. Because every unit of the pool carries a prediction, ψ_i is known for all $i \in U$ and the proxy total $F = \sum_{i \in U} \psi_i$ is a *known constant*, not an estimate: the pool is the finite population of interest, so, unlike in prediction-powered inference with an unlabelled sample from a superpopulation (Angelopoulos et al., 2023, 2026), there is no sampling error in the proxy total and no $(1 + n/N_u)$ factor. For any label-free vector ψ_i (not necessarily a proxy mean) the *difference estimator* of the totals is

$$\hat{T}_\psi = \sum_{i \in U} \psi_i + \sum_{h=1}^{H} N_h \big(\bar{t}_{s_h} - \bar{\psi}_{s_h} \big) = \sum_{h=1}^{H} \big\{ F_h + N_h \bar{\xi}_{s_h} \big\}, \qquad F_h = \sum_{i \in U_h} \psi_i, \qquad \hat{\theta}_\psi = f(\hat{T}_\psi). \tag{13}$$

Componentwise, $\hat{T}_{\psi,c} = \sum_{i \in U} \psi_{ic} + \sum_h N_h \, \text{mean}_{s_h} \{ t_{ic}(Y_i) - \psi_{ic} \}$. With $\psi \equiv 0$ it is the plain estimator (10); with the proxy mean (12) it is the stratified difference estimator, the estimator of PATTERN-DIF (it is a model-assisted, generalised difference estimator in the sense of Cassel et al. 1976 and Särndal et al. 1992, Ch. 6, and the stratified version of the difference estimator of prediction-powered inference). The residual influence value, its sample version and the linearised

variance are

$$e_i = \nabla f(T)^\top \xi_i = z_i - \tilde{z}_i, \qquad \tilde{z}_i = \nabla f(T)^\top \psi_i,$$
$$\hat{V}_\psi = \sum_h N_h^2 \frac{1-f_h}{n_h} s^2_{\hat{e},h}, \qquad \hat{e}_i = \nabla f(\hat{T}_\psi)^\top \xi_i, \qquad V_\psi(\theta; n) = \sum_h c_h S^2_{e,h}. \tag{14}$$

Thus $z_i = \tilde{z}_i + e_i$: the plain influence value splits into the part $\tilde{z}_i$ that the proxy predicts and the residual e_i that the labels must supply. The plain estimator is the case $\tilde{z} \equiv 0$, and $V_0 = V(\theta; n)$ in (11).

5.2 Unbiasedness, consistency and asymptotic normality

Asymptotic statements refer to a sequence of pools U_ν, $\nu = 1, 2, \ldots$, with $N_\nu \to \infty$ and a fixed number H of strata; the index ν is suppressed. Let $r^2 = \sum_h W_h^2/n_h$, which is the order of $\mathrm{Var}(\hat{\tau}_c)$ for a generic total. Throughout, ψ is either 0 or a proxy mean (12), or any label-free vector with $\|\psi_i\|_\infty \le 1$.

Assumption 1. (A1) *Design.* Strata, sample sizes and the proxy π (hence ψ) are functions of $(\hat{y}_{mi}, \hat{p}_{mi}, G_i)_{i \in U}$ only, in particular not refitted on the labelled sample; $2 \le n_h \le N_h$; $\min_h n_h \to \infty$; and $f_h \le 1 - \varepsilon$ for some $\varepsilon > 0$ and all strata that are not fully enumerated (strata with $n_h = N_h$ are known exactly and contribute no variance).

(A2) *Limits.* $\tau_\nu \to \tau_\infty$, and $P_g/N \ge c$, $Q_g/N \ge c$ for some $c > 0$, $g \in \{0, 1\}$.

(A3) *Non-degeneracy.* The linearised variance of the estimator in use, $V_\psi(\theta; n)$ in (14) ($= V(\theta; n)$ for $\psi = 0$), satisfies $V_\psi(\theta; n) \ge c\, r^2$ for some $c > 0$.

(A4) *Lindeberg-type condition.* For every stratum h, either $n_h(1 - f_h) S^2_{e,h} N^2 \to \infty$, or the stratum's share of the variance, $N_h^2(1 - f_h) S^2_{e,h} / \{n_h V_\psi(\theta; n)\}$, tends to zero.

(A1) excludes the use of labels in the design and in the proxy. (A2) asks that each group contains a non-vanishing fraction of positives and negatives, so that the group rates are well defined. (A3) excludes estimands whose influence values are (nearly) identically zero, for instance two models that make identical predictions, in which case $\hat{\theta} = \theta = 0$ trivially; for the difference estimator it also excludes a proxy that is (nearly) exact, which can make $S^2_{e,h}$ vanish (Remark 1). (A4) is a simple sufficient form of the condition of Hájek (1960) for the central limit theorem under simple random sampling without replacement; because the contributions are bounded, Hájek's condition holds in every stratum whose scaled variance diverges, and (A4) only rules out strata that carry a non-vanishing share of the variance while containing a bounded effective number of observations. Asymptotics for stratified sampling with a fixed or growing number of strata are treated by Bickel and Freedman (1984); for the linearisation variance estimator see Krewski and Rao (1981), whose regime (many strata, few draws per stratum) is complementary to ours (few strata, many draws per stratum).

Proposition 1 (Validity). *Let $\hat{T}_\psi$, $\hat{\theta}_\psi$, $\hat{V}_\psi$ and $V_\psi(\theta; n)$ be as in (13)–(14); $\psi \equiv 0$ gives $\hat{T}, \hat{\theta}, \hat{V}, V(\theta; n)$ of (10)–(11).*

(a) If $n_h \geq 1$ for all h, then $E(\hat{T}_\psi) = T$ and $\mathrm{Cov}(\hat{T}_\psi) = \sum_h N_h^2(1-f_h)n_h^{-1}\Sigma_h^\xi$, with Σ_h^ξ the population covariance matrix of $\xi_i = t_i(Y_i) - \psi_i$ in stratum h; if $n_h \geq 2$, the sample covariance matrix of ξ_i gives an exactly unbiased estimator of $\mathrm{Cov}(\hat{T}_\psi)$. This holds for any strata and any allocation that do not depend on the labels, and for any label-free ψ, in particular for any proxy.

(b) Under (A1)-(A2), $\hat{\theta}_\psi - \theta = O_p(r)$; in particular $\hat{\theta}_\psi$ is consistent.

(c) Under (A1)-(A4), $(\hat{\theta}_\psi - \theta)/V_\psi(\theta; n)^{1/2} \to N(0,1)$ in distribution.

(d) Under (A1)-(A3), $\hat{V}_\psi / V_\psi(\theta; n) \to 1$ in probability. Consequently, under (A1)-(A4) the Wald interval $\hat{\theta}_\psi \pm z_{1-\alpha/2}\hat{V}_\psi^{1/2}$ has asymptotic coverage $1-\alpha$.

Proof sketch. (a) is the classical unbiasedness of the stratified expansion estimator (Cochran, 1977, Ch. 5) applied to ξ_i, because $\hat{T}_\psi - T = \sum_h N_h(\bar{\xi}_{s_h} - \bar{\xi}_h)$ and the known term F cancels exactly. For (b), each coordinate of $\hat{\tau}_\psi - \tau$ has variance at most $2r^2$ because the contributions are bounded; a second-order Taylor expansion of ϕ around τ, valid by (A2), gives $\hat{\theta}_\psi - \theta = \nabla\phi(\tau)^\top(\hat{\tau}_\psi - \tau) + O_p(r^2)$. For (c), the leading term is a sum of H independent stratum means; each is asymptotically normal by Hájek's theorem, and (A3) makes the $O_p(r^2)$ remainder negligible relative to the standard deviation. For (d), the stratum sample covariance matrices are consistent for Σ_h^ξ and $\nabla\phi(\hat{\tau}_\psi) \to \nabla\phi(\tau_\infty)$; (A3) turns the resulting $o_p(r^2)$ error into a ratio statement. □

The estimator $\hat{\theta}_\psi$ of a ratio-type estimand is not exactly unbiased; its bias is $O(r^2)$, which by (A3) is of smaller order than its standard deviation. The joint version for the vector of all comparison estimands follows from the Cramér–Wold device, provided (A3) and (A4) hold for every fixed linear combination. The proof uses only that the contributions are bounded and that the proxy is label-free; it makes no use of the proxy being accurate, calibrated or even a probability.

5.3 Structure of the influence values

For a pair (m, j) let $D_i = \hat{y}_{mi} - \hat{y}_{ji} \in \{-1, 0, 1\}$, and write $\delta_g = TPR_{mg} - TPR_{jg}$ and $\gamma_g = FPR_{mg} - FPR_{jg}$ for the group-wise components, so that $\theta_{mj}^{TPR} = \delta_1 - \delta_0$ and $\theta_{mj}^{FPR} = \gamma_1 - \gamma_0$.

Proposition 2 (Influence values)**.** *For every instance i with $G_i = g$,*

$$z_i^{TPR} = s_g \frac{Y_i(D_i - \delta_g)}{P_g}, \qquad z_i^{FPR} = s_g \frac{(1-Y_i)(D_i - \gamma_g)}{Q_g}, \qquad z_i^{ERR} = \frac{D_i(1-2Y_i)}{N}, \tag{15}$$

with $s_1 = 1$, $s_0 = -1$. The influence values of θ^{TPR} and θ^{FPR} sum to zero within each group. For a stratum h inside group g, let $\pi_h = \bar{Y}_h$, $\rho_h = N_h^{-1}\sum_{U_h}|D_i|$ (the pair's disagreement rate in the stratum), $r_h = N_h^{-1}\sum_{U_h} Y_i|D_i|$ and $c_h = N_h^{-1}\sum_{U_h} Y_i D_i$ (not to be confused with the factor c_h of (11), which is not used in this proposition). Then, with $\kappa_h = N_h/(N_h - 1)$,

$$S^2_{z^{TPR},h} = \frac{\kappa_h}{P_g^2}\Big\{(r_h - c_h^2) + \delta_g^2\pi_h(1-\pi_h) - 2\delta_g\, c_h(1-\pi_h)\Big\}, \tag{16}$$

$$S^2_{z^{ERR},h} = \frac{\kappa_h}{N^2}\Big\{\rho_h - e_h^2\Big\}, \qquad e_h = N_h^{-1}\textstyle\sum_{U_h} D_i(1-2Y_i), \tag{17}$$

and $S^2_{z^{FPR},h}$ *is* (16) *with* Y *replaced by* $1-Y$, P_g *by* Q_g *and* δ_g *by* γ_g. *In particular, in a stratum in which* m *and* j *agree on every instance (always the case for* $K_i \in \{0, M\}$*),*

$$S^2_{z^{TPR},h} = \kappa_h \Big(\frac{\delta_g}{P_g}\Big)^2 \pi_h(1-\pi_h), \qquad S^2_{z^{FPR},h} = \kappa_h \Big(\frac{\gamma_g}{Q_g}\Big)^2 \pi_h(1-\pi_h), \qquad S^2_{z^{ERR},h} = 0. \tag{18}$$

The proof is a direct calculation (Appendix A). Equation (18) is the structural fact behind the design. Where all models agree, the numerators of the comparison estimands receive nothing, and the only remaining influence is the denominator term $-s_g\delta_g Y_i/P_g$: label variability $\pi_h(1-\pi_h)$ scaled by the square of the group-wise target component. Comparisons between competitive models have $|\delta_g|$ of a few hundredths, so these strata, which typically contain most of the pool, carry a variance that is smaller by two to three orders of magnitude than the per-instance variance in the disagreement strata, where (16) is dominated by r_h, the share of positives on which the pair disagrees. For the error difference the agreement strata carry no variance at all.

The residual influence values of the difference estimator have an equally explicit form, and a feature that the plain values lack: the label enters only through $Y_i - \pi_i$.

Proposition 3 (Residual influence values)**.** *Let* $d_i = \nabla f(T)^\top (u_i - v_i)$*, which does not depend on the labels. Then* $e_i = (Y_i - \pi_i)\, d_i$*, and for* $G_i = g$

$$\begin{aligned} e_i^{TPR} &= s_g \frac{(Y_i - \pi_i)(D_i - \delta_g)}{P_g}, \qquad e_i^{FPR} = s_g \frac{\{(1-Y_i) - (1-\pi_i)\}(D_i - \gamma_g)}{Q_g}, \\ e_i^{ERR} &= -\frac{2(Y_i - \pi_i)D_i}{N}, \end{aligned} \tag{19}$$

and $\tilde{z}_i = z_i - e_i$ *equals* (15) *with* Y_i *replaced by* π_i*. In a stratum where the pair agrees,* $e_i^{ERR} = 0$ *for every* i*: the stratum contributes neither variance nor error to the difference estimator of* θ^{ERR}*. Under the working model that* $Y_1, \ldots, Y_N$ *are independent with* $P(Y_i = 1) = \pi_i$ *(the proxy is calibrated), with expectation* E_π*, the gradient evaluated at the non-random model-expected totals* $T_\pi = \sum_{i \in U} \psi_i$ *(as in the design; this changes* d_i *by a relative* $O(r)$ *only), and with* $\bar{\sigma}^2_h = N_h^{-1} \sum_{U_h} \pi_i(1-\pi_i) d_i^2$ *and* $C_h = \mathrm{Cov}_h(z, \tilde{z})$ *the stratum covariance,*

$$E_\pi(e_i) = 0, \quad \mathrm{Var}_\pi(e_i) = \pi_i(1-\pi_i)d_i^2, \qquad E_\pi S^2_{e,h} = \bar{\sigma}^2_h, \quad E_\pi S^2_{z,h} = S^2_{\tilde{z},h} + \bar{\sigma}^2_h, \quad E_\pi C_h = S^2_{\tilde{z},h}. \tag{20}$$

5.4 The difference estimator against the plain estimator

This subsection answers four questions: when does $\hat{\theta}_\psi$ beat $\hat{\theta}$, by how much, what happens if the proxy is wrong, and what do tuning of the coefficient and zero-variance strata do to inference. Write $\beta_h = \mathrm{Cov}_h(z, \tilde{z})/S^2_{\tilde{z},h}$ for the within-stratum regression slope of the plain influence value on its proxy mean, and R_h for their within-stratum correlation, so that $\beta_h = R_h S_{z,h}/S_{\tilde{z},h}$ (terms with $S^2_{\tilde{z},h} = 0$ are dropped). For a vector $\lambda = (\lambda_h)$ of label-free constants consider the family $\hat{T}_{\lambda\psi}$ obtained by replacing ψ by $\lambda_h\psi$ in stratum h; $\lambda \equiv 0$ is the plain estimator and $\lambda \equiv 1$ the difference estimator. By Proposition 1(a) every member is unbiased for T.

Proposition 4 (Variance of the difference estimator, design-based)**.** *Under the notation above, for any label-free proxy and any allocation:*

(a) Identity. $S^2_{e,h} = S^2_{z,h} - 2\,\mathrm{Cov}_h(z,\tilde{z}) + S^2_{\tilde{z},h}$, *hence*

$$V_0 - V_\psi = \sum_h c_h\, S^2_{\tilde{z},h}\,(2\beta_h - 1) = \sum_h c_h\, S_{\tilde{z},h}\,(2R_h S_{z,h} - S_{\tilde{z},h}). \tag{21}$$

The difference estimator has the smaller linearised variance if and only if the right side is positive. A sufficient condition is $R_h > S_{\tilde{z},h}/(2S_{z,h})$ in every stratum, that is $\beta_h > 1/2$: the within-stratum correlation of z_i with its proxy mean must exceed half the ratio of their standard deviations.

(b) Coefficient. $V(\lambda) = \sum_h c_h\{S^2_{z,h}(1-R^2_h) + (\lambda_h - \beta_h)^2 S^2_{\tilde{z},h}\}$. *It is minimised at $\lambda_h = \beta_h$, with $V(\beta) = \sum_h c_h S^2_{z,h}(1-R^2_h)$; moreover $V_0 = V(\beta) + \sum_h c_h \beta_h^2 S^2_{\tilde{z},h}$ and $V_\psi = V(\beta) + \sum_h c_h(1-\beta_h)^2 S^2_{\tilde{z},h}$. With the clipped coefficient $\lambda_h = \min\{1, \max(0,\beta_h)\}$, $V(\lambda) \le \min(V_0, V_\psi)$.*

(c) Calibrated proxy, design-based form. *If $\mathrm{Cov}_h(e,\tilde{z}) = 0$ in every stratum (the residual is uncorrelated with the proxy influence within strata), then $\beta_h = 1$ and $V_0 - V_\psi = \sum_h c_h S^2_{\tilde{z},h} \ge 0$ exactly.*

Part (a) shows that, unlike what is often assumed for prediction-powered estimators, the sign of the gain is not automatic: $\lambda = 1$ is the right coefficient only if the proxy influence is neither over- nor under-dispersed relative to its regression on z; the harm from an over-dispersed proxy is quantified in Proposition 6. When the proxy is calibrated, however, the gain is guaranteed and can be computed without labels.

Proposition 5 (Design-effect ratio under a calibrated proxy)**.** *Under the working model of Proposition 3, let $\nu_h = S^2_{\tilde{z},h}/\bar{\sigma}^2_h$ (signal-to-noise ratio of the proxy influence in stratum h; strata with $\bar{\sigma}^2_h = S^2_{\tilde{z},h} = 0$ are omitted from the sums, and $\bar{\sigma}^2_h > 0$ is assumed otherwise) and $a_h = c_h\bar{\sigma}^2_h$. Then*

(a) $E_\pi V_0 - E_\pi V_\psi = \sum_h c_h S^2_{\tilde{z},h} \ge 0$, *with equality if and only if $\tilde{z}$ is constant within every stratum that is not fully enumerated.*

(b) For the same strata and sample sizes the design effect of the difference estimator relative to the plain estimator is

$$\frac{E_\pi V_\psi}{E_\pi V_0} = \frac{\sum_h a_h}{\sum_h a_h(1+\nu_h)} = \frac{1}{G}, \qquad G = \frac{\sum_h a_h\,(1+\nu_h)}{\sum_h a_h} = \frac{\sum_h a_h/(1-R^2_h)}{\sum_h a_h}, \tag{22}$$

where $R^2_h = \nu_h/(1+\nu_h)$ is the (expected) squared within-stratum correlation of z_i with $\tilde{z}_i$. The gain G is the average of the stratum gains $1/(1-R^2_h)$ weighted by the strata's shares of the residual variance; $G = 1 + \nu$ if $\nu_h \equiv \nu$.

(c) For $\theta^{TPR}, \theta^{FPR}$ and θ^{ERR} the stratum gains depend on the proxy as follows. Let $\overline{\pi(1-\pi)}$ denote the stratum mean of $\pi_i(1-\pi_i)$. In a disagreement stratum *($|D_i| = 1$ on U_h, D_i*

and π_i uncorrelated within the stratum, $\bar{a}$ the stratum mean of D_i, and δ_g, γ_g negligible),

$$1+\nu^{TPR} = \frac{\bar{\pi}(1-\bar{\pi}\bar{a}^2)}{\overline{\pi(1-\pi)}}, \qquad 1+\nu^{FPR} = \frac{(1-\bar{\pi})\{1-(1-\bar{\pi})\bar{a}^2\}}{\overline{\pi(1-\pi)}},$$
$$1+\nu^{ERR} = \frac{1-\bar{a}^2(1-2\bar{\pi})^2}{4\,\overline{\pi(1-\pi)}}. \tag{23}$$

In an agreement stratum *($D_i = 0$) the error difference has $e_i = \tilde{z}_i = 0$, and for the rate differences $1+\nu = \{\overline{\pi(1-\pi)} + S^2_{\pi,h}\}/\overline{\pi(1-\pi)}$, the inverse of the share of the Bernoulli variance that the proxy does* not *explain.*

Four consequences. First, calibration alone guarantees $G \geq 1$: with a calibrated proxy the difference estimator is never worse than the plain one in expectation, for any allocation. Second, the gain is a within-stratum R^2, so it is governed by what the proxy influence $\tilde{z}_i$ explains of z_i *inside* a stratum, and the strata are already built to remove between-stratum variation. Third, $\tilde{z}_i$ contains two kinds of information: the proxy π_i and the known structure of u_i, v_i (the signs and the pattern of D_i). The second kind is free and is the main source of the gain: in a two-sided disagreement stratum with a constant proxy $\pi_i \equiv \bar{\pi}$ and $\bar{a} = 0$, (23) gives gains $1/(1-\bar{\pi})$ for θ^{TPR}, $1/\bar{\pi}$ for θ^{FPR} and $1/\{4\bar{\pi}(1-\bar{\pi})\}$ for θ^{ERR}, i.e. 1.4, 3.3 and 1.2 for $\bar{\pi} = 0.3$ and exactly 2, 2 and 1 for $\bar{\pi} = 1/2$. Only the error difference needs a sharp proxy: its gain is $\{1-\overline{(1-2\pi)^2}\}^{-1}$ and equals one for $\pi \equiv 1/2$, which makes $\tilde{z} \equiv 0$ and the two estimators identical. Fourth, the gain is bounded for proxies of realistic sharpness, because the disagreement strata are the strata in which the compared models, and hence typically the averaged probabilities π_i, are least certain: a large G requires within-stratum R^2_h close to one, that is a sharp proxy (π_i near 0 or 1) and a stratum in which the models still disagree. Where the proxy is that sharp the predictions would be nearly exact and few labels would be needed in any case. For disagreement strata with $\bar{\pi}$ between 0.2 and 0.5 the closed forms give gains of the order of one to three, on top of the stratification gain; Section 6.4 compares the prediction with the Monte Carlo ratios.

Gain and stratification multiply: for the same sample the efficiency of the difference estimator relative to SRS is, to the order of the approximations of Section 5.6, $\mathrm{Eff}_t \cdot G_t$, with Eff_t from (29) and G_t from (22). They are not independent: a stratifier that removes more of the proxy's within-stratum signal (a finer partition of the proxy range) reduces G and raises Eff_t.

Proposition 6 (Misspecified proxy for the difference estimator)**.** *(a) For every proxy, $\hat{T}_\psi$ is design-unbiased and $\hat{V}_\psi$ is a consistent variance estimator (Proposition 1). Misspecification can affect efficiency only.*

(b) Let the labels be independent with $P(Y_i = 1) = q_i$, where q_i may differ from π_i, and let $\tilde{z}^q_i$, $\tilde{z}^\pi_i$ be the proxy influence values for q and π. With E_q the expectation under this model,

$$E_q(V_0 - V_\psi) = \sum_h c_h \Big\{ S^2_{\tilde{z}^q,h} - S^2_{\tilde{z}^q - \tilde{z}^\pi,h} \Big\}, \tag{24}$$

so the difference estimator wins if and only if, within strata and on the c_h-weighted scale, the proxy error in influence units is smaller than the signal it should capture. If $\tilde{z}^\pi_i - \bar{\tilde{z}}^\pi_h = \kappa(\tilde{z}^q_i - \bar{\tilde{z}}^q_h)$ in every stratum (a proxy that is sharper by the factor κ than the truth), the

estimator wins if and only if $0 < \kappa < 2$; for $\kappa < 0$ (anti-correlated proxy) it loses.

(c) *In all cases $V_\psi \le (1 + \max_h S_{\tilde{z},h}/S_{z,h})^2 V_0$, with equality only if $\tilde{z}_i = -s\, z_i$ within each stratum for a constant $s > 0$ (perfect anti-correlation). Hence the possible loss is bounded by a function of the dispersion of the proxy influence relative to the plain one.*

(d) *With the clipped coefficient of Proposition 4(b) the variance never exceeds that of the plain estimator (to first order). In particular, an uninformative proxy, in the sense $\beta_h = 0$ ($\tilde{z}$ uncorrelated with z within strata), has optimal coefficient $\lambda_h = 0$ and then $V_\psi = V_0 + \sum_h c_h S^2_{\tilde{z},h}$: the fixed coefficient $\lambda = 1$ adds the proxy's own variance.*

Two clarifications. The statement that the difference estimator can be worse than the plain estimator *only* if the proxy is anti-correlated is too strong: by (21) it is worse whenever $\beta_h < 1/2$, so a weakly aligned or over-dispersed proxy ($\kappa > 2$) also hurts, while a modestly noisy but correctly scaled proxy does not. And "uninformative" must be read in terms of $\tilde{z}_i$, not of π_i: a constant proxy $\pi_i \equiv \bar{\pi}$ still has $\tilde{z}_i$ varying with the known pattern of D_i, and for the rate differences it still gains, as the closed forms (23) show; only for θ^{ERR} with $\bar{\pi} = 1/2$ is $\tilde{z} \equiv 0$, and only an inverted proxy ($\kappa < 0$) makes $\beta_h < 0$ for the error difference. Section 6.4 confirms both points numerically.

Fixed versus tuned coefficient. In prediction-powered inference (Angelopoulos et al., 2026; Fisch et al., 2024) and in the regression-estimator literature (Cassel et al., 1976; Särndal et al., 1992) the coefficient is estimated from the labelled sample, here per stratum and per estimand: $\hat{\lambda}_h = \widehat{\mathrm{Cov}}(\hat{z}, \hat{\tilde{z}})/\widehat{\mathrm{Var}}(\hat{\tilde{z}})$ clipped to $[0, 1]$.

Proposition 7 (Estimated coefficient)**.** *Suppose $\hat{\lambda}_h$ is a function of the labelled sample of stratum h with $\hat{\lambda}_h - \lambda_h^\infty \to 0$ in probability, for deterministic λ_h^∞ (for the clipped sample slope, $\lambda_h^\infty = \min\{1, \max(0, \beta_h)\}$ whenever β_h converges). Under (A1)–(A4) for the fixed coefficient λ^∞, $\hat{\theta}_{\hat{\lambda}\psi} - \hat{\theta}_{\lambda^\infty\psi} = o_p(V^{1/2})$; hence the tuned estimator has the same asymptotic normal limit and linearised variance $V(\lambda^\infty)$ as the one with the optimal fixed coefficient, and is never asymptotically worse than the plain estimator. At second order, estimating $\hat{\lambda}_h$ adds to the variance of stratum h approximately $c_h(1 - R_h^2)S^2_{z,h}/n_h$, a relative increase of order $1/n_h$ in a stratum that is already optimally tuned, and the plug-in estimator $\hat{V}_\psi$ computed with the fitted $\hat{\lambda}_h$ does not account for it (it is biased downwards by a factor of the same order).*

The consequence is a trade-off that is independent of the particular data. The best that tuning can add to $\lambda = 1$ is the factor $V_\psi/V(\lambda^*) = 1 + \sum_h c_h(1 - \beta_h)^2 S^2_{\tilde{z},h}/V(\beta)$ (21), which is small when the proxy is nearly calibrated ($\beta_h \approx 1$): then tuning gains little in variance but costs an $O(1/n_h)$ inflation in each of the H strata and a downward-biased variance estimator, which matters when n_h is small, as at the budgets of this paper (n_h of the order of 10). With a miscalibrated proxy the gain from tuning can be real (see the $\kappa \neq 1$ cases in Section 6.4), but it is paid for in coverage. The calibrated difference estimator with $\lambda = 1$ has no estimated parameters, exact unbiasedness and a plug-in variance that is unbiased for the totals (Proposition 1(a)), which is why we adopt it as the main estimator. Section 6.4 reports the predicted and the observed gain of tuning together with the coverage of each version.

Remark 1 (Zero-variance strata and the variance floor)**.** The Wald interval needs $\hat{V}_\psi > 0$ and $\hat{V}_\psi \approx V_\psi$, and both fail when a stratum's sample shows no variation in $\hat{e}_i$ although $S^2_{e,h} > 0$. For the difference estimator this has three sources. (i) *Structural zeros*: strata with $d_i \equiv 0$ (the error difference where the pair agrees, Proposition 3), where $e_i \equiv 0$ in the whole stratum; the estimate is exact there and $\hat{V}_\psi$ correctly contributes zero. (ii) *Sampling zeros*: a stratum with few positives (or negatives) in which the sample has none. For the plain estimator $\hat{z}_i = 0$ for every sampled unit, so $s^2_{\hat{z},h} = 0$ exactly; for the difference estimator $\hat{e}_i = -\pi_i \hat{d}_i$ varies with $\pi_i \hat{d}_i$, so exact zeros are rarer, but the sample variance is still far below the population variance $\bar{\sigma}^2_h = N_h^{-1} \sum_{U_h} \pi_i(1-\pi_i) d_i^2$ whenever the missed positives matter. With a calibrated proxy of level π and constant d, the probability of a sample without positives is $(1-\pi)^{n_h}$, which is the zero-count failure of the Wald interval for a binomial proportion (Brown et al., 2001). (iii) *Exact proxies*: if $\pi_i \in \{0, 1\}$ equals Y_i then $e_i \equiv 0$ and (A3) fails; the estimator is then exact, but the proxy is unlikely to be exact on the whole pool. The floor of Section 9 carries over: replace $s^2_{\hat{e},h}$ by the anticipated residual variance

$$\tilde{s}^2_h = N_h^{-1} \sum_{i \in U_h} \pi_i(1-\pi_i) \hat{d}_i^2, \qquad \hat{d}_i = \nabla f(\hat{T}_\psi)^\top (u_i - v_i), \tag{25}$$

whenever $s^2_{\hat{e},h} = 0$ (or, more conservatively, use $\max\{s^2_{\hat{e},h}, \eta \tilde{s}^2_h\}$ with $\eta \in (0, 1]$). Because $\tilde{s}^2_h = 0$ exactly when $d_i \equiv 0$ on U_h, structural zeros are untouched. The floor is label-free, never decreases $\hat{V}_\psi$, leaves the point estimate unchanged, and (Appendix A) does not affect Proposition 1(c)–(d): the floor is active with probability tending to zero in every stratum with $S^2_{e,h}$ bounded away from zero, and in strata with vanishing variance it is, if anything, conservative. It is a finite-sample safeguard, not an asymptotic correction, and Section 6.4 reports its effect on coverage.

5.5 Allocation

Let $w_t \geq 0$ be weights on a finite set of targets t with linearised variances $V_t(n) = \sum_h N_h^2 (1/n_h - 1/N_h) S^2_{ht}$, where $S^2_{ht} = S^2_{z,ht}$ for the plain estimator and $S^2_{ht} = S^2_{e,ht}$ for the difference estimator, and let $A_h = (\sum_t w_t S^2_{ht})^{1/2}$. The weighted objective is

$$\Phi(n) = \sum_t w_t V_t(n) = \sum_h \frac{N_h^2 A_h^2}{n_h} - \sum_h N_h A_h^2. \tag{26}$$

Proposition 8 (Weighted Neyman allocation)**.** *(a) Over real $n_h > 0$ with $\sum_h n_h = L$, Φ is minimised by $n^*_h = L\, N_h A_h / \sum_k N_k A_k$, with $\Phi(n^*) = (\sum_h N_h A_h)^2 / L - \sum_h N_h A_h^2$.*

(b) With bounds $\ell_h \leq n_h \leq N_h$ (ℓ_h a minimum per stratum, $\sum_h \ell_h \leq L \leq N$), the minimiser is $n_h = \min\{N_h, \max(\ell_h, N_h A_h / \lambda)\}$ with $\lambda > 0$ chosen so that $\sum_h n_h = L$.

(c) If an integer allocation $\tilde{n}$ satisfies $\tilde{n}_h \geq n_h - 1$ and $n_h \geq n_0 > 1$ for all h, then $\Phi_0(\tilde{n}) \leq \{n_0/(n_0-1)\} \Phi_0(n)$, where $\Phi_0(n) = \sum_h N_h^2 A_h^2 / n_h$ is the part of Φ that depends on n.

(d) Ignore finite-population corrections and compute variances with divisors N_h and N. With

$w_t = 1/\sigma_t^2$, σ_t^2 *the population variance of target* t*'s influence values,*

$$\Phi_0(n)\, L/N^2 = \sum_t 1/\mathrm{Eff}_t(n), \qquad \mathrm{Eff}_t = V_t^{SRS}/V_t.$$

At the optimum the harmonic mean efficiency over the $|\mathcal{T}|$ *targets equals* $|\mathcal{T}|/(\sum_h W_h A_h)^2 \geq 1$.

Part (a) is Neyman's allocation (Neyman, 1934) for a weighted sum of variances; the proof is one application of the Cauchy–Schwarz inequality, $(\sum_h N_h A_h)^2 = (\sum_h \{N_h A_h n_h^{-1/2}\} n_h^{1/2})^2 \leq \Phi_0(n)\, L$, with equality if and only if $n_h \propto N_h A_h$. Part (b) follows from the Karush–Kuhn–Tucker conditions of a convex problem; strata that would receive more than N_h labels are enumerated and the remainder is re-allocated, which is what the iterative capping in our implementation does. Part (c) bounds the cost of rounding: with the minimum of 10 labels per stratum it is at most a factor 10/9 on the weighted variance, and much less in practice; exact integer optimisation is possible (Wright, 2012) but unnecessary. Part (d) explains the default weights: with $w_t = 1/S_t^2$ the allocation minimises the sum of inverse efficiencies over all comparison estimands, so no target is neglected because it happens to be measured on a small scale (the inequality in (d) is for the plain estimator, whose influence values define σ_t^2). Because the proposition holds for any non-negative S_{ht}^2, it applies verbatim to the difference estimator with the residual variances.

In the design the S_{ht}^2 are unknown. They are replaced by their expectations under the working model $Y_i \sim \mathrm{Bernoulli}(\pi_i)$ independently, with the gradient evaluated at the model-expected totals; up to terms of order $1/N_h$ this is the anticipated variance of Isaki and Fuller (1982). By (20) the anticipated stratum variances are

$$\begin{aligned} \tilde{A}_h^{0\,2} &= \sum_t w_t \big(\bar{\sigma}_{ht}^2 + S_{\tilde{z},ht}^2\big) \quad \text{(plain)}, \\ \tilde{A}_h^{\psi\,2} &= \sum_t w_t \bar{\sigma}_{ht}^2 \quad \text{(difference)}, \qquad \bar{\sigma}_{ht}^2 = N_h^{-1} \sum_{U_h} \pi_i(1-\pi_i) d_{it}^2. \end{aligned} \tag{27}$$

The plain rule is what the design uses (Section 4). The residual rule is the Neyman allocation for the prediction-powered residual (Fisch et al., 2024); it depends on the proxy only through $\pi_i(1-\pi_i)$ and the label-free d_i, so a proxy of any sharpness gives a well-defined allocation.

Proposition 9 (Allocation for the difference estimator)**.** *Let* n^ψ *be the optimal allocation for the difference estimator* ($A_h = A_h^\psi$, $A_h^{\psi\,2} = \sum_t w_t S_{e,ht}^2$) *and* $\tilde{n}^0$ *the allocation obtained from the plain anticipated variances. Put* $\tilde{\nu}_h = \sum_t w_t S_{\tilde{z},ht}^2 / \sum_t w_t \bar{\sigma}_{ht}^2$, $x_h = (1+\tilde{\nu}_h)^{1/2}$ *and assume the anticipated values are exact for the residual variances* ($\tilde{A}_h^\psi = A_h^\psi$)*. Ignoring caps, minima and rounding, the weighted variance of the difference estimator with the* wrong *(plain) allocation satisfies*

$$\frac{\Phi_\psi(\tilde{n}^0)}{\Phi_\psi(n^\psi)} = \Big(\sum_h \omega_h x_h\Big)\Big(\sum_h \omega_h / x_h\Big) \leq \frac{(x_{\max}+x_{\min})^2}{4 x_{\max} x_{\min}}, \qquad \omega_h = \frac{N_h A_h^\psi}{\sum_k N_k A_k^\psi}, \tag{28}$$

with equality to one iff $\tilde{\nu}_h$ *is constant over the strata with* $\omega_h > 0$*. The same expressions hold, with* x_h *replaced by* $1/x_h$ *and* $\omega_h \propto N_h A_h^0$*, for the plain estimator allocated with the residual*

variances. Consequently, if $\tilde{\nu}_h \in [0, \nu_{\max}]$*, the loss from using the wrong rule is at most* $\{1+(1+\nu_{\max})^{1/2}\}^2/\{4(1+\nu_{\max})^{1/2}\}$ *in either direction, for instance* $9/8$ *for* $\nu_{\max} = 3$*.*

So the choice between the two rules is a second-order matter wherever the proxy's signal-to-noise ratio $\tilde{\nu}_h$ is similar across strata, which is the case for strata with comparable disagreement structure, and it matters when $\tilde{\nu}_h$ is large in some strata and small in others (for example, very sharp proxies in the agreement strata of the rate differences). Neither rule is optimal for the other estimator, and the plain rule remains the sensible default for the difference estimator: it is exactly Neyman for the plain estimator, is robust to a proxy whose calibration is not trusted (the residual rule depends on $\pi_i(1-\pi_i)$ and gives zero weight to strata in which the proxy is extreme), and its efficiency loss for the difference estimator is bounded by (28). Section 6.4 reports the predicted and the observed loss.

5.6 Design effect and the disagreement rate

Let $\sigma_{ht}^2 = (1 - 1/N_h)S_{ht}^2$ and $\sigma_t^2 = (1 - 1/N)S_t^2$ be the variances with divisor N_h and N. For a single target allocated by its own Neyman rule, and with $L/N \to 0$, the efficiency relative to SRS is

$$\mathrm{Eff}_t = \frac{\sigma_t^2}{(\sum_h W_h \sigma_{ht})^2} \;\geq\; 1, \tag{29}$$

because $\sigma_t^2 = \sum_h W_h \sigma_{ht}^2 + \sum_h W_h (\bar{z}_h - \bar{z})^2 \geq (\sum_h W_h \sigma_{ht})^2$; equality holds only if all within-stratum standard deviations are equal and the stratum means coincide. Combined with Proposition 2, (29) gives explicit formulas for the plain estimator. We state them for an idealised design in which, within each group, the strata separate the instances on which m and j disagree (set $\mathcal{D}_g$, share ρ_g of group g) from those on which they agree (set $\mathcal{A}_g$). This is exactly the pattern design when $M = 2$; for $M > 2$ it describes a design stratified for the pair (m, j), and is discussed below. Finite-population corrections are ignored ($L/N \to 0$ and $n_h/N_h \to 0$); they are retained in the exact linearised efficiency used in Section 6.

Corollary 1 (Design effect as a function of the disagreement rate)**.** *Let* $\zeta_g = N_g/N$*,* $p_g = P_g/N_g$*,* $r_g = N_g^{-1} \sum_{G_i = g} Y_i |D_i|$ *(positives on which the pair disagrees, as a share of the group),* $\pi_{\mathcal{D},g} = r_g/\rho_g$*,* $\pi_{\mathcal{A},g} = (p_g - r_g)/(1 - \rho_g)$ *and* $a_g = c_{\mathcal{D},g}/\pi_{\mathcal{D},g}$*, where* $c_{\mathcal{D},g} = \delta_g p_g/\rho_g$ *is the mean of* $Y_i D_i$ *over* $\mathcal{D}_g$*;* $a_g \in [-1, 1]$ *is the net share of disputed positives on which* m *is right and* j *wrong.*

(a) For θ_{mj}^{TPR}*, with own-target Neyman allocation on the strata* $\{\mathcal{A}_g, \mathcal{D}_g\}_{g=0,1}$ *and* $L/N \to 0$*,*

$$\mathrm{Eff}^{TPR} = \frac{\sum_g (r_g - \delta_g^2 p_g)/(\zeta_g p_g^2)}{\Big[\sum_g \big\{\rho_g \sigma_{\mathcal{D},g} + (1 - \rho_g)|\delta_g| \sqrt{\pi_{\mathcal{A},g}(1 - \pi_{\mathcal{A},g})}\big\}/p_g\Big]^2}, \tag{30}$$

$\sigma_{\mathcal{D},g}^2 = \pi_{\mathcal{D},g} - c_{\mathcal{D},g}^2 + \delta_g^2 \pi_{\mathcal{D},g}(1 - \pi_{\mathcal{D},g}) - 2\delta_g c_{\mathcal{D},g}(1 - \pi_{\mathcal{D},g})$*. The same holds for* θ_{mj}^{FPR} *with* Y *replaced by* $1 - Y$*. For* θ_{mj}^{ERR}*, with* $\rho = \sum_g \zeta_g \rho_g$ *and* μ_g *the mean of* $D_i(1 - 2Y_i)$ *over* $\mathcal{D}_g$*,*

$$\mathrm{Eff}^{ERR} = \frac{\rho - (\theta_{mj}^{ERR})^2}{\big(\sum_g \zeta_g \rho_g \sqrt{1 - \mu_g^2}\big)^2}. \tag{31}$$

Any further split of $\mathcal{A}_g$ or $\mathcal{D}_g$ (for example $\mathcal{A}_g$ into $K = 0$ and $K = M$) can only increase (30) *and* (31).

(b) Small targets. *If $\delta_g = 0$ and $\rho_g = \rho$ for both groups, then*

$$\mathrm{Eff}^{TPR} = \frac{1}{\rho}\,\Gamma, \qquad \Gamma = \frac{\sum_g r_g/(\zeta_g p_g^2)}{(\sum_g \sqrt{r_g}/p_g)^2} \geq 1, \tag{32}$$

where Γ is the efficiency of Neyman allocation between the two groups alone; and if $\theta_{mj}^{ERR} = 0 = \mu_g$, then $\mathrm{Eff}^{ERR} = 1/\rho$.

(c) Target size. *For one group (drop g), with $\pi_{\mathcal{D}}$, $\pi_{\mathcal{A}}$ and a held fixed, as $\rho \to 0$,*

$$\mathrm{Eff}^{TPR} = \frac{1 + O(\rho)}{\rho\,(\beta + \lambda)^2}, \qquad \beta = \sqrt{1 - a^2\pi_{\mathcal{D}}}, \qquad \lambda = \frac{|a|\sqrt{\pi_{\mathcal{D}}\,\pi_{\mathcal{A}}(1 - \pi_{\mathcal{A}})}}{p}. \tag{33}$$

Since $\delta = \rho\pi_{\mathcal{D}}a/p$, the target is proportional to ρ and to a. A more one-sided disagreement set (larger $|a|$, hence a larger target) changes the gain by the bounded factor $(\beta + \lambda)^{-2}$, which is below one when the agreement region is noisy ($\lambda > 1 - \beta$), but it does not change the order $1/\rho$.

(d) When the gain vanishes. *As $\rho_g \to 1$ the agreement strata disappear and* (30) *and* (31) *tend to the efficiency of stratification by group alone (for the rate differences with $\delta_g = 0$ this is Γ; for the error difference it equals 1 when $\mu_0 = \mu_1$). The gain beyond group stratification is therefore governed by $1 - \rho$: it is large when the compared models disagree on a small share of the pool, it is reduced by a constant factor when the pair's disagreements are one-sided ($|a_g|$ near 1) and the agreement region has substantial label noise, and it disappears when the models disagree almost everywhere.*

The heuristic *efficiency* $\approx 1/\rho$ is thus exact for an error difference with a two-sided disagreement set ($\mu_g = 0$), and correct up to the group factor Γ and the target-size factor $(\beta + \lambda)^{-2}$ for the rate differences. For a pair that disagrees on 5% of the pool it promises a gain of the order of 20 relative to SRS, and of 2.5 for a pair that disagrees on 40%. These are gains of the stratification; the difference estimator multiplies them by the factor G of (22), which by (23) is of order one to three and does not grow as $\rho \to 0$: the difference estimator removes the part of the label noise that a proxy of moderate sharpness can explain, whereas the stratification removes the unlabelled agreement region.

For $M > 2$ the pattern strata (G_i, K_i) do not separate the disagreements of a particular pair: an instance with $K_i = 4$ out of nine may or may not be disputed by m and j. The general formula (29) still applies with the stratum variances of Proposition 2. For an error difference whose disagreements are two-sided within every stratum ($e_h = 0$ for all h, hence $\theta_{mj}^{ERR} = 0$), $\sigma_{ht}^2 \propto \rho_h$ and (29) becomes

$$\mathrm{Eff}^{ERR} = \frac{\rho}{(\sum_h W_h\sqrt{\rho_h})^2}, \qquad 1 \leq \mathrm{Eff}^{ERR} \leq \frac{1}{\rho}, \tag{34}$$

since $\rho_h \leq \sqrt{\rho_h}$ and $(\sum_h W_h\sqrt{\rho_h})^2 \leq \sum_h W_h\rho_h = \rho$. The bound $1/\rho$ is attained only when

every stratum is either entirely disputed or entirely agreed; the pattern strata approach it to the extent that the pair's disagreements concentrate in a few values of K. A lower bound of the same type holds for the pattern strata. Let ρ_{any} be the share of instances on which not all M models agree, i.e. with $0 < K_i < M$. The pair can disagree only there, so $\rho_h = 0$ in the strata with $K \in \{0, M\}$, and Cauchy–Schwarz over the remaining strata gives $(\sum_h W_h \sqrt{\rho_h})^2 \le \rho_{\text{any}}\, \rho$, hence

$$\frac{1}{\rho_{\text{any}}} \;\le\; \text{Eff}^{ERR} \;\le\; \frac{1}{\rho}, \tag{35}$$

with equality on the left when the pair's disagreements are spread evenly over the strata with $0 < K < M$ ($\rho_h = \rho/\rho_{\text{any}}$ there). With nine models ρ_{any} is large (between 0.18 and 0.73 in our pools) while the pairwise ρ is small, and the position of a pair within this interval measures how well the counts K locate its disagreements. Two further losses separate (30) and (31) from the efficiency of the implemented design: a single allocation serves all $3\binom{M}{2}$ comparison estimands (Proposition 8(d)), and it is computed from the proxy (Section 5.7). The exact linearised efficiency of the implemented design is $N^2(1/L - 1/N)S_t^2/V_t(n)$ with V_t from (11) (or (14) for the difference estimator); Section 6 compares it, and the closed forms above, with Monte Carlo results.

5.7 Misspecified proxy

Let $\tilde{n}$ be the allocation computed from the anticipated values $\tilde{A}_h$, and A_h the values computed from the true labels, both with the same weights. This subsection concerns the proxy in the allocation; its role in the difference estimator is Proposition 6.

Proposition 10 (Imperfect proxy). *(a) Proposition 1 holds for $\tilde{n}$ whatever the proxy, provided $\tilde{n}_h \ge 2$ in every stratum (guaranteed by the minimum of 10 labels).*

(b) Ignoring caps, minima and rounding, the relative loss in the weighted variance is

$$\frac{\Phi_0(\tilde{n})}{\Phi_0(n^*)} = \frac{(\sum_h N_h \tilde{A}_h)(\sum_h N_h A_h^2/\tilde{A}_h)}{(\sum_h N_h A_h)^2} = \Big(\sum_h \omega_h x_h\Big)\Big(\sum_h \omega_h / x_h\Big) \ge 1, \tag{36}$$

with $\omega_h = N_h A_h / \sum_k N_k A_k$ and $x_h = \tilde{A}_h / A_h$. The loss is zero if and only if $\tilde{A}_h \propto A_h$; a proxy that is wrong by a common factor costs nothing.

(c) If $x_{\min} \le x_h \le x_{\max}$, the loss is at most $(x_{\max} + x_{\min})^2/(4 x_{\max} x_{\min})$ (Kantorovich inequality). For instance, if every anticipated value $\tilde{A}_h$ is within a factor 2 of A_h up to a common scale ($x_{\max}/x_{\min} \le 4$), the weighted variance is at most $25/16$ times the optimum on the same strata, so the design keeps at least 64% of the efficiency of the optimal allocation, whatever the number of strata.

(d) Proportional allocation is the special case $\tilde{A}_h \equiv 1$. Neither allocation dominates the other for every population: (36) is smaller for the proxy allocation if and only if

$$\Big(\sum_h \omega_h x_h\Big)\Big(\sum_h \omega_h / x_h\Big) \le \Big(\sum_h \omega_h / A_h\Big)\Big(\sum_h \omega_h A_h\Big).$$

A proxy that is badly wrong in a high-variance stratum can make the design worse than proportional allocation, and, through the minimum of ℓ labels in that stratum, worse than SRS.

(e) A defensive mixture $n_h = \eta\, n_h^{\mathrm{prop}} + (1-\eta)\tilde{n}_h$, $0 < \eta < 1$, satisfies $\Phi_0(n) \le \Phi_0(n^{\mathrm{prop}})/\eta$ and $\Phi_0(n) \le \Phi_0(\tilde{n})/(1-\eta)$; since proportional allocation is never worse than SRS up to the factor $(1 - 1/\min_h N_h)^{-1}$, the mixture is never worse than $1/\eta$ times SRS in the weighted variance, up to the same factor.

What can be guaranteed is therefore validity (part (a)) and a loss that is controlled by how far the anticipated stratum standard deviations are from proportional to the true ones (parts (b) and (c)). What cannot be guaranteed for the implemented rule is improvement over SRS for every population; that requires a defensive component as in part (e). Two features make large losses unlikely in our setting. The dominant part of the structure, namely which strata carry the numerators of the comparison estimands, is determined by the predictions and does not depend on the proxy (Proposition 2); and the proxy enters $\tilde{A}_h$ only through $\pi_h(1-\pi_h)$ and r_h, which are smooth in the proxy probabilities. Section 6 quantifies the loss for good, uninformative and adversarial proxies.

6 Checking the theory by simulation

Section 5 gives three predictions of the efficiency $\mathrm{Eff}_t = \mathrm{MSE}_{\mathrm{SRS}}/\mathrm{MSE}_{\mathrm{PATTERN}}$ of a comparison estimand t, in decreasing order of generality and increasing order of simplicity:

LIN the exact linearised design effect of the allocation actually used, $N^2(1/L - 1/N)S_t^2/V_t(n)$ with V_t from (11); it is exact up to the linearisation of Proposition 1 and includes the compromise between targets, the proxy-based allocation, the minimum of 10 labels per stratum and the finite-population corrections;

OWN the own-target Neyman efficiency (29) on the pattern strata, which removes the compromise and the proxy but keeps the actual strata;

FORM the closed form of Corollary 1, (30) and (31), which assumes strata that separate the pair's agreements from its disagreements, and its small-target version $1/\rho$.

All three are computed from the full pool, using the labels only to evaluate the formulas, and are compared with Monte Carlo efficiencies. Everything in this section is produced by `experiments/theory_check.py` and stored in `results/theory_check.json`.

6.1 Real pools

We re-ran the SRS and PATTERN arms of the main study (Section 7) on the 11 cached pools, at the same budgets, with $R = 200$ replications and the same random seeds; the GROUP and ORACLE samples are drawn as well, so that the random stream is identical. The resulting median efficiencies coincide with those in `results/label_efficiency.json` for all 35 pool–budget combinations, so the per-target efficiencies below are those underlying the main results.

Table 1: Real pools: predicted and observed efficiency of PATTERN relative to SRS (medians over the 108 comparison estimands; $L = 500$, $L = 250$ for Spambase). ρ_{any}: share of the pool on which not all nine models agree; ρ: median pairwise disagreement rate. Obs: Monte Carlo ($R = 200$; identical to `results/label_efficiency.json`). LIN, OWN, FORM: predictions defined in the text. Last two columns: per-target ratio Obs/LIN, median and 10%–90% range, and the correlation of log Obs and log LIN across targets. Source: `results/theory_check.json`.

Pool	N	ρ_{any}	ρ	Obs	LIN	OWN	FORM	$1/\rho$	Obs/LIN	corr
Adult-sex	19537	0.51	0.069	4.08	3.97	5.43	13.22	14.5	1.03 (0.83, 1.27)	0.95
Adult-race	19537	0.51	0.070	6.02	6.68	7.47	18.71	14.2	0.98 (0.77, 1.20)	0.98
Bank	18085	0.18	0.053	7.87	7.17	9.08	22.47	18.8	1.02 (0.83, 1.25)	0.96
Credit card	12000	0.56	0.055	3.79	3.88	4.48	12.96	18.3	0.99 (0.84, 1.18)	0.97
Law school	8320	0.19	0.001	15.33	13.79	25.74	3.54	924	1.06 (0.96, 1.22)	1.00
MAGIC	7608	0.40	0.166	2.76	2.83	3.31	5.68	6.0	0.96 (0.78, 1.16)	0.82
EEG	5992	0.73	0.329	1.83	1.75	2.24	3.19	3.0	1.03 (0.88, 1.27)	0.85
Phoneme	2162	0.39	0.162	3.60	3.61	2.91	5.53	6.2	1.00 (0.84, 1.19)	0.81
Spambase	1841	0.32	0.086	5.55	5.63	5.13	12.02	11.6	1.00 (0.85, 1.24)	0.92
Churn	2000	0.19	0.084	32.36	32.15	5.95	11.29	11.9	1.02 (0.89, 1.21)	1.00
Electricity	12000	0.34	0.128	2.84	2.93	3.34	6.69	7.8	0.96 (0.82, 1.18)	0.61

A target is compared when its influence values are not identically zero and its design variance is positive. This removes the 45 targets of the 15 pairs of models with identical predictions in Law school, a further six targets in that pool, and all targets of Law school at $L = 2000$, where the strata that carry the comparisons are fully enumerated and both the observed and the predicted efficiency are infinite (the json reports 4×10^{23}). That leaves 3519 target–budget pairs.

The linearised design effect is accurate. Over the 3519 pairs the ratio of observed to predicted efficiency has median 0.998 and 10%–90% range 0.83–1.22; 77% of the ratios are within 20% and 99% within a factor 1.5, and the correlation of the logarithms is 0.988 (Figure 1, left). The spread is what Monte Carlo error alone produces: each MSE is an average of 200 squared errors, with a relative standard error of about $\sqrt{2/200} = 0.10$ for approximately normal errors, so the ratio of two independent MSEs has a relative standard error of about 0.14 and a 10%–90% range of roughly 0.84–1.20. The medians per pool and budget agree as well: the ratio of the observed median efficiency to the median LIN prediction lies between 0.88 and 1.20 for the 34 pool–budget combinations, and for each type of estimand separately the median of these ratios is 0.99 (TPR), 0.98 (FPR) and 1.00 (error). The agreement includes the cases with very large gains: for Churn at $L = 500$ the observed efficiency of 32.4 (predicted 32.2) arises because the disagreement strata are fully enumerated, which only the finite-population corrections in LIN capture. The lower correlations for Electricity, Phoneme and MAGIC reflect the narrow range of efficiencies across targets in those pools (most lie between 2 and 4), not a larger error. We conclude that, for the purpose of predicting efficiency, the linearised variance of Section 5.2 is adequate already at $L = 250$.

The closed form overstates the gain on real pools. FORM is a much poorer predictor: the median ratio of observed efficiency to FORM is 0.47 (10%–90%: 0.17–1.17), and the rule of thumb $1/\rho$ is more optimistic still (Table 1). The reason is the one anticipated in Section 5.6: with nine models the strata (G, K) do not isolate the disagreements of a particular pair. The

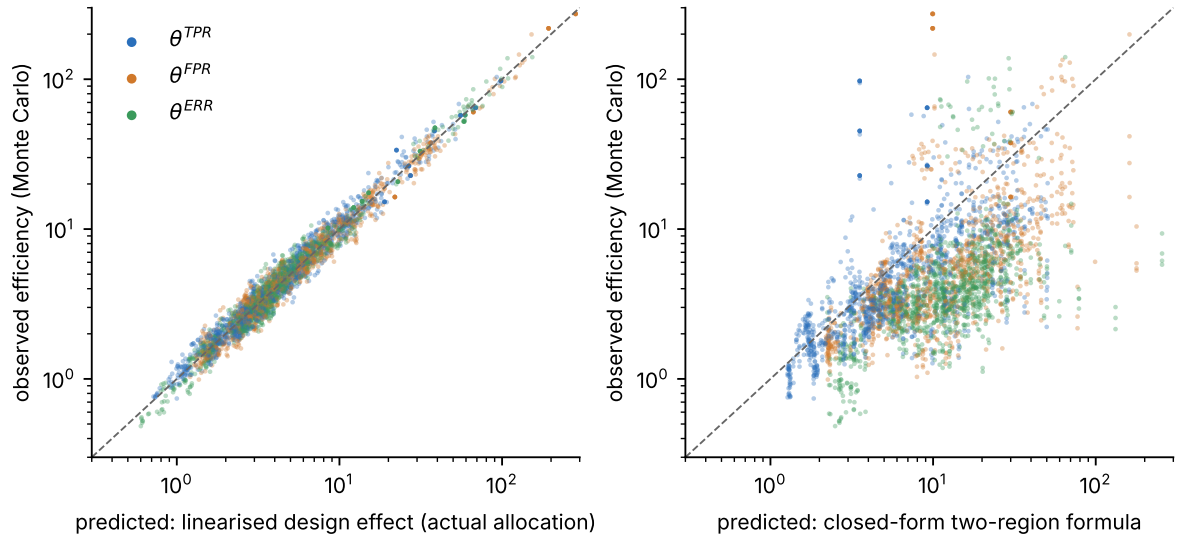


Figure 1: Real pools, all budgets: observed efficiency of PATTERN relative to SRS for each of the 3519 compared target–budget pairs against the linearised prediction LIN (left) and the closed-form two-region formula FORM (right). Dashed: equality. Source: `results/theory_check.json`.

share of the pool on which some model disagrees, ρ_{any}, is between 0.18 and 0.73, while the median pairwise rate is between 0.001 and 0.33, and by (35) the attainable gain lies between $1/\rho_{\text{any}}$ and $1/\rho$. OWN, which uses the actual strata, is much closer (median ratio of observed to OWN 0.87): the remaining 13% is the price of serving all 108 estimands with one allocation, which Proposition 8(d) makes explicit. Law school is a degenerate case for FORM, since the median pair disagrees on 0.1% of the pool and many targets have nearly zero variance; FORM and the observed efficiencies are negatively correlated there. FORM is therefore useful as an explanation of where the gain comes from and as an upper guide for a pair-specific design, but not as a numerical prediction for the nine-model design; for prediction, LIN should be used, and it requires nothing beyond the pool and a pilot or proxy estimate of the stratum variances.

6.2 Synthetic pools with a controlled disagreement rate

To test the closed form where its assumptions hold, and to vary ρ over a wide range, we generated pools of $N = 200\,000$ with two groups ($\zeta_1 = 0.3$, $p_0 = 0.30$, $p_1 = 0.20$). A share $\rho \in \{1, 2, 5, 10, 20, 40\}\%$ of instances is placed in a disagreement region. Outside it, all models predict the same label, equal to Y with probability 0.85. Inside it, with two models, model 1 is right with probability q_g and model 2 predicts the opposite label; we use $q = (0.65, 0.35)$ (moderate group-wise gaps) and $q = (0.95, 0.05)$ (large gaps, one-sided disagreements). With five models, each model is right independently with a model- and group-specific probability between 0.40 and 0.70, conditionally on not all models agreeing. Each model's probability is $\text{logit}^{-1}\{1.73(2\hat{y} - 1) + \varepsilon\}$ with $\varepsilon \sim N(0, 1)$, and the proxy is their average, as in the real pools. The budget is $L = 1000$ and $R = 1000$, so the Monte Carlo relative standard error of an efficiency is about 0.06. With two models, the pattern strata separate agreement from disagreement exactly, and we also ran, for each target, its own Neyman allocation computed from the true

Table 2: Synthetic pools: observed (Obs), linearised (LIN) and closed-form (FORM) efficiency of PATTERN relative to SRS for θ^{TPR} and θ^{ERR} (median over pairs for five models), as a function of the disagreement rate ρ ($N = 200\,000$, $L = 1000$, $R = 1000$). Source: `results/theory_check.json`.

	Two models, moderate gaps						Two models, large gaps						Five models					
	θ^{TPR}			θ^{ERR}			θ^{TPR}			θ^{ERR}			θ^{TPR}			θ^{ERR}		
ρ	Obs	LIN	FORM	Obs	LIN	FORM	Obs	LIN	FORM	Obs	LIN	FORM	Obs	LIN	FORM	Obs	LIN	FORM
1%	190.3	216.0	102.2	174.1	181.1	115.5	56.2	55.8	46.2	665.4	608.0	517.0	172.9	198.3	170.7	158.1	162.9	203.2
2%	54.3	57.4	43.7	56.7	58.2	55.7	22.2	26.0	23.4	251.5	240.7	261.2	44.9	64.2	82.9	49.8	55.3	99.7
5%	14.9	18.6	16.6	23.2	20.2	22.1	10.0	9.7	9.5	88.1	83.1	103.7	17.4	21.0	32.9	19.5	19.6	40.1
10%	7.2	8.9	8.4	8.6	9.6	10.9	4.4	5.1	5.1	45.6	42.5	53.5	9.2	11.0	17.5	8.9	9.4	20.1
20%	4.4	4.7	4.6	5.0	4.8	5.5	2.4	2.8	2.8	16.7	20.3	25.6	4.9	5.4	9.0	4.7	4.6	10.0
40%	2.3	2.5	2.5	2.4	2.4	2.7	1.7	1.6	1.8	9.5	10.1	12.4	2.5	2.8	4.7	2.3	2.3	5.0

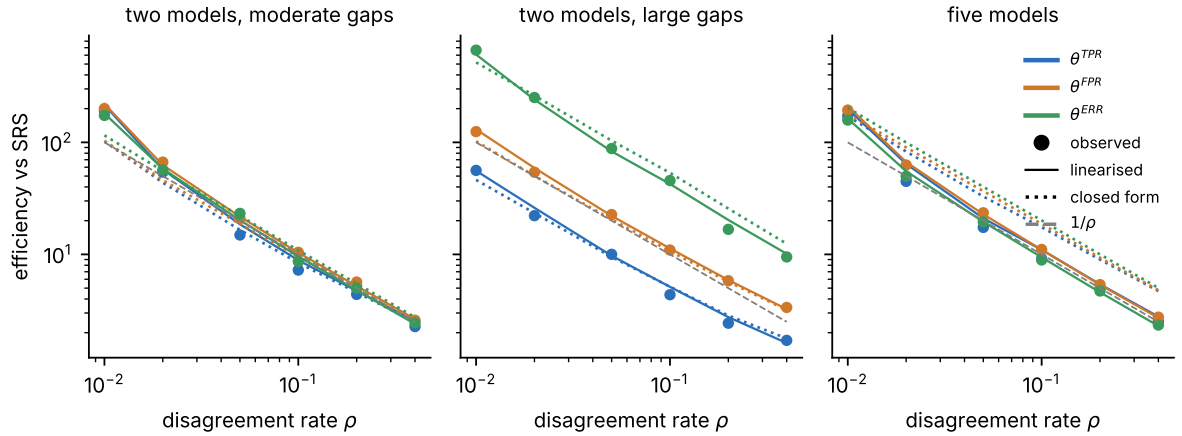


Figure 2: Synthetic pools: efficiency of PATTERN relative to SRS against the disagreement rate ρ (share of the pool in the disagreement region), median over pairs for each type of estimand. Points: Monte Carlo ($R = 1000$); solid: linearised prediction LIN; dotted: closed form FORM; grey dashed: $1/\rho$. Source: `results/theory_check.json`.

stratum variances, which is the design FORM describes.

Efficiency scales as $1/\rho$. In all three families the efficiency falls in proportion to $1/\rho$ over a forty-fold range of ρ (Figure 2). With two models and moderate gaps, ρ times the median observed efficiency is between 0.86 and 1.13 for $\rho \geq 2\%$, so the rule of thumb is accurate to about 15%; at $\rho = 1\%$ it is 1.9, because the budget of 1000 labels is then about half the size of the disagreement region and the finite-population correction roughly doubles the gain. LIN agrees with the observed efficiencies throughout (median ratio per scenario between 0.88 and 1.04), with a tendency to be optimistic by a few per cent that we attribute to the second-order terms neglected by the linearisation.

The closed form is accurate, and conservative, when its assumptions hold. With two models, the observed efficiency of the own-target Neyman design divided by FORM is between 0.90 and 1.25 for moderate gaps and $\rho \geq 5\%$, and between 0.95 and 1.47 for large gaps; the LIN predictions for the same designs give the same ratios, so the discrepancy is in the closed form and not in the simulation. FORM is below the truth for two reasons stated in Corollary 1(a): it

Table 3: Proxy misspecification (five models, $\rho = 5\%$, $L = 1000$, $R = 1000$). Loss: weighted variance $\sum_t w_t V_t$ relative to the oracle allocation, predicted by the linearisation (pred.) and observed with MSEs (obs.). Eff: harmonic mean over the 30 estimands of the efficiency relative to SRS, predicted and observed, and the smallest observed efficiency. Kant.: Kantorovich bound of Proposition 10(c) evaluated with the actual allocation. Source: `results/theory_check.json`.

Allocation	Loss pred.	Loss obs.	Eff pred.	Eff obs.	min Eff obs.	Kant.
Default proxy	1.00	1.01	22.0	20.5	15.4	1.24
Oracle	1.00	1.00	21.9	20.6	17.1	1.07
Uninformative	1.00	1.01	21.9	20.5	16.6	1.23
Inverted	1.00	0.98	22.0	21.2	17.2	1.24
Adversarial	1.00	0.98	21.9	21.0	15.9	1.21
Proportional	8.48	7.85	2.6	2.6	2.1	56.21

treats the agreement region as one stratum, whereas the pattern design splits it by the common prediction ($K = 0$ versus $K = M$, which have very different positive rates), and it ignores the finite-population correction, which matters when the disagreement region is small (at $\rho = 1\%$ the ratio is 1.5 to 2.5). The large-gap family also shows the role of the target size in Corollary 1(c): the TPR gaps are large and the disagreements one-sided ($|a_g| = 0.9$), and the gain for θ^{TPR} is only 0.45 to 0.7 times $1/\rho$, in line with the factor $(\beta + \lambda)^{-2}$, which is about 0.4 for group 0 ($\pi_{\mathcal{D}} \approx \pi_{\mathcal{A}} \approx 0.3$) before the group factor Γ is applied, while for the error difference, whose disagreements are nearly deterministic within each group ($|\mu_g| = 0.9$), the gain is several times $1/\rho$, as (31) predicts.

With more models, the relevant rate is between ρ_{any} and the pairwise rate. In the five-model family the pairwise disagreement rate is about half of ρ, and FORM, which assumes that each pair's disagreements are isolated, overstates the gain by a factor of about 1.9 (median ratio of observed to FORM between 0.53 and 0.56 for $\rho \geq 2\%$). The observed efficiency is instead close to $1/\rho_{\text{any}}$ (ρ_{any} times the median efficiency is between 0.94 and 1.03 for $\rho \geq 2\%$), the lower end of (35). This is expected: in this family the models err independently inside the disagreement region, so a pair's disagreements are spread evenly over the values of K. In the real pools, where models make correlated errors, the median OWN efficiency lies inside the interval $[1/\rho_{\text{any}}, 1/\rho]$ for every pool (Table 1), mostly in its lower half.

6.3 Misspecified proxy

Proposition 10 bounds the loss from a wrong proxy by the ratio (36). We took the five-model pool with $\rho = 5\%$ and compared allocations computed from the default proxy, from the true labels (oracle), from a constant proxy equal to the overall positive rate (uninformative), from one minus the default proxy (inverted), and from an adversarial proxy that sets $\pi_i = 0.5$ where all models agree and $\pi_i = 0.001$ where they do not, with proportional allocation on the same strata as a reference.

None of the four wrong proxies costs a measurable amount of efficiency (Table 3): the predicted loss is at most 0.2%, the observed differences (between -2.4% and $+1.2\%$) are within Monte Carlo error, and every one of the 30 estimands is estimated at least 15 times more efficiently than under SRS. Proportional allocation on the same strata, by contrast, has eight

times the weighted variance. The explanation is Proposition 2: the proxy enters the anticipated stratum variances only through label rates, while the dominant feature, namely that the comparison estimands receive almost no variance from the strata where all models agree, is fixed by the predictions. With a constant or adversarial proxy the agreement strata are still anticipated to have variance of order δ_g^2, which is small, and the disagreement strata still have variance of order one because D_i varies within them; so all these allocations put almost the whole budget into the disagreement strata and differ only in how it is divided among them. The Kantorovich bounds (1.21–1.24) are valid but loose here. This robustness is specific to the structure of the problem and should not be read as a general guarantee: Proposition 10(d) shows that a proxy that misjudges a high-variance stratum can do arbitrarily badly in other stratified problems, and the empirical comparison with the oracle allocation in Section 7 is the relevant check for real data.

6.4 The difference estimator (PATTERN-DIF)

Section 5.4 makes four predictions about the difference estimator (13) with the proxy mean (12): it is exactly unbiased for the totals for any proxy and its plug-in variance is unbiased (Proposition 1); its linearised variance is $V_\psi = \sum_h c_h S_{e,h}^2$ with residual influence values $e_i = z_i - \tilde{z}_i$; its gain over the plain estimator on the same sample is V_0/V_ψ, which a label-free calculation forecasts under calibration (Proposition 5); and a miscalibrated proxy can lose to the plain estimator (Proposition 6). We check them with `experiments/theory_check_dif.py` (results in `results/theory_check_dif.json`). Every comparison is between the plain and the difference estimator on the *same* PATTERN sample, so the ratio MSE(plain)/MSE(DIF) isolates the estimator. Three predictions are used: **LIN**, $\sum_h c_h S_{z,h}^2 / \sum_h c_h S_{e,h}^2$, computed with the true labels; **ANT**, the label-free forecast $\sum_h c_h(\bar{\sigma}_h^2 + S_{\tilde{z},h}^2)/\sum_h c_h \bar{\sigma}_h^2$ of (22), which uses the proxy only; and **TUNE**, $V_\psi/V(\lambda)$ with $\lambda_h = \min\{1, \max(0, \beta_h)\}$, the most that tuning the coefficient could add to $\lambda = 1$.

Exactness and the variance formulae. On a pool of 14 units in two strata (6 and 8 units, 2 and 3 labels) we enumerated all 840 samples for three proxies (a random mixture of u_i, v_i, an arbitrary bounded vector, and the anti-correlated vector $1-t_i$) and three coefficients ($\lambda = 0, \frac{1}{2}, 1$). The mean of $\hat{T}_{\lambda\psi}$, its covariance matrix and the mean of the plug-in variance estimator agree with T, the formula of Proposition 1(a) and each other to within 3×10^{-13} in absolute terms (relative to the variances, 10^{-14}), that is machine precision, for every proxy. By Monte Carlo ($R = 2000$) the totals are also unbiased on the four real pools and six synthetic pools: the largest standardised bias over the 29–49 coordinates is $|z| = 2.7$ for the difference estimator and 2.6 for the plain estimator, which share their samples and have strongly correlated coordinates, and the largest bias is 0.06 standard deviations (the Monte Carlo resolution is 0.022). The variance formula holds: the median over estimands of the Monte Carlo variance divided by the linearised V_ψ is between 0.99 and 1.02 on the real pools (Table 4) and 1.01 to 1.07 on the synthetic pools (up to 1.10 for θ^{TPR}, a second-order effect of the random denominators at $L = 600$), the mean of $\hat{V}_\psi$ divided by the Monte Carlo variance is between 0.98 and 1.02, and the Wald coverage of the difference estimator (0.943 to 0.951 on the real pools, 0.947 to 0.953 on the synthetic pools)

Table 4: Four real pools, $R = 2000$: efficiency of the difference estimator relative to the plain estimator on the same PATTERN sample (median over the 108 estimands). Obs: Monte Carlo; LIN, ANT, TUNE: predictions defined in the text; Obs/LIN: per-target ratio, median and 10%–90% range, and correlation of the logarithms; Var: Monte Carlo variance of $\hat{\theta}_{\mathrm{DIF}}$ over V_ψ; $\hat{V}$: mean estimated over Monte Carlo variance; Cov: Wald coverage of plain and difference estimator; max $|z|$: largest standardised bias of the 46–49 totals. Source: `results/theory_check_dif.json`.

Pool	L	Obs	LIN	ANT	TUNE	Obs/LIN	corr	Var	$\hat{V}$	Cov plain	Cov DIF	max $\lvert z\rvert$
Adult-sex	500	1.50	1.46	1.47	1.01	1.01 (0.99, 1.04)	0.998	0.99	1.02	0.949	0.951	1.6
Bank	500	1.45	1.48	1.46	1.01	0.99 (0.96, 1.03)	0.994	1.02	0.99	0.942	0.943	2.0
EEG	500	1.90	1.91	1.43	1.04	1.00 (0.97, 1.03)	0.997	1.00	1.00	0.947	0.948	2.2
Spambase	250	2.30	2.34	1.97	1.02	1.00 (0.96, 1.03)	0.998	1.00	0.99	0.944	0.945	1.4

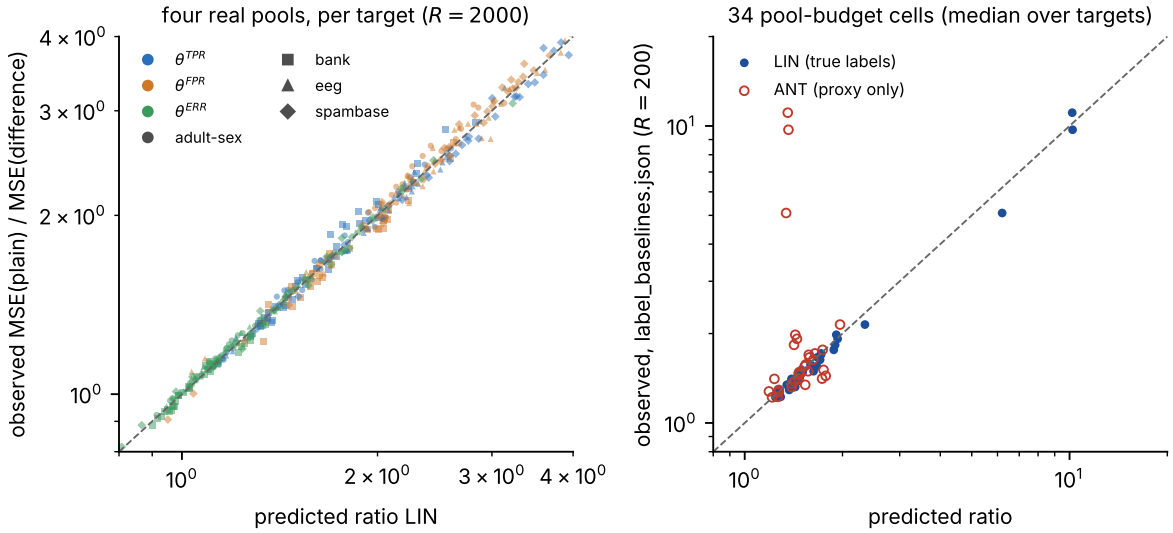


Figure 3: Left: observed against linearised (LIN) gain of the difference estimator over the plain estimator, per estimand, on four real pools ($R = 2000$; 432 estimands). Right: median over estimands in each of 34 pool–budget cells, observed in `results/label_baselines.json` (PATTERN-DIF against PATTERN, $R = 200$) against LIN (filled) and ANT (open). Dashed: equality. Source: `results/theory_check_dif.json`.

is that of the plain estimator.

Predicted against observed gain. On the four real pools the observed per-estimand gain divided by LIN has median 1.00 and 10%–90% range 0.97–1.04 (432 estimands, correlation of the logarithms 0.997; Figure 3, left), which is Monte Carlo error. We then compared LIN with the gains stored in `results/label_baselines.json`, which were produced independently by `experiments/label_baselines.py` with $R = 200$. In the 34 cells with a defined gain (all 35 except Law school at $L = 2000$, where both estimators are exact) the observed median gain is 1.47 (10%–90% range 1.28–2.10, extremes 1.22 and 11.1), LIN has median 1.49, and the ratio of observed to LIN has median 0.98 (10%–90% range 0.93–1.01, 97% of cells within 20%, correlation of the logarithms 0.996, 0.978 without the three Law school cells); Figure 3 (right). LIN uses the labels, so it is not a forecast, but it confirms that the residual-variance formula V_ψ describes the estimator. The label-free forecast ANT has median ratio 1.01 (0.90–1.37) over cells, and on the four real pools ratio 1.12 (0.86–1.53) with correlation 0.78 across estimands; the observed

Table 5: Synthetic pools ($N = 60\,000$, five models, $Y_i \sim \text{Bernoulli}(\pi_i)$, $L = 600$, $R = 2000$): gain of the difference estimator over the plain estimator, observed (Obs), linearised (LIN) and label-free (ANT), median over the ten pairs. Top: calibrated proxy, sharpness a of the latent score. Bottom: $a = 2$, proxy $\pi_i^{\text{proxy}} = \text{logit}^{-1}(\kappa s_i)$ ($\kappa = 1$ calibrated, $\kappa = 0$ constant 0.5, $\kappa < 0$ inverted). Source: `results/theory_check_dif.json`.

	θ^{TPR}			θ^{FPR}			θ^{ERR}		
	Obs	LIN	ANT	Obs	LIN	ANT	Obs	LIN	ANT
$a = 0.5$	1.76	1.76	1.76	2.40	2.44	2.47	1.10	1.09	1.10
$a = 1$	2.03	1.97	1.98	2.33	2.33	2.36	1.09	1.09	1.10
$a = 2$	2.11	2.12	2.14	2.21	2.32	2.31	1.11	1.12	1.11
$a = 4$	2.23	2.23	2.24	2.26	2.27	2.26	1.11	1.12	1.12
$\kappa = -1$	1.43	1.48	2.29	1.52	1.56	2.15	0.76	0.76	1.12
$\kappa = 0$	1.88	1.94	2.00	2.04	2.05	2.00	1.00	1.00	1.00
$\kappa = 0.5$	2.11	2.11	2.02	2.23	2.22	2.09	1.10	1.09	1.03
$\kappa = 1$	2.21	2.16	2.15	2.28	2.27	2.29	1.12	1.12	1.12
$\kappa = 2$	2.09	2.06	2.59	2.22	2.17	2.89	1.08	1.07	1.39
$\kappa = 4$	1.81	1.80	3.81	1.89	1.90	4.45	0.96	0.94	2.11

gain is 1.30 and 1.37 times ANT in EEG and Spambase. It fails in Law school (forecast 1.3–1.4, observed 5.1–11.1), where the disagreement sets of most pairs contain a handful of units per stratum: the realised residual variance is then a single draw from the label distribution, and it is an order of magnitude below its expectation under the proxy (we checked the stratum variances for one estimand). Without Law school the correlation of ANT with the observed gain is 0.62. The gain is therefore not a doubling in general: it is 1.2 to 1.7 in most cells, about 1.8 to 2.1 in EEG and Spambase, and 5 to 11 only in Law school. By estimand type the order is that of (23): the gain is largest for θ^{FPR}, next for θ^{TPR} and smallest for θ^{ERR} (Adult-sex: 2.25, 1.67 and 1.13; Table 5 for the synthetic pools).

Tuning the coefficient. TUNE is small: the median over the 34 cells is 1.02 (10%–90% 1.01–1.04), and 1.01 to 1.04 on the four real pools, so a perfectly tuned coefficient would add at most a few per cent to $\lambda = 1$ except in Law school (2.3–3.5). The observed estimated-coefficient version (PATTERN-PPI) has median gain 1.03 over $\lambda = 1$ (10%–90% 0.99–1.10), and it costs coverage: mean coverage over the 35 cells is 0.946 for PATTERN-DIF and 0.929 for PATTERN-PPI, and 8 of 35 cells fall below 0.93 for PATTERN-PPI against 1 of 35 for PATTERN-DIF and for the plain estimator. This is the trade-off of Proposition 7.

Allocation. The residual-variance allocation of Proposition 9 does not improve on the design's plain-variance allocation for the difference estimator. The linearised prediction of the ratio of V_ψ under the plain allocation to V_ψ under the residual allocation has median 0.99 (10%–90% 0.95–1.00) over the 34 cells, and on four real pools the observed ratio of mean squared errors is 0.98, 0.98, 0.98 and 1.01, against linearised predictions 1.00, 1.00, 0.98 and 1.00, with the same coverage. Using the "wrong" variances therefore costs at most about 2% here, as the bound (28) allows when the signal-to-noise ratio $\tilde{\nu}_h$ is similar across strata.

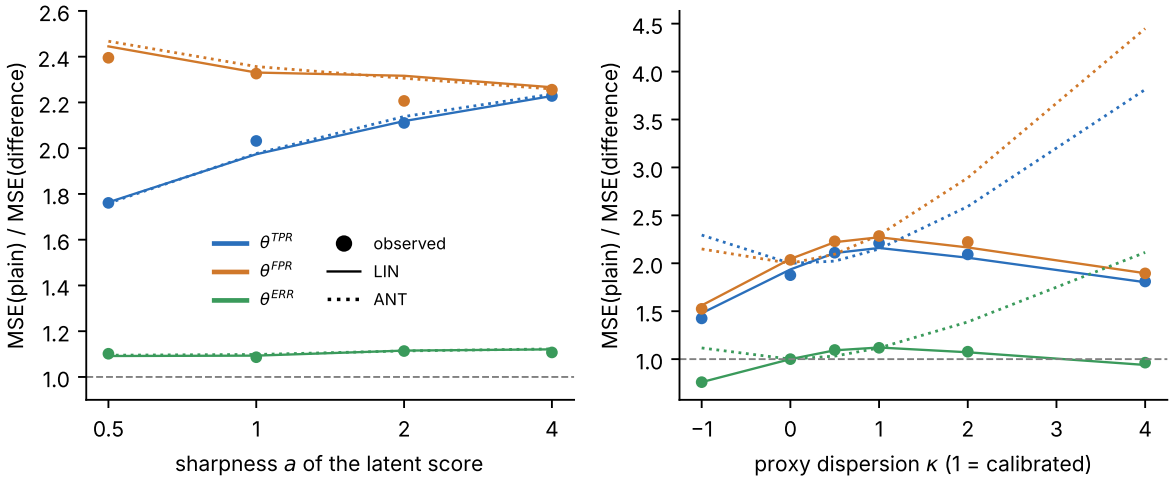


Figure 4: Synthetic pools, gain of the difference estimator over the plain estimator (median over pairs, by estimand). Left: calibrated proxy, sharpness a of the latent score. Right: $a = 2$, proxy dispersion κ ($\kappa = 1$ calibrated). Points: Monte Carlo ($R = 2000$); solid: LIN; dotted: ANT, which assumes calibration. Dashed: no gain. Source: `results/theory_check_dif.json`.

Calibrated and miscalibrated proxies. In pools whose labels follow the proxy, ANT and LIN agree with each other and with the Monte Carlo gain within 2% (median observed/LIN between 0.99 and 1.00; Table 5, top), which verifies Proposition 5: the gain is 1.8 to 2.4 for the rate differences and 1.1 for the error difference, because in the disagreement strata the proxy is uncertain ($\overline{\pi(1-\pi)}$ is large) and the closed forms (23) are bounded; a sharper latent score hardly changes them. For miscalibrated proxies (bottom), unbiasedness and coverage are unaffected (coverage 0.947–0.953), LIN keeps tracking the Monte Carlo gain (median ratio 0.99 to 1.01) and ANT, which assumes calibration, does not. The theory's two warnings appear. First, an inverted proxy ($\kappa = -1$) makes the difference estimator *worse* than the plain estimator for the error difference (gain 0.76, that is 1.3 times the mean squared error, as LIN predicts), while the rate differences still gain 1.4 to 1.5. The design-weighted slope $\bar{\beta} = \sum_h c_h \mathrm{Cov}_h(z, \tilde{z}) / \sum_h c_h S^2_{\tilde{z},h}$ of the error difference is -1.02, the clipped coefficient is 0 and TUNE, which here recovers the plain estimator, is $1.32 = 1/0.76$. Second, a constant proxy $\pi_i \equiv \frac{1}{2}$ ($\kappa = 0$) is uninformative for the error difference (gain exactly 1, since $\tilde{z} \equiv 0$) but not for the rate differences (1.9 and 2.0, TUNE 1.03 and 1.04), because $\tilde{z}_i$ still carries the known pattern of D_i. Third, the threshold $\beta < \frac{1}{2}$ of Proposition 4(a) is visible: for the error difference $\bar{\beta}$ is 1.02 at $\kappa = 1$, 0.62 at $\kappa = 2$ (gain 1.08) and 0.44 at $\kappa = 4$ (gain 0.96), so an over-sharp proxy gains little for the rate differences (1.8–1.9) and crosses to a loss for the error difference.

Zero-variance strata and the floor. The residual floor (25) was evaluated on four pools at $L = 250$ ($R = 1000$). At Churn the difference estimator alone raises the worst-estimand coverage from 0.52 (plain estimator, `results/label_baselines.json`) to 0.84 (0.87 in this run), because $\hat{e}_i = -\pi_i \hat{d}_i$ varies even when no positive is sampled; at Law school it does not (0.64 for both). Where the problem persists the floor repairs it: for Law school mean coverage rises from 0.913 to 0.954 and the worst-estimand coverage from 0.65 to 0.89; for Adult-sex and Credit card default

the mean rises from 0.942 to 0.946 and from 0.949 to 0.955 with the worst estimand unchanged (0.85 and 0.92); at Churn the floor is active in 0.2% of (replication, estimand) pairs and changes nothing (0.938). The floor widens the intervals by a median factor of 1.00 (90th percentile 1.08, 1.15, 1.33 for Adult-sex, Credit card and Law school). It does not alter the point estimate. As in Section 9 it is a safeguard chosen after seeing the failure, and these numbers are not an out-of-sample test.

6.5 Summary

The linearised design effect predicts the efficiency of the implemented design to within Monte Carlo error on all 11 real pools and in the synthetic families (per-target median ratio 1.00, 10%–90% range 0.83–1.22 on the real pools). The closed form of Corollary 1 is accurate and conservative when the strata separate a pair's agreements from its disagreements (the own-target efficiency is 0.90 to 1.47 times the formula for $\rho \geq 5\%$), and it captures the $1/\rho$ scaling and the effect of the target size. With nine models it overstates the gain on real pools by a median factor of about two, because the counts K locate any disagreement but not that of a specific pair; the bounds $1/\rho_{\text{any}} \leq \text{Eff} \leq 1/\rho$ of (35) describe this gap. The gain vanishes only as the disagreement rate approaches one, as in EEG ($\rho_{\text{any}} = 0.73$, efficiency 1.6–1.9). For the difference estimator the linearised residual variance predicts the observed gain over the plain estimator within Monte Carlo error (median ratio 0.98 over 34 pool–budget cells; Section 6.4), the label-free forecast does so except in Law school, and the gain is moderate (median 1.47) rather than a doubling.

7 Empirical study on eleven data settings

7.1 Data, models and protocol

We use eleven settings built from ten public binary classification data sets available on OpenML (Table 6; details in Appendix C). In five settings the group is a protected attribute: sex or race in Adult, age below 30 in Bank Marketing, sex in the credit-card default data and race in the law-school admission data. In the other six the group is an arbitrary split of the first numeric feature at its median, removed from the features; these settings have no fairness meaning but test the design on a wider range of prediction structures. A twelfth candidate (Nomao) was excluded because its arbitrary group contains no positive instances.

Each data set is split once into a training part (60%) and a pool (40%); data sets above 30,000 rows were first subsampled to 30,000. Nine classifiers are trained on the training part with scikit-learn: logistic regression with default and with strong regularisation, decision trees of depth 3 and 12, a random forest, histogram gradient boosting, 25-nearest neighbours, Gaussian naive Bayes and a multilayer perceptron. Their predictions and predicted probabilities on the pool, together with the group attribute, are all the design uses. The pool labels play two roles: they answer label queries, and they define the true values of the estimands, so that the error of every estimate is known exactly.

For each setting and budget $L \in \{250, 500, 1000, 2000\}$ with $L \leq N/4$, we draw 200 independent labelled samples under each design and compute the $3\binom{9}{2} = 108$ comparison estimands of (4). The efficiency of a design is the ratio of the mean squared error under SRS to that

Table 6: Data settings. Pool: size of the held-out pool from which labels are drawn. P_0, P_1: positives in each group. Disagreement: share of pool instances on which two of the three most accurate models disagree, averaged over the three pairs. Groups marked "arbitrary" are median splits of a numeric feature.

Setting	Group	Pool N	Positives P_0	Positives P_1	Disagreement
Adult (sex)	sex	19537	3963	712	0.053
Adult (race)	race	19537	4249	426	0.047
Bank marketing	age	18085	1753	363	0.045
Credit card	sex	12000	1148	1506	0.030
Law school	race	8320	2038	640	0.000
MAGIC	arbitrary	7608	1084	1591	0.049
EEG eye state	arbitrary	5992	1183	1506	0.086
Phoneme	arbitrary	2162	388	247	0.095
Spambase	arbitrary	1841	490	235	0.043
Churn	arbitrary	2000	149	134	0.041
Electricity	arbitrary	12000	2582	2488	0.086

under the design, computed for each estimand and summarised by the median over estimands; an efficiency of 4 means that SRS needs about four times as many labels to reach the same accuracy. Efficiencies are computed over the samples in which both designs give finite estimates; SRS produced an undefined estimate (no sampled positive in a group) in at most 0.5% of its samples, and PATTERN and GROUP never did. The code and all results are in the repository (`experiments/label_efficiency.py`, `results/label_efficiency.json`).

7.2 Efficiency

Table 8 and Figure 5 summarise the results; Table 7 gives every setting. Across settings, the median efficiency of PATTERN relative to SRS was 3.7 at $L = 250$ (eleven settings), 3.9 at $L = 500$ (ten), 4.4 at $L = 1000$ (eight) and 7.3 at $L = 2000$ (six); without the degenerate law-school setting the medians are 3.6, 3.8, 4.4 and 5.4. Excluding the degenerate law-school setting discussed below, it ranged from 1.7 to 7.2 at $L = 250$ and from 4.1 to 15.7 at $L = 2000$. The efficiency grows with the budget because larger budgets allow the design to label a larger share of the small disagreement strata, where the variance of the comparisons is concentrated.

The GROUP design had a median efficiency of 1.0 at every budget. Oversampling the minority group, on its own, therefore did not improve the comparison of models in these settings: the variance of a contrast between two models is driven by the instances on which they disagree, and these are spread over both groups. Relative to GROUP, PATTERN was 3.8 to 5.8 times as efficient (medians by budget). The allocation computed from the true labels (ORACLE-ALLOC) was only marginally better than the proxy allocation (medians 3.8, 4.1, 4.1 and 7.8), and at $L = 1000$ the proxy allocation was slightly ahead; the average predicted probability of nine models is an adequate proxy for computing the allocation.

Efficiency differed by estimand. At $L = 1000$ it was 6.8 for θ^{TPR}, 4.1 for θ^{FPR} and 4.0 for θ^{ERR} in Adult with sex as the group, and 13.3, 8.7 and 3.5 in Adult with race; contrasts of true positive rates, which depend on the rarest cells (positives in the minority group), gained most in these two settings. For the pairs formed by the three most accurate models, which are the comparisons a practitioner is most likely to care about, the median efficiency was 2.7, 3.8, 3.6

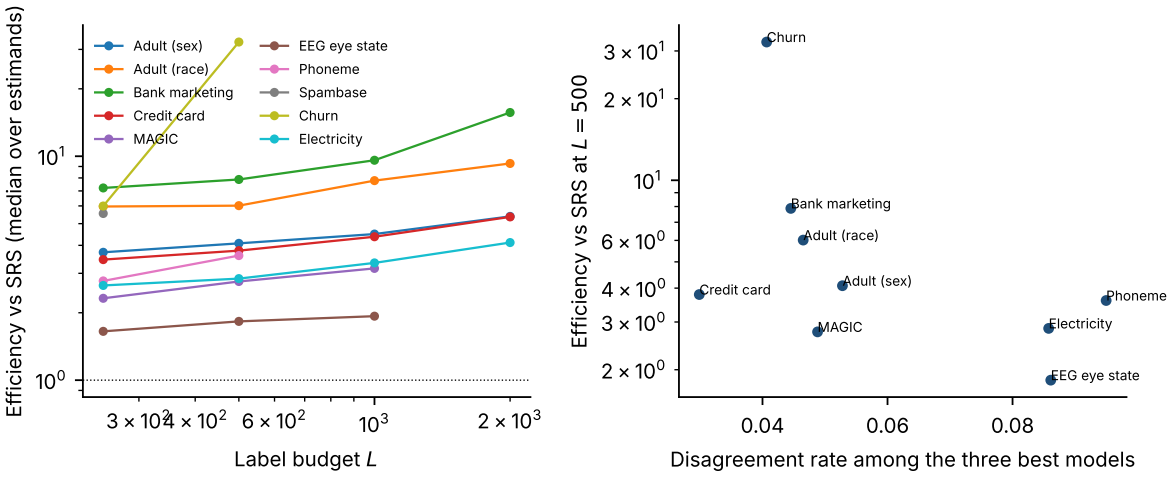


Figure 5: Left: efficiency of PATTERN relative to SRS (median over the 108 comparison estimands) by label budget, one line per setting (law school omitted). Right: efficiency at $L = 500$ against the disagreement rate among the three most accurate models.

and 5.6 at the four budgets.

7.3 Efficiency and disagreement

The gain is larger when the models disagree less (Figure 5, right). The Spearman correlation between the disagreement rate and the efficiency at $L = 500$ was -0.65 over nine settings ($p = 0.06$), and -0.62 at $L = 250$ over ten settings ($p = 0.05$). The three settings with the highest disagreement (EEG, electricity and phoneme, 8.6% to 9.5%) had the smallest gains, 1.7 to 3.6. The relation is not exact, because efficiency also depends on the size of the minority-positive cells and on the magnitude of the contrasts; Section 5 makes this precise and Section 6 checks it.

Two settings are atypical. In the law-school data all nine models predict almost identically (the three best models disagree on fewer than 0.01% of instances); PATTERN then labels almost every disagreement instance and the contrasts are estimated nearly exactly, so the efficiency ratios (10 to 21 at budgets up to 1,000, and numerically unbounded at $L = 2000$) say little beyond the fact that SRS wastes labels on agreements. In the churn data (a pool of 2,000) the efficiency at $L = 500$ was 32 overall, driven by contrasts of false positive rates and of errors (31 and 62) that are concentrated in very few instances.

7.4 Inference and decisions

Table 8 reports the coverage of nominal 95% intervals and the probability of estimating the correct sign of a contrast whose true value exceeds 0.02 in absolute value. The mean coverage of the PATTERN intervals was 0.944 at $L = 250$ and between 0.947 and 0.951 at larger budgets. Under SRS it was 0.931 to 0.949, and the worst-covered estimand had coverage as low as 0.11 at $L = 250$, because SRS samples often contain no disagreement instance in a small cell and then produce a zero standard error. The worst-covered estimand under PATTERN had coverage 0.53 at $L = 250$ (in the churn setting), 0.83 at $L = 500$ and 0.89 at $L \geq 1000$; small budgets in small

Table 7: Efficiency (ratio of mean squared errors, median over estimands) by setting and budget. "vs GROUP": efficiency of PATTERN relative to the GROUP design. The last three columns are medians over the estimands of each type. Values above 1000 arise when PATTERN labels essentially all disagreement instances.

Setting	L	Efficiency vs SRS			vs GROUP	PATTERN by estimand		
		GROUP	PATTERN	ORACLE	PATTERN	θ^{TPR}	θ^{FPR}	θ^{ERR}
Adult (sex)	250	1.0	3.7	3.8	3.8	5.7	3.7	3.1
	500	1.0	4.1	4.2	4.3	5.6	4.1	3.7
	1000	1.0	4.5	4.4	4.5	6.8	4.1	4.0
	2000	1.0	5.4	5.2	5.4	8.2	5.4	4.5
Adult (race)	250	1.6	6.0	5.6	4.1	10.0	6.9	2.6
	500	1.5	6.0	6.4	4.6	10.8	6.6	3.1
	1000	1.7	7.8	7.8	5.3	13.3	8.7	3.5
	2000	1.7	9.3	9.4	6.2	28.0	11.7	3.9
Bank marketing	250	2.2	7.2	6.7	4.0	7.2	8.7	4.2
	500	1.9	7.9	7.7	5.6	7.5	14.3	5.8
	1000	2.1	9.6	10.3	7.2	9.6	18.9	6.3
	2000	2.2	15.7	14.9	10.5	15.6	37.2	8.0
Credit card	250	1.1	3.5	3.4	3.1	2.8	3.7	4.5
	500	1.0	3.8	4.0	3.5	3.7	3.8	4.3
	1000	1.0	4.4	3.9	4.4	4.4	3.7	4.9
	2000	1.0	5.4	6.2	5.3	4.7	6.8	6.4
Law school	250	1.0	10.2	9.3	8.5	15.2	0.0	10.2
	500	0.8	15.3	17.8	16.8	26.4	0.0	15.3
	1000	0.7	20.7	29.5	24.0	64.5	0.0	20.7
	2000	0.7	>1000	>1000	>1000	>1000	0.0	>1000
MAGIC	250	1.0	2.3	2.4	2.4	2.2	2.3	2.5
	500	1.1	2.8	2.8	2.5	2.7	2.8	2.8
	1000	1.1	3.2	3.4	3.0	3.0	3.2	3.4
EEG eye state	250	1.0	1.7	1.8	1.6	1.7	1.6	1.8
	500	1.0	1.8	1.9	1.8	2.0	1.7	1.9
	1000	1.0	1.9	2.1	1.9	2.0	1.8	2.2
Phoneme	250	1.0	2.8	2.7	2.5	2.1	3.1	2.7
	500	1.0	3.6	3.6	3.3	2.9	3.9	3.7
Spambase	250	1.0	5.6	5.8	4.8	5.9	7.2	3.9
Churn	250	0.9	6.0	6.0	6.5	3.8	6.0	7.8
	500	1.0	32.4	50.5	30.7	4.6	31.0	62.4
Electricity	250	1.0	2.6	2.7	2.7	2.7	2.7	2.6
	500	1.0	2.8	2.8	2.9	2.7	2.8	3.0
	1000	1.0	3.3	3.3	3.4	3.1	3.2	3.7
	2000	1.0	4.1	4.5	4.0	3.3	4.3	5.0

Table 8: Quality of inference by budget (medians over settings unless stated). Coverage: share of nominal 95% intervals containing the true value, averaged over estimands (mean) or for the worst estimand over all settings (worst). Correct sign: share of estimates with the sign of the true contrast, among contrasts with absolute value above 0.02. SRS failures: largest share of SRS samples with an undefined estimate.

L	Cells	Coverage, mean		Coverage, worst target		Correct sign		SRS failures
		SRS	PATTERN	SRS	PATTERN	SRS	PATTERN	(max)
250	11	0.931	0.944	0.10	0.53	0.85	0.92	0.005
500	10	0.941	0.947	0.28	0.82	0.88	0.95	0.000
1000	8	0.947	0.951	0.71	0.89	0.92	0.97	0.000
2000	6	0.949	0.948	0.53	0.89	0.91	0.98	0.000

pools are where the Wald intervals are least reliable, and Section 9 examines alternatives. The probability of getting the direction of a contrast right rose from 0.85 to 0.92 at $L = 250$ and from 0.92 to 0.97 at $L = 1000$ (medians over settings), so that with 1,000 labels the design almost always ranks two models correctly on a fairness criterion whenever they differ by more than two percentage points.

8 Comparison with existing labelling designs

PATTERN combines three ingredients that are each known: sampling where models disagree, stratification on a model-based proxy with Neyman-type allocation, and design-based ratio estimation. This section asks whether a reviewer's natural alternatives, built from one or two of these ingredients, already do as well. We implemented five families of baselines (B1–B5 below; code in `tcs/labeling_baselines.py`, study in `experiments/label_baselines.py`) and ran them with SRS and PATTERN on the eleven settings of Section 7, at $L \in \{250, 500, 1000, 2000\}$ with $L \leq N/4$ (35 setting–budget cells) and $R = 200$ replications per cell.

Common ground. To make the comparison about the choice of labels and estimators and nothing else, every method estimates the same vector of totals T (Section 3), plugs it into the same map $\hat{\theta} = f(\hat{T})$ for all $3\binom{M}{2} = 108$ comparison estimands $\theta^{\mathrm{TPR}}_{mj}, \theta^{\mathrm{FPR}}_{mj}, \theta^{\mathrm{ERR}}_{mj}$, and reports 95% Wald intervals from a linearisation variance. Every method that uses a proxy uses the same $\pi_i = M^{-1}\sum_m \hat{p}_{mi}$, and every allocation over several estimands uses the same weights $w_k = 1/S_k^2$ as (8). The budget is counted in distinct labelled instances. Pool labels are used only to answer label queries, to compute the truth and, for reporting, to pick the three most accurate models. Estimands whose numerator is identically zero in the pool (the two models never disagree on the relevant instances) are estimated exactly by every design and are left out of all summaries. This removes 51 of the 108 estimands in the law-school setting, including all those among its three most accurate models, and none elsewhere.

8.1 Baselines

B1: importance sampling for active comparison. Sawade et al. (2012) compare the risks of two fixed models by drawing n instances with replacement from an instrumental distribution q

and estimating each risk by the self-normalised weighted mean $\sum_k \ell_k/q_k / \sum_k 1/q_k$ (with $p(x) = 1/N$ on a pool). Their Theorem 1 shows that the asymptotic variance of the estimated risk difference is minimised by $q^*(x) \propto p(x)\{\mathbb{E}[(\ell_1 - \ell_2 - \Delta)^2 \mid x]\}^{1/2}$. They replace $p(y \mid x)$ by an equal mixture of the models' predictive distributions, and for k models they use the equal mixture of the pairwise-optimal distributions. We keep all of this and change only the quantity being compared. With replacement draws $i_1, \ldots, i_n \sim q$, our totals are the Hájek (self-normalised) estimators

$$\hat{T}^{\mathrm{IS}} = N \frac{\sum_{k=1}^n t_{i_k}(Y_{i_k})/q_{i_k}}{\sum_{k=1}^n 1/q_{i_k}}, \qquad \hat{\theta} - \theta \approx \frac{1}{n}\sum_{k=1}^n \frac{a_{i_k}}{q_{i_k}}, \quad a_i = \nabla f(T)^\top \{t_i(Y_i) - T/N\}.$$

For $\theta^{\mathrm{ERR}}_{mj}$, $a_i = \{|\hat{y}_{mi} - Y_i| - |\hat{y}_{ji} - Y_i| - \theta^{\mathrm{ERR}}_{mj}\}/N$, which is exactly Sawade et al.'s centred loss difference. For $\theta^{\mathrm{TPR}}_{mj}$, the self-normalisation is automatic because f is a ratio, and $a_i = \sum_g s_g \mathbf{1}\{G_i = g\} Y_i\{(\hat{y}_{mi} - \hat{y}_{ji}) - (\mathrm{TPR}_{mg} - \mathrm{TPR}_{jg})\}/P_g$. The asymptotic variance is $n^{-1}\sum_i \mathbb{E}[a_i^2]/q_i$ (Hansen and Hurwitz, 1943), so the single-estimand optimum is $q_i \propto \{\mathbb{E}_\pi a_i^2\}^{1/2}$, with the expectation taken under $Y_i \sim \mathrm{Bernoulli}(\pi_i)$ and a evaluated at the proxy totals. This is our adaptation to TPR and FPR gaps. For a pair (m, j) we combine its three estimands as $q_i^{(mj)} \propto \{\sum_{k\in(m,j)} w_k \mathbb{E}_\pi a_{ik}^2\}^{1/2}$. The design distribution is the equal mixture over the 36 pairs, as in Sawade et al., mixed with the uniform distribution with weight 0.05 so that every q_i is bounded away from zero. Draws continue until L distinct instances are labelled; repeated draws reuse the label. The median number of draws was $1.12L$ (range $1.02L$–$1.48L$), so B1 effectively gets more than L draws for L labels. The variance is estimated by n^{-1} times the sample variance of $\hat{a}_{i_k}/q_{i_k}$.

B2: proxy-score stratification. The classic approach to estimating a classifier's metrics with few labels stratifies on the classifier's score and allocates optimally (Bennett and Carvalho, 2010b; Kumar and Raj, 2018b). We form strata as group × within-group quantile bins of π_i (10 bins, Score-10; and 5 bins, Score-5), merge strata smaller than 50 as for PATTERN, and use the same estimator (6), variance (7) and proxy-Neyman allocation (8). B2 therefore differs from PATTERN only in the strata. It uses the proxy strongly but ignores which models disagree.

B3: disagreement-focused two-phase design. Labelling only where models disagree is the "active $\neq$" baseline of Sawade et al. (2012) and the discordant-pair approach of Musgrove et al. (2023b) and of the screening-test literature (Pepe and Alonzo, 2001). On its own it identifies only ratios, because the denominators P_g, Q_g need labels in the agreement set. Let $D = \{i : 0 < K_i < M\}$ and $A = U \setminus D$. Phase 1 draws a simple random sample s_0 of size n_0 from U. Phase 2 draws a simple random sample of $L - n_0$ from $D \setminus s_0$. Conditionally on $n_A = |s_0 \cap A|$, the labelled units in A are a simple random sample from A. Conditionally on $|s_0 \cap D|$, the union of $s_0 \cap D$ and the phase-2 sample is a simple random sample of size $n_D = |s_0 \cap D| + L - n_0$ from D. The combined estimator is therefore the two-stratum estimator $\hat{T} = N_A \bar{t}_A + N_D \bar{t}_D$ with variance (7) at the realised (n_A, n_D), which is conditionally design-unbiased. We take $n_0 = L/4$ (Dis-25) and $n_0 = L/2$ (Dis-50).

B4: prediction-powered inference. PPI (Angelopoulos et al., 2023) corrects the predictions on the whole pool with the labelled residuals. With the prediction of t_i taken as its proxy mean $\tilde{t}_i = \pi_i u_i + (1-\pi_i)v_i$, the PPI estimator of a total under SRS is the difference estimator

$$\hat{T}_\lambda^{\mathrm{PPI}} = \lambda \sum_{i \in U} \tilde{t}_i + \frac{N}{n} \sum_{i \in s} \{t_i(Y_i) - \lambda \tilde{t}_i\},$$

with $\lambda = 1$ (PPI). PPI++ (Angelopoulos et al., 2026) tunes λ for each estimand. In our finite-pool setting the predictions are known for every unit, so the $(1 + n/N)$ factor of PPI++ drops out and the variance-minimising value is the regression coefficient of $\nabla f^\top t_i$ on $\nabla f^\top \tilde{t}_i$ in the sample. We clip it to $[0, 1]$. StratPPI (Fisch et al., 2024) applies PPI++ within strata with a stratum-specific λ_{hk} and allocates the budget by a Neyman rule on the residual standard deviation, estimated from the predictor's own confidence scores. We use the B2 strata (group $\times$ 10 proxy bins), per-stratum and per-estimand λ_{hk} clipped to $[0, 1]$, and the allocation $n_h \propto N_h \{\sum_k w_k \overline{\pi(1-\pi)(A_k - B_k)^2}_h\}^{1/2}$. Here $\pi_i(1-\pi_i)(A_{ik} - B_{ik})^2$ is the proxy variance of the residual $z_{ik} - \mathbb{E}_\pi z_{ik}$, so this allocation is Fisch et al.'s confidence-based rule for our estimands. In all three versions the variance is the stratified formula (7) applied to the residuals $z_{ik} - \lambda \nabla f^\top \tilde{t}_i$.

B5 (extension): PATTERN with a model-assisted estimator. The PPI correction is a difference or GREG-type estimator (Särndal et al., 1992, Chapter 6) and can be applied to the PATTERN sample without changing the design. PATTERN-DIF uses the stratified difference estimator with $\lambda = 1$. PATTERN-PPI tunes λ_{hk} per stratum as in StratPPI. B5 is an extension of our method, not a competitor, and it is reported separately.

Not run. Active testing with LURE (Kossen et al., 2021), adaptive variance-minimising selection (Hara et al., 2024), and active inference (Zrnic and Candès, 2024; Li et al., 2025) select labels sequentially and target one estimand of one model. Their fixed-budget, one-shot analogue for our vector of estimands is B1, which uses the same proxy-optimal inclusion principle. Model-based estimators that need a fitted joint model of labels and scores (for example SSME) are outside the design-based comparison made here.

8.2 Results

Table 9 summarises all methods by budget. PATTERN is the most efficient of the designs that use the same plain estimator as SRS (B1 to B3), at every budget: its median efficiency against SRS is 3.75, 3.95, 4.35 and 7.43 at $L = 250, 500, 1000, 2000$. It is not the most efficient method overall. StratPPI, which combines proxy strata with a model-assisted estimator, matches or exceeds it in the paired medians at $L = 250$, 500 and 1000 (ratios 1.01, 1.12, 1.08) and has the higher median efficiency against SRS at $L = 500$, 1000 and 2000 (4.45, 4.58, 8.50), and the extensions B5, which add the same kind of correction to the PATTERN sample, are better than both. Among the competing designs the closest are importance sampling (B1: 3.09, 2.77, 3.48, 4.30 against SRS; 0.54–0.79 of PATTERN) and StratPPI (B4: 3.07, 4.45, 4.58, 8.50; 1.01, 1.12, 1.08, 0.83 of PATTERN). Proxy-score strata alone (B2) reach 1.45–4.16 against SRS, the disagreement-only designs (B3) 1.7–2.5 and PPI/PPI++ with simple random labels 1.8–2.1, all

Table 9: Baselines by label budget L (number of settings in brackets). Efficiency: ratio of mean squared errors, median over the non-degenerate comparison estimands within a setting, then median over settings; "vs SRS" is MSE(SRS)/MSE(method), "vs PATTERN" is MSE(PATTERN)/MSE(method), so values above 1 favour the method. Coverage: nominal 95% Wald intervals, mean over estimands and settings; "worst": lowest single-estimand coverage over all settings and budgets. $R = 200$ replications per setting and budget. Source: `results/label_baselines.json`.

	$L = 250$ (11)		$L = 500$ (10)		$L = 1000$ (8)		$L = 2000$ (6)		All (s, L)		
Method	vs SRS	vs PAT.	vs SRS	vs PAT.	vs SRS	vs PAT.	vs SRS	vs PAT.	Cover.	Worst	ms
SRS	1	0.27	1	0.25	1	0.23	1	0.14	0.925	0.16	8
PATTERN	3.75	1	3.95	1	4.35	1	7.43	1	0.947	0.52	8
IS (B1)	3.09	0.79	2.77	0.74	3.48	0.67	4.30	0.54	0.950	0.60	9
Score-10 (B2)	1.45	0.40	2.09	0.52	2.24	0.56	4.16	0.50	0.941	0.17	8
Score-5 (B2)	1.97	0.54	2.24	0.57	2.15	0.52	3.18	0.43	0.943	0.37	8
Dis-25 (B3)	2.25	0.42	2.27	0.55	2.04	0.35	2.48	0.26	0.942	0.45	9
Dis-50 (B3)	1.84	0.43	1.94	0.37	1.68	0.34	1.93	0.25	0.940	0.36	9
PPI (B4)	1.82	0.49	1.82	0.47	1.86	0.44	1.81	0.27	0.940	0.27	34
PPI++ (B4)	2.10	0.49	1.97	0.48	1.93	0.44	1.89	0.27	0.929	0.16	49
StratPPI (B4)	3.07	1.01	4.45	1.12	4.58	1.08	8.50	0.83	0.924	0.14	44
Extension (B5): PATTERN sample, model-assisted estimator											
PATTERN-DIF (B5)	6.57	1.63	6.23	1.48	6.64	1.48	15.5	1.32	0.946	0.64	35
PATTERN-PPI (B5)	6.71	1.68	6.37	1.55	6.83	1.54	15.5	1.33	0.929	0.20	51

clearly below PATTERN (medians 0.25–0.57 of its efficiency). PPI and PPI++ with SRS labels barely move with L (1.8–2.1): they use the proxy but not the structure that the numerators vanish where models agree. The disagreement-only designs also improve little with the budget (1.7–2.5). A plausible reason is that their agreement-region labels are drawn by SRS at a fixed share of the budget, so the denominators remain noisy, but we did not isolate this effect.

Where a baseline beats PATTERN. Table 10 and Figure 6 show where the ranking changes. First, StratPPI beats PATTERN at $L = 500$ in seven of the ten settings of the table (ratios 1.01–1.57 in Bank, Credit-card, Law school, MAGIC, EEG, Phoneme and Electricity, and 1.13 in Spambase at $L = 250$), and is below it in Adult-sex (0.97), Adult-race (0.93) and Churn (0.24). Over all cells its ratio exceeds 1 in 19 of 35 (median 1.12 at $L = 500$). The margin is small in most settings, and it is not a failure of the stratification: StratPPI uses the same sample size but a model-assisted estimator, and B5 shows that the same correction applied to the PATTERN sample improves PATTERN itself by a median factor of 1.3–1.7. Second, Dis-25 beats PATTERN on Churn at $L = 500$ (ratio 1.52); there the pool has a 19% disagreement set, PATTERN strata are mostly small, and Dis-25 labels most of its sample inside the disagreement set. Its intervals are not reliable in that cell: mean coverage 0.93 but worst-estimand coverage 0.66, so the apparent gain should be read with caution. Third, PPI++ beats PATTERN on EEG (ratio 1.19 at $L = 500$), the setting with the largest disagreement set (73%), where agreement cells hold little and the proxy is informative. Importance sampling never beats PATTERN in the all-estimand median (ratio below 1 in all 35 cells), but on the pairs among the three most accurate models it is competitive: its median efficiency against SRS on those pairs is 4.2, 4.1, 4.2 and 6.3 at $L = 250, \ldots, 2000$, against 3.4, 3.6, 4.0 and 6.8 for PATTERN, and it beats PATTERN on those pairs in 10 of 35 cells (Adult-sex: 1.49 at $L = 250$ and 1.10 at $L = 500$,

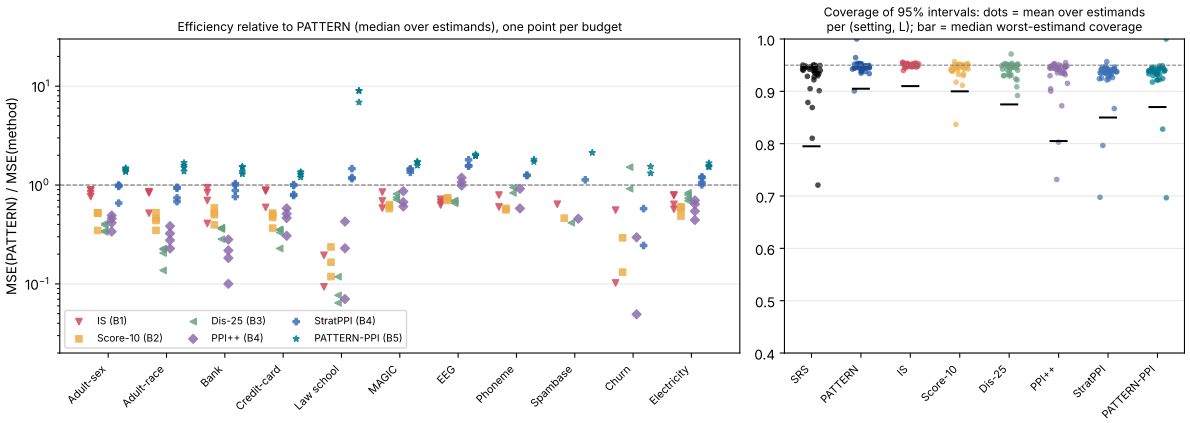


Figure 6: Left: MSE(PATTERN)/MSE(method) by setting, one point per budget, median over comparison estimands (values above 1 mean that the method beats PATTERN; the axis is clipped to $[0.02, 30]$ and law school at $L = 2000$ is not shown in range because PATTERN labels the whole disagreement set there). Right: mean coverage of the nominal 95% interval over estimands for each (setting, L), with the median over cells of the worst-estimand coverage as a bar. Source: `results/label_baselines.json`.

while its all-estimand ratio there is 0.89). Where the question is only about the best few models, importance sampling is a serious alternative.

Where PATTERN wins clearly. In the two settings with the lowest disagreement rate, Law school and Churn, PATTERN is 6–84 times more efficient than SRS at $L \leq 1000$ and PATTERN labels essentially every disagreement instance at $L = 2000$ in Law school (all non-degenerate estimands then have error close to zero, ratio above 10^{20}, so these cells are shown as >100 and excluded from the figure range). IS, Dis-50, score strata and PPI++ reach only 0.04–0.7 of PATTERN's efficiency there (Dis-25 on Churn is the exception discussed above), because none of them isolates the agreement region exactly: PATTERN gives it a stratum of its own that can be labelled sparsely, while the other designs either draw it at random or leave the disagreement set under-sampled.

Coverage and failures. Over all 35 cells the mean 95% coverage is 0.925 (SRS), 0.947 (PATTERN), 0.950 (IS), 0.941 (Score-10), 0.942 (Dis-25), 0.929 (PPI++), 0.924 (StratPPI), 0.946 (PATTERN-DIF) and 0.929 (PATTERN-PPI). The shortfalls concentrate in the smallest cells. The lowest worst-estimand coverages over all cells are 0.14 for StratPPI (Churn, $L = 250$), 0.16 for SRS and for PPI++, 0.17 for Score-10, 0.20 for PATTERN-PPI, 0.45 for Dis-25 and 0.52 for PATTERN (Churn, $L = 250$). The tuned estimators (PPI++, StratPPI, PATTERN-PPI) undercover when per-stratum tuning parameters are estimated from a few labels, notably in Law school where PATTERN-PPI covers 0.70 at $L = 250$ and StratPPI 0.70–0.87 below $L = 2000$. PATTERN-DIF, which fixes $\lambda = 1$, keeps the coverage of PATTERN (0.946). Undefined estimates (a group with no sampled positive) occurred in at most 1% of replications for any method (Dis-25, Adult-race, $L = 250$: 1%; SRS 0.5%; PATTERN and the others none). Runtimes per replicate (design computation excluded) are 11–16 ms for the stratified and importance-sampling designs and 45–68 ms for the PPI-type estimators, which tune per estimand and stratum. Design

Table 10: Efficiency relative to PATTERN, MSE(PATTERN)/MSE(method), median over estimands, by setting at $L = 500$ ($L = 250$ for Spambase, whose pool is too small for $L = 500$). Values above 1 mean that the method beats PATTERN. Second column: efficiency of PATTERN relative to SRS. Last two columns: mean coverage of the 95% intervals of StratPPI and of PATTERN-PPI. Source: `results/label_baselines.json`.

	PAT.	Baselines (B1–B4) vs PATTERN						B5 vs PATTERN		Coverage	
Setting	vs SRS	IS	Score-10	Dis-25	Dis-50	PPI++	StratPPI	PATTERN-DIF	PATTERN-PPI	StratPPI	PAT-PPI
Adult-sex	4.19	0.89	0.52	0.39	0.37	0.46	0.97	1.45	1.49	0.935	0.939
Adult-race	7.27	0.86	0.46	0.23	0.22	0.32	0.93	1.44	1.58	0.939	0.939
Bank	8.26	0.84	0.50	0.36	0.38	0.22	**1.03**	1.47	1.52	0.941	0.933
Credit-card	3.71	0.89	0.52	0.36	0.35	0.51	**1.01**	1.28	1.35	0.937	0.933
Law school	39.0	0.09	0.24	0.08	0.08	0.23	**1.47**	11.1	8.97	0.797	0.828
MAGIC	2.93	0.69	0.63	0.75	0.63	0.67	**1.43**	1.70	1.73	0.942	0.943
EEG	1.70	0.67	0.73	0.70	0.66	**1.19**	**1.57**	1.98	2.05	0.926	0.939
Phoneme	3.53	0.60	0.58	0.83	0.67	0.58	**1.26**	1.71	1.72	0.946	0.941
Spambase*	5.77	0.64	0.46	0.42	0.37	0.46	**1.13**	2.14	2.12	0.925	0.933
Churn	34.6	0.10	0.13	**1.52**	0.28	0.05	0.24	1.28	1.32	0.930	0.932
Electricity	2.86	0.79	0.59	0.80	0.73	0.64	**1.21**	1.49	1.53	0.942	0.933

computation is at most 0.8 s per cell for every method.

The model-assisted extension. PATTERN-DIF and PATTERN-PPI (bottom block of Table 9) improve on PATTERN in every cell except Law school at $L = 2000$, where PATTERN is already exact, by a median factor of 1.3–1.6 (by budget) and up to 11 in Law school at $L \leq 500$; they are the best methods in this study, with efficiency against SRS of 6.6, 6.2, 6.6 and 15.5 for PATTERN-DIF. The estimated-λ version adds little over $\lambda = 1$ (PATTERN-PPI is within 3% of PATTERN-DIF in the median) and costs coverage, so we recommend the fixed-λ version if the extension is used. Because the extension changes the estimator rather than the design, its variance expressions are those of the difference estimator under stratified sampling (Särndal et al., 1992); its validity as a design-based method in our setting follows from the known proxy totals, but we did not extend the theory of Section 5 to it, and it is reported here as an empirical result.

Limitations. The comparison is restricted to fixed-budget, one-shot designs on cached pools of nine models with logistic-type probabilities; the proxy is the same for every method, so a better proxy for any of them (for example a calibrated or learned one) could change the ranking, especially for StratPPI and PATTERN-PPI which rely on it directly. The importance-sampling baseline draws with replacement and so spends more than L draws for L labels; counting draws instead of labels would lower its efficiency. $R = 200$ replications give Monte Carlo errors of about 10% on a single mean squared error, so differences between 0.9 and 1.1 in Table 10 should not be read as ranking. Medians over estimands hide estimand-level differences: on the three types, PATTERN's median efficiency against SRS is 6.2 (TPR), 4.8 (FPR) and 3.8 (ERR), against 3.0, 3.5 and 3.0 for IS and 5.3, 5.2 and 2.3 for StratPPI.

9 Ablations and robustness

This section varies one ingredient of the PATTERN design at a time and asks two questions: does the estimator remain valid, and how much efficiency is gained or lost relative to the default? The default is the design of the main study: strata (G_i, K_i) with strata smaller than 50 merged within group, weighted generalised Neyman allocation (Neyman, 1934) computed with the ensemble-mean proxy π_i, a floor of 10 labels per stratum, and 95% Wald intervals from the linearisation variance. All results are from `experiments/label_ablations.py` (helpers in `tcs/labeling_variants.py`) and are stored in `results/label_ablations.json`.

Protocol. We use the eleven cached pools and budgets $L \in \{250, 500, 1000\}$ with $L \leq N/4$, which gives 29 (setting, budget) cells; $R = 150$ replications per cell and design. With $M = 9$ models there are 108 comparison targets ($\theta_{mj}^{TPR}, \theta_{mj}^{FPR}, \theta_{mj}^{ERR}$ for 36 pairs). Efficiency is the per-target ratio $\mathrm{MSE}_{\mathrm{SRS}}/\mathrm{MSE}_{\mathrm{design}}$ computed on the replications in which both estimates are defined, summarised by its median over targets and then by its median over cells; per-target ratios are clipped to $[10^{-4}, 10^{4}]$ because some designs make a target exact. "Relative to default" is the median over targets of $\mathrm{MSE}_{\mathrm{PATTERN}}/\mathrm{MSE}_{\mathrm{variant}}$, so values above one favour the variant. Coverage is reported as the mean over targets and as the worst target. With $R = 150$ the Monte Carlo standard error of a single coverage is about 0.018, and the minimum over 108 independent targets that all cover exactly 95% has mean 0.900 (5% quantile 0.88) in a simulation stored with the results. A worst-target coverage near 0.90 is therefore consistent with nominal behaviour; values well below 0.88 are not. A design that uses more strata than the budget can feed at 10 labels each gets a smaller floor ($\lfloor 0.8L/H \rfloor$, at least 2). Undefined estimates (a denominator total estimated as zero) occurred in at most 0.7% of replications for any design.

9.1 Stratifier and proxy (A1, A2)

Table 11 compares four alternative stratifiers and eight alternative allocation proxies; per-cell values are in Table 15 of the appendix. The binary stratifier (group × unanimous/not) keeps 71% of the default's precision (median), and the joint prediction pattern of the three models agreeing most often with the thresholded ensemble mean keeps only 37%. That stratifier isolates disagreement among three models, but most of the 36 pairs involve the other six models. Refining the default by splitting each (G, K) cell at the median of π (K×PROXY-2) is a tie: median relative efficiency 1.00, better in 15 of 29 cells, 0.98 at $L = 250$ and 1.05 at $L = 1000$. Tertile bins do better only at $L = 1000$ (1.02) and lose at $L = 500$ (0.89), because the extra strata eat the budget through the per-stratum floor. The count K_i is therefore the part of the stratifier that matters; within-K proxy bins add little.

The allocation is remarkably insensitive to the proxy. Replacing π_i by a single model's probabilities (logistic regression or Gaussian naive Bayes), by the constant 0.5, by π shuffled within group, by logit noise with standard deviation 2, or even by the reversed proxy $1 - \pi_i$ changes the median efficiency by at most 4% at any budget, and every variant beats the default in about a quarter to a half of the cells. The allocation computed with the true labels (ORACLE-ALLOC) is only 1% better in the median. This is consistent with the structure of the estimands. The influence values z_i vary within a stratum mainly through the differences $\hat{y}_{mi} - \hat{y}_{ji}$, which are

Table 11: Stratifier (A1) and proxy (A2) ablations. Efficiency vs SRS: median over cells of the median over targets of $\mathrm{MSE}_{\mathrm{SRS}}/\mathrm{MSE}$. Rel.: median over all 29 cells of the median-over-targets $\mathrm{MSE}_{\mathrm{PATTERN}}/\mathrm{MSE}_{\mathrm{variant}}$; Win: share of cells with Rel. > 1. Coverage: mean over cells of the mean over targets (mean) and of the worst target (worst), nominal 0.95. Source: `results/label_ablations.json`.

	Efficiency vs SRS					Coverage	
Variant	$L = 250$	500	1000	Rel.	Win	mean	worst
PATTERN (default)	3.63	3.93	4.34	1.00	–	0.947	0.858
A1: stratifier (ensemble-mean proxy)							
Group × agree/disagree	2.44	2.61	2.60	0.71	0.00	0.949	0.832
Group × pattern of 3 models	1.40	1.51	1.61	0.37	0.00	0.940	0.759
Group $\times K\times$ 2 proxy bins	3.77	3.68	4.47	1.00	0.52	0.946	0.856
Group $\times K\times$ 3 proxy bins	3.25	3.63	4.55	0.97	0.41	0.948	0.863
A2: allocation proxy (default strata)							
Single model (logistic)	3.69	3.82	4.22	0.99	0.41	0.946	0.862
Single model (naive Bayes)	3.67	3.97	4.34	1.01	0.52	0.941	0.854
Constant 0.5	3.71	3.96	4.31	0.98	0.45	0.941	0.849
π shuffled within group	3.40	3.83	4.10	0.98	0.38	0.944	0.852
π with logit noise (sd 2)	3.20	3.98	4.11	0.98	0.24	0.945	0.858
Reversed, $1 - \pi$	3.55	3.82	4.50	1.00	0.48	0.947	0.857
True labels (ORACLE-ALLOC)	3.79	3.80	4.30	1.01	0.52	0.947	0.865
No proxy (proportional)	2.38	2.06	1.96	0.52	0.03	0.945	0.818
References							
GROUP	1.04	1.01	1.06	0.26	0.00	0.941	0.723
PATTERN, gap-targeted allocation (A5)	2.85	2.78	2.91	0.74	0.00	0.944	0.836

known without labels, so any proxy that is not degenerate puts labels where models disagree. What does matter is the Neyman step itself: proportional allocation over the same strata keeps only 52% of the default's precision. Coverage is unaffected by the proxy in all but one cell (mean coverage 0.941–0.947 for every proxy). The exception is Churn at $L = 500$, where the constant and naive Bayes proxies label the small disagreement strata exhaustively. This makes many θ^{ERR} targets exact (efficiency over 1000 relative to SRS), but it leaves the large unanimous strata with only the floor of 10 labels, and for 36–37% of (replication, target) pairs the estimated standard error is exactly zero. Mean coverage in that cell falls to 0.78–0.80. Validity is preserved in expectation (the totals remain unbiased whatever the proxy), but very aggressive allocations amplify the degenerate-variance problem studied in A4.

9.2 Number of compared models (A3)

For $M \in \{2, 3, 5\}$ we drew three random subsets of the nine models per size (fixed seed), rebuilt strata, proxy and estimands from the subset only, and compared PATTERN with SRS on the $3\binom{M}{2}$ comparison targets. Subsets in which all models agree on every instance (two Law school subsets) are dropped, since every target is then exactly zero. Table 12 and Figure 7 show that the gain is governed by the share of the pool on which the compared models disagree, not by M as such. Across configurations the Spearman correlation between log efficiency and the disagreement share is -0.81, -0.79 and -0.78 at $L = 250$, 500 and 1000, and within settings at $L = 250$ it is negative in all eleven settings (median -0.50). At $L = 250$, comparing two models

Table 12: Number of compared models (A3). Configs: (setting, budget, subset) combinations; disagreement: median share of the pool on which not all M models agree. Efficiency: median over configurations of the median-over-targets $\mathrm{MSE}_{\mathrm{SRS}}/\mathrm{MSE}_{\mathrm{PATTERN}}$. Coverage: PATTERN, Wald.

			Efficiency vs SRS			Coverage	
M	Configs	Disagreement	$L = 250$	500	1000	mean	worst
2	87	0.133	7.04	11.21	7.84	0.942	0.915
3	84	0.193	6.14	7.47	4.93	0.944	0.903
5	84	0.294	4.10	4.92	4.68	0.948	0.898
9	29	0.396	3.63	3.93	4.34	0.947	0.858

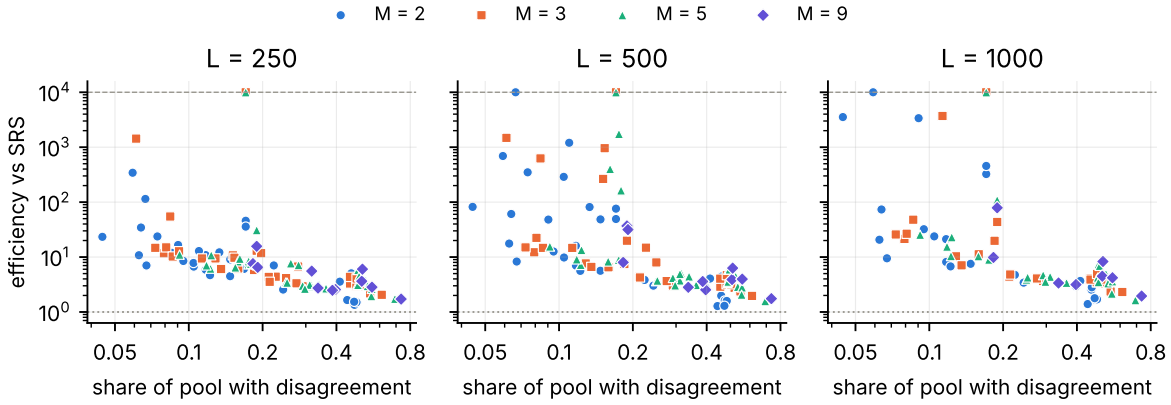


Figure 7: Efficiency of PATTERN relative to SRS (median over targets) against the share of the pool on which the compared models do not all agree, for random subsets of M models in every setting (A3). Both axes are logarithmic; the dotted line marks parity and the dashed line the clipping value 10^4.

that disagree on 4–10% of the pool gives efficiencies between 7 and 343. When the compared models disagree on 40–80% of instances (EEG, Credit card default, Adult) the gain falls to 1.3–6. Smaller M helps only because fewer models disagree less often. The points at the clipping line (10^4) are Law school subsets in which some compared models predict identically within a group, so some targets are estimated exactly by PATTERN but not by SRS. Coverage is not affected by M (mean 0.942–0.948). The worst-target figures in Table 12 cannot be compared across rows, because the number of targets grows from 3 to 108.

9.3 Small-sample inference (A4)

The main study reports worst-target coverage down to about 0.5 at $L = 250$ in Churn and Law school. The same pattern appears here: worst-target Wald coverage 0.473 for Churn and 0.633 for Law school at $L = 250$, and 0.753 for Adult-sex (Table 16). In Churn, Law school, Credit card default and Phoneme the worst target has a true value close to zero ($|\theta| \leq 0.002$). The mechanism is a stratum in which the few sampled units show no variation in z_i, typically because no sampled positive (or negative) falls where the two models disagree. The stratum's sample variance is then exactly zero, the interval collapses, and it misses a small non-zero truth. Neither of the standard small-sample corrections addresses this. A t quantile with Satterthwaite

(1946) degrees of freedom widens intervals by about 1% and leaves the worst cells unchanged. The stratified bootstrap percentile interval (Efron, 1979), which resamples within strata with $B = 999$ and so ignores the finite-population correction (a rescaled version is discussed by Rao and Wu, 1988), is also unchanged in the worst cells and is worse than Wald in Credit card default and MAGIC at $L = 250$ (Figure 8). A grid over the label floor $(5, 10, 20)$ and the collapse threshold $(25, 50, 100)$ does not help either. No combination raises the minimum over cells above 0.573 at $L = 250$, and the threshold 25 makes the worst Law school target much worse at $L \geq 500$ (0.26–0.33 at $L = 500$, about 0.66 at $L = 1000$; Table 17). The grid does not move efficiency much (median relative efficiency 0.92–1.05).

We therefore also evaluated a label-free remedy that targets the mechanism directly. We call it the proxy variance floor. In a stratum whose sample variance $s^2_{z,h}$ of the influence values is exactly zero, we replace it by $\tilde{S}^2_{z,h} = \nabla f(\hat{T})^\top \tilde{\Sigma}_h \nabla f(\hat{T})$, where $\tilde{\Sigma}_h$ is the within-stratum covariance of $t_i(Y_i)$ when the Y_i are independent Bernoulli(π_i), the same quantity the allocation already uses. Strata with $s^2_{z,h} > 0$ are untouched, and the point estimate does not change. The floor applies only on an event whose probability vanishes as n_h grows whenever the true within-stratum variance is positive, so the asymptotic theory is unaffected. Table 13 shows that it raises the mean worst-target coverage at $L = 250$ from 0.807 to 0.915 and the minimum over cells from 0.473 to 0.893, at an average interval-width cost of 21% (median 13%). At $L \geq 500$ it raises the minimum from 0.800 to 0.860. Mean coverage moves from 0.945–0.951 to 0.958–0.965, so the intervals become mildly conservative on average. They are clearly conservative in Law school (mean coverage 0.99, widths 1.8–2.4 times Wald), where most strata are unanimous and the proxy overstates the label variance there. The floor gives similar coverage on the binary, K×PROXY-2 and grid designs (Table 13), and it also repairs part of the much worse SRS coverage (minimum 0.140 to 0.567). We designed the floor after seeing the failure on these pools, so these numbers are in-sample. As a partial out-of-sample check we used sixteen synthetic pools that played no role in its design (four random designs of the non-smooth simulation model, pool sizes 2,000 and 8,000, $L \in \{250, 500\}$, nine models, 200 samples each; `experiments/floor_check.py`). There the Wald intervals already had mean coverage 0.942 to 0.952 and worst-target coverage of at least 0.89, no stratum ever had a zero sample variance, and the floor left the intervals unchanged. The check therefore shows that the floor costs nothing when it is not needed, but it does not test the situation it was built for, because the synthetic models disagree on 21% to 29% of instances and do not produce unanimous strata. We use the floor as an optional safeguard and report it alongside the Wald intervals rather than as the default.

9.4 The equalised-odds gap (A5)

The estimands of the method are smooth comparison contrasts. Practitioners also look at each model's equalised-odds gap $EO_m = \max(|\Delta^{TPR}_m|, |\Delta^{FPR}_m|)$ (Hardt et al., 2016) and rank models by it. We computed the plug-in $\widehat{EO}_m = f_{EO}(\hat{T})$ from the same samples and recorded its RMSE, the probability that the model with the smallest estimated gap is a model with the smallest true gap, and Kendall's τ (Kendall, 1938) between estimated and true gaps. We also included a variant of PATTERN whose allocation targets the $2M$ per-model gaps $\Delta^{TPR}_m, \Delta^{FPR}_m$ instead of the comparison contrasts ("gap-targeted"). Table 14 summarises the results; per-cell values are

Table 13: Small-sample inference (A4), default design unless stated. Mean: mean over cells and targets; worst: mean over cells of the worst-target coverage; min: minimum over cells of the worst-target coverage. Small pools: Churn, Spambase, Phoneme, Law school. Width: average interval width relative to Wald over all cells.

	$L = 250$, all			$L = 250$, small		$L \geq 500$		
Method	mean	worst	min	worst	min	worst	min	Width
Wald (default)	0.945	0.807	0.473	0.728	0.473	0.890	0.800	1.00
t, Satterthwaite df	0.948	0.812	0.473	0.728	0.473	0.891	0.800	1.01
Stratified bootstrap	0.953	0.801	0.473	0.753	0.473	0.919	0.800	–
Wald + proxy floor	0.965	0.915	0.893	0.923	0.907	0.906	0.860	1.19
Proxy floor on other designs								
Group × agree/disagree	0.957	0.899	0.880	0.900	0.880	0.907	0.873	1.04
Group $\times K\times$ 2 proxy bins	0.968	0.921	0.887	0.920	0.887	0.906	0.853	1.23
min n_h 20, collapse 100	0.962	0.912	0.893	0.915	0.893	0.901	0.840	1.15
SRS for reference								
SRS, Wald	0.922	0.576	0.140	0.505	0.140	0.735	0.260	1.00
SRS, Wald + proxy floor	0.936	0.788	0.567	0.787	0.567	0.841	0.593	1.01

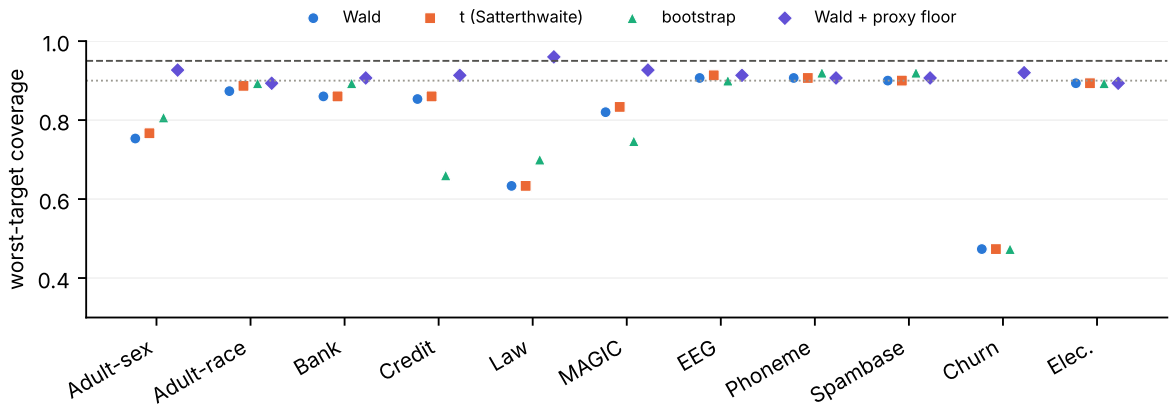


Figure 8: Worst-target coverage of the default design at $L = 250$ in each setting under four interval methods (A4). Dashed line: nominal 0.95; dotted line: expected minimum over 108 targets of 150-replicate coverage if every target covered exactly 95% (0.90).

in Table 18.

The plug-in $\widehat{EO}_m$ is biased upwards, as a maximum of absolute values must be (mean bias over cells and models 0.042 under SRS and 0.026 under PATTERN). PATTERN is more accurate than SRS in the median (efficiency 1.80–1.94) and identifies the best model more often (mean 0.46 vs 0.35 at $L = 250$; 0.40 vs 0.26 when Law school and EEG are excluded). Those two settings are easy: in Law school six models share the true minimum $EO_m = 0$, and in EEG the best model leads by 0.039. Ranking agreement improves from $\tau = 0.32$–0.48 to 0.45–0.62. The gain is not uniform, however. In 8 of 29 cells (all MAGIC and Electricity cells, Credit card default at $L = 250$, Phoneme at $L = 500$) PATTERN estimates EO_m less precisely than SRS, by a factor down to 0.51. The default allocation spends labels on disagreement, whereas the per-model gap of a single model depends on all of its positive predictions. The gap-targeted allocation is never worse than SRS for EO_m (median efficiency 2.05–2.53, more than the default), but it

Table 14: Equalised-odds gap (A5). Eff.: median over cells of the median over models of $\mathrm{MSE}_{\mathrm{SRS}}/\mathrm{MSE}$ for $\widehat{EO}_m$; P: mean probability that the estimated smallest-gap model has the smallest true gap; τ: mean Kendall τ between estimated and true EO_m.

	$L = 250$			$L = 500$			$L = 1000$		
Design	Eff.	P	τ	Eff.	P	τ	Eff.	P	τ
SRS	1.00	0.35	0.32	1.00	0.42	0.39	1.00	0.49	0.48
GROUP	1.07	0.37	0.34	1.10	0.43	0.42	1.63	0.51	0.52
PATTERN	1.83	0.46	0.45	1.80	0.51	0.49	1.94	0.62	0.62
PATTERN, gap-targeted	2.05	0.45	0.43	2.49	0.53	0.52	2.53	0.59	0.62
ORACLE-ALLOC	2.25	0.46	0.43	1.64	0.54	0.51	2.12	0.62	0.62

pays for this on the comparison contrasts, keeping only 74% of the default's precision (Table 11). Identifying the fairest model remains hard in every design when the true margin between the two best models is 0.002 or less (Adult-sex, Bank, Credit card default, Churn; Law school also has a margin of 0.002 but six models tie at zero): there the probability of picking the right model is at most 0.55 even at the largest budget.

Summary. The design is robust where robustness was in doubt and fragile where it was expected to be robust. Efficiency hardly depends on the quality of the proxy: a constant, a corrupted or even a reversed proxy loses at most a few per cent, and allocating with the true labels gains about 1%. It depends mainly on the K-count stratifier, on Neyman rather than proportional allocation, and on how rarely the compared models disagree. None of the variants beat the default stratifier or proxy consistently, so we keep them. The weak point is small-sample inference for near-zero contrasts. Neither t quantiles nor the bootstrap nor a different minimum stratum size fixes it. A label-free proxy variance floor for strata with zero sample variance does fix it on these pools, at a modest cost in width outside Law school. Finally, if the equalised-odds gaps themselves are of primary interest, an allocation that targets the per-model gaps is preferable to the default.

10 Practical guidance

This section turns the results into a procedure that an analyst can follow when comparing a small set of candidate classifiers on fairness and accuracy with a limited labelling budget.

Step 1: describe the agreement structure before labelling. Compute K_i and the stratum sizes N_h on the unlabeled pool, and the pairwise disagreement rates of the models to be compared. These numbers are free and predict the benefit of the design: in our data, disagreement rates below 5% went with efficiencies of 3.2 to 9.6 at $L = 1000$, while rates of 8.6% to 9.5% went with efficiencies of 1.9 to 3.6. If the candidate models disagree on a large share of the pool, the design still helps, but the gain is modest.

Step 2: check that the minority cells are not empty. Every group-wise rate needs positive and negative instances in each group. The proxy probabilities give an expected count of positives

per group and stratum before any label is collected; if the expected number of minority positives in the pool is very small, no design can estimate the corresponding rates, and the estimand should be reconsidered (for example by merging groups).

Step 3: choose the estimands and allocate. Use the signed contrasts (4) for the pairs of interest. If only a few pairs matter, restricting the estimand set to them concentrates labels further. Compute the allocation (8) with the average predicted probability as proxy, at least ten labels per stratum, and label the drawn instances.

Step 4: estimate and report uncertainty. Report the contrasts with standard errors and intervals. For budgets below about 500 labels or pools below a few thousand instances, use the more conservative interval options of Section 9 and report the stratum sample sizes, so that readers can judge how many minority positives informed each rate.

Step 5: decide. A contrast whose interval excludes zero supports a statement that one model has a smaller gap than the other on the stated metric. When no interval excludes zero, the honest conclusion is that the budget did not resolve the comparison; the stratum-level variances computed in Step 3 indicate how many additional labels would be needed, since the variance of every contrast decreases in proportion to the budget for a fixed allocation.

When not to use the design. The design compares fixed models; it does not help to estimate a single model's fairness more precisely than a group-stratified design does, because a single model's group-wise rates do not depend on disagreements with other models. It also assumes that the group attribute is observed for the whole pool. When the attribute is missing for some instances, it can be added as a third stratum level, but the fairness estimands then refer only to instances with a recorded group.

11 Discussion

Summary. Comparing classifiers on group-fairness criteria is expensive in labels because the relevant information sits in small cells: positives and negatives of the minority group, and instances on which the models disagree. A design that stratifies the pool by group and by the number of models predicting the positive class, and that allocates labels with a proxy-based generalised Neyman rule, reduced the labels needed for a given accuracy by a median factor of about 4 at budgets of 250 to 1,000 labels and about 7 at 2,000 labels, across eleven data settings. Oversampling the minority group alone gave no gain. The design is simple, needs no label to set up, keeps the usual design-based guarantees of stratified sampling, and its benefit can be predicted from the disagreement rate before any label is collected.

Relation to earlier work. Sampling where models disagree is not new. It is the central idea of active comparison of prediction models (Sawade et al., 2012), and verification designs that oversample discordant test results have been used to compare diagnostic tests in medicine for decades. What is specific to the present setting is that the estimands are ratios whose numerators

depend on disagreements and whose denominators depend on group prevalence everywhere, that many contrasts across several models must be estimated from one sample, and that the group is known before labelling. The allocation rule handles all three at once, and the experiments quantify the gain on public data.

Limitations. Several limitations qualify the results. First, the experiments use one split, one set of nine models and pools of 1,841 to 19,537 instances per setting; larger model collections with more diverse predictions would reduce the gain. Second, in six of the eleven settings the group is arbitrary, so these settings test the statistics of the design rather than any fairness question. Third, the intervals are asymptotic: at budgets of 250 labels in small pools, the worst-covered contrast had coverage well below the nominal level, and the alternatives studied in Section 9 only partly remove this. Fourth, the primary estimands are signed gaps; the equalised-odds gap itself is a non-smooth function of them, and inference for it needs separate treatment. Fifth, the inference is about the fixed models on the given pool; variation from retraining is not addressed. Finally, labels are assumed to be correct; label noise that depends on the group would bias every design, including this one.

Extensions. The design extends directly to more than two groups (one stratum dimension per group level), to other ratio metrics such as positive predictive value or calibration within groups, and to sequential labelling, in which the allocation is updated as labels arrive by replacing the proxy moments with estimated ones. Combining the stratification with model-assisted or prediction-powered estimators, which use the predicted probabilities also in the estimator rather than only in the design, is a natural next step (Section 8).

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A Proofs

Throughout, $x_i = a^\top \xi_i$ with $\xi_i = t_i(Y_i) - \psi_i$ for a fixed or sequence-dependent vector a and a label-free vector ψ_i with $\|\psi_i\|_\infty \le 1$ (the plain estimator is $\psi \equiv 0$, so $\xi_i = t_i(Y_i)$), and $\bar{x}_h$, $S^2_{x,h}$, $\bar{x}_{s_h}$, $s^2_{x,h}$ are the stratum population and sample means and variances. We use repeatedly two classical facts on simple random sampling without replacement of n units from N (Cochran, 1977, Ch. 2): $E(\bar{x}_s) = \bar{x}_U$ and $\mathrm{Var}(\bar{x}_s) = (1-n/N)S^2_x/n$; the same formulas hold coordinatewise and for covariances of vector-valued variables. Because all coordinates of ξ_i lie in $[-1,1]$, any population variance of a coordinate or of a product of two coordinates satisfies $S^2 \le N/(N-1) \le 2$ for $N \ge 2$ (for $\psi \equiv 0$ the sharper bound $N/\{4(N-1)\} \le 1/2$ holds).

A.1 Proof of Proposition 1

(a) Since $\sum_{i\in U} \psi_i = \sum_h N_h \bar{\psi}_h$ and $T = \sum_h N_h \bar{t}_h$, definition (13) gives $\hat{T}_\psi - T = \sum_h N_h(\bar{\xi}_{s_h} - \bar{\xi}_h)$: the known proxy total cancels exactly, which is where the fact that ψ_i is known for every unit of the pool is used. Within stratum h, $E(\bar{\xi}_{s_h}) = \bar{\xi}_h$ and $\mathrm{Cov}(\bar{\xi}_{s_h}) = (1-f_h)\Sigma^\xi_h/n_h$. The strata are sampled independently and N_h, n_h and ψ_i are constants because, by (A1), they are functions of the predictions, probabilities and groups only. Hence $E(\hat{T}_\psi) = T$ and $\mathrm{Cov}(\hat{T}_\psi) = \sum_h N_h^2(1-f_h)\Sigma^\xi_h/n_h$. For $n_h \ge 2$ the sample covariance matrix of ξ_i in stratum h is unbiased for Σ^ξ_h (Cochran, 1977, Ch. 2), which gives the unbiased estimator of $\mathrm{Cov}(\hat{T}_\psi)$. Only $n_h \ge 1$ is used for unbiasedness, and nothing is assumed about the proxy beyond its being label-free. The case $\psi \equiv 0$ is the plain estimator.

(b) Let $\hat{\tau} = \hat{T}_\psi/N$. By (a), each coordinate satisfies $\mathrm{Var}(\hat{\tau}_c) = \sum_h W_h^2(1-f_h)S^2_{\xi_c,h}/n_h \le 2r^2$, so $\|\hat{\tau} - \tau\| = O_p(r)$ by Chebyshev's inequality, and $r^2 \le \sum_h W_h^2/\min_k n_k \le 1/\min_k n_k \to 0$ by (A1). Let $K = \{x \in [0,1]^C : x_{P_g} \ge c/2,\ x_{Q_g} \ge c/2,\ g = 0,1\}$, where x_{P_g}, x_{Q_g} denote the coordinates of the denominators. K is compact and convex, and ϕ is a rational function whose denominators are bounded away from zero on K, so its first and second derivatives are bounded on K, say by B. By (A2) $\tau \in K$ with the denominators at least c; on the event $\mathcal{E} = \{\|\hat{\tau} - \tau\|_\infty \le c/2\}$, whose probability tends to one, the segment from τ to $\hat{\tau}$ lies in K, and Taylor's theorem gives

$$\left|\hat{\theta}_\psi - \theta - \nabla\phi(\tau)^\top(\hat{\tau} - \tau)\right| \le \tfrac{B}{2}\, C\, \|\hat{\tau} - \tau\|^2 = O_p(r^2). \tag{37}$$

Therefore $\hat{\theta}_\psi - \theta = O_p(r) + O_p(r^2) = O_p(r) \to 0$.

(c) Let $a = \nabla\phi(\tau)$ (it depends on ν and is bounded by B), $x_i = a^\top \xi_i$, and $\Lambda = a^\top(\hat\tau - \tau) = \sum_h W_h(\bar x_{s_h} - \bar x_h)$. Since $e_i = x_i/N$, $\mathrm{Var}(\Lambda) = \sum_h W_h^2(1-f_h)S_{x,h}^2/n_h = V_\psi(\theta; n)$, abbreviated V. By (37) and (A3), $(\hat\theta_\psi - \theta - \Lambda)/V^{1/2} = O_p(r^2)/(c^{1/2}r) = O_p(r) \to 0$, so by Slutsky's lemma it suffices to show $\Lambda/V^{1/2} \to N(0,1)$.

Since the limit does not depend on the subsequence, it suffices to show that every subsequence has a further subsequence along which the convergence holds. Passing to a subsequence, we may therefore assume that the set $\mathcal{H}_1$ of strata satisfying the first alternative of (A4), and its complement $\mathcal{H}_2$ (second alternative), do not depend on ν, and that $\lambda_h^2 = W_h^2(1-f_h)S_{x,h}^2/(n_h V)$ converges to some $\lambda_{h,\infty}^2$ for every h (the λ_h^2 lie in $[0,1]$ and sum to one). For $h \in \mathcal{H}_2$, $\lambda_{h,\infty} = 0$, so $\sum_{h\in\mathcal{H}_2} W_h(\bar x_{s_h} - \bar x_h)/V^{1/2}$ has variance $\sum_{h\in\mathcal{H}_2}\lambda_h^2 \to 0$ and is $o_p(1)$. For $h \in \mathcal{H}_1$, put $\chi_h = (\bar x_{s_h} - \bar x_h)/\{(1-f_h)S_{x,h}^2/n_h\}^{1/2}$. Hájek's condition for asymptotic normality of a simple random sample mean (Hájek, 1960) requires $n_h \to \infty$, $N_h - n_h \to \infty$ and, for every $\eta > 0$,

$$\frac{1}{(N_h-1)S_{x,h}^2} \sum_{i\in U_h:\ |x_i - \bar x_h| > \eta\{n_h(1-f_h)\}^{1/2} S_{x,h}} (x_i - \bar x_h)^2 \to 0.$$

The first two hold by (A1) ($N_h - n_h \ge \varepsilon N_h \ge \varepsilon n_h$). Because $|x_i - \bar x_h| \le 2\|a\|_1 \le 2CB$ and $n_h(1-f_h)S_{x,h}^2 \to \infty$ by (A4), the index set of the sum is empty for all large ν, so the third holds as well, and $\chi_h \to N(0,1)$. The χ_h are independent, and $\Lambda/V^{1/2} = \sum_{h\in\mathcal{H}_1}\lambda_h\chi_h + o_p(1)$. By Lévy's continuity theorem the characteristic functions φ_h of χ_h converge to $e^{-u^2/2}$ uniformly on compact sets, hence $E\exp(iu\sum_{\mathcal{H}_1}\lambda_h\chi_h) = \prod_{\mathcal{H}_1}\varphi_h(\lambda_h u) \to \prod_{\mathcal{H}_1} e^{-\lambda_{h,\infty}^2 u^2/2} = e^{-u^2/2}$, using $\sum_{\mathcal{H}_1}\lambda_{h,\infty}^2 = 1$. This proves (c).

(d) Let $\hat a = \nabla\phi(\hat\tau)$ and let $\hat\Sigma_h$ be the sample covariance matrix of ξ_i in the sample from stratum h. Since $\hat e_i = \hat a^\top \xi_i/N$,

$$\hat V_\psi = \sum_h W_h^2 \frac{1-f_h}{n_h}\hat a^\top \hat\Sigma_h \hat a, \qquad V_\psi = \sum_h W_h^2 \frac{1-f_h}{n_h} a^\top \Sigma_h^\xi a,$$

and strata with $f_h = 1$ contribute zero to both. Hence $|\hat V_\psi - V_\psi| \le r^2 \max_h |\hat a^\top\hat\Sigma_h\hat a - a^\top\Sigma_h^\xi a|$.

Lemma 1. *Under (A1), $\hat\Sigma_h - \Sigma_h^\xi \to 0$ in probability for every h.*

Proof. For coordinates c, d, $\hat\Sigma_{h,cd} = \frac{n_h}{n_h-1}\{\overline{(\xi_c\xi_d)}_{s_h} - \bar\xi_{c,s_h}\bar\xi_{d,s_h}\}$ and $\Sigma_{h,cd}^\xi = \frac{N_h}{N_h-1}\{\overline{(\xi_c\xi_d)}_h - \bar\xi_{c,h}\bar\xi_{d,h}\}$. The variables ξ_c, ξ_d and $\xi_c\xi_d$ take values in $[-1,1]$, so each sample mean differs from its population mean by $O_p(n_h^{-1/2})$. The factors $n_h/(n_h-1)$ and $N_h/(N_h-1)$ tend to one, and the products are of bounded variables, so the difference tends to zero in probability. □

By part (b), $\hat\tau - \tau \to 0$ in probability, and $\nabla\phi$ is uniformly continuous on K, so $\hat a - a \to 0$ in probability. All entries of $\hat a$, a, $\hat\Sigma_h$, Σ_h are bounded on $\mathcal{E}$, and with the lemma, $\max_h|\hat a^\top\hat\Sigma_h\hat a - a^\top\Sigma_h^\xi a| = o_p(1)$. With (A3), $|\hat V_\psi/V_\psi - 1| \le r^2 o_p(1)/(cr^2) = o_p(1)$. Finally $(\hat\theta_\psi - \theta)/\hat V_\psi^{1/2} = \{(\hat\theta_\psi - \theta)/V_\psi^{1/2}\}(V_\psi/\hat V_\psi)^{1/2} \to N(0,1)$ by (c) and Slutsky's lemma, which gives the coverage statement. □

A.2 Proof of Proposition 2

Fix g and the pair (m, j). The estimand $\theta^{TPR}_{mj} = \sum_{g'} s_{g'}(TP_{mg'} - TP_{jg'})/P_{g'}$ depends on T through $TP_{mg'}$, $TP_{jg'}$ and $P_{g'}$, with partial derivatives $s_{g'}/P_{g'}$, $-s_{g'}/P_{g'}$ and $-s_{g'}(TP_{mg'} - TP_{jg'})/P^2_{g'} = -s_{g'}\delta_{g'}/P_{g'}$. Instance i with $G_i = g$ contributes $Y_i\hat{y}_{mi}$, $Y_i\hat{y}_{ji}$ and Y_i to the totals of its own group and nothing to the totals of the other group. By (9), $z^{TPR}_i = s_g\{Y_i\hat{y}_{mi} - Y_i\hat{y}_{ji} - \delta_g Y_i\}/P_g = s_g Y_i(D_i - \delta_g)/P_g$. Summing over the group gives $s_g\{(TP_{mg} - TP_{jg}) - \delta_g P_g\}/P_g = 0$. The FPR case is identical with $1 - Y_i$, Q_g and γ_g. For the error, $|\hat{y} - Y| = \hat{y} + Y - 2\hat{y}Y$ for binary values, so $|\hat{y}_{mi} - Y_i| - |\hat{y}_{ji} - Y_i| = (\hat{y}_{mi} - \hat{y}_{ji})(1 - 2Y_i)$ and $z^{ERR}_i = D_i(1 - 2Y_i)/N$.

For the stratum variances let $u_i = Y_i(D_i - \delta_g)$, so $z^{TPR}_i = s_g u_i/P_g$. Because $Y^2_i = Y_i$, $\bar{u}_h = c_h - \delta_g\pi_h$ and $\overline{u^2}_h = r_h - 2\delta_g c_h + \delta^2_g\pi_h$, using $D^2_i = |D_i|$. Therefore

$$\overline{u^2}_h - \bar{u}^2_h = r_h - 2\delta_g c_h + \delta^2_g\pi_h - c^2_h + 2\delta_g c_h\pi_h - \delta^2_g\pi^2_h = (r_h - c^2_h) + \delta^2_g\pi_h(1 - \pi_h) - 2\delta_g c_h(1 - \pi_h),$$

and multiplying by κ_h/P^2_g gives (16). For the error, $\{D_i(1 - 2Y_i)\}^2 = |D_i|$, so the variance with divisor N_h is $\rho_h - e^2_h$, which gives (17). If $D_i = 0$ on U_h then $r_h = c_h = \rho_h = e_h = 0$, and (18) follows; for FPR the label variance of $1 - Y$ is again $\pi_h(1 - \pi_h)$. When $K_i \in \{0, M\}$ all models predict the same label, so every pair agrees. □

A.3 Proof of Proposition 3

By (12), $\xi_i = Y_i u_i + (1 - Y_i)v_i - \pi_i u_i - (1 - \pi_i)v_i = (Y_i - \pi_i)(u_i - v_i)$, so $e_i = \nabla f(T)^\top \xi_i = (Y_i - \pi_i)d_i$. The partial derivatives of θ^{TPR}_{mj} for an instance of group g are those listed in the proof of Proposition 2: s_g/P_g and $-s_g/P_g$ for the coordinates TP_{mg} and TP_{jg}, and $-s_g\delta_g/P_g$ for P_g. For these coordinates $u_i - v_i$ equals $\hat{y}_{mi}$, $\hat{y}_{ji}$ and 1 (they have $v_i = 0$), so $d_i = s_g(\hat{y}_{mi} - \hat{y}_{ji} - \delta_g)/P_g = s_g(D_i - \delta_g)/P_g$. For θ^{FPR}_{mj} the coordinates FP_{mg}, FP_{jg}, Q_g have $u_i = 0$, $v_i = \hat{y}_{mi}, \hat{y}_{ji}, 1$, so $u_i - v_i$ is the negative of these values and $d_i = -s_g(D_i - \gamma_g)/Q_g$, which gives $e^{FPR}_i = (Y_i - \pi_i)\{-s_g(D_i - \gamma_g)/Q_g\} = s_g\{(1 - Y_i) - (1 - \pi_i)\}(D_i - \gamma_g)/Q_g$. For the error, E_m has $u_i - v_i = (1 - \hat{y}_{mi}) - \hat{y}_{mi} = 1 - 2\hat{y}_{mi}$, the gradient is $1/N$ and $-1/N$ on E_m and E_j, so $d_i = \{(1 - 2\hat{y}_{mi}) - (1 - 2\hat{y}_{ji})\}/N = -2D_i/N$. The influence value $z_i = B_i + Y_i d_i$ with $B_i = \nabla f(T)^\top v_i$ is affine in Y_i, so $\tilde{z}_i = z_i - e_i = B_i + \pi_i d_i$ is z_i with Y_i replaced by π_i. If $D_i = 0$ for all $i \in U_h$ then $d^{ERR}_i = 0$ and $e^{ERR}_i = 0$.

For (20) treat $d_i, \pi_i, \tilde{z}_i$ as constants (gradient at T_π). The $e_i = (Y_i - \pi_i)d_i$ are independent with mean zero and variances $\sigma^2_i = \pi_i(1 - \pi_i)d^2_i$. Then $E_\pi \sum_{U_h}(e_i - \bar{e}_h)^2 = (1 - N^{-1}_h)\sum_{U_h}\sigma^2_i$, so $E_\pi S^2_{e,h} = N^{-1}_h \sum_{U_h}\sigma^2_i = \bar{\sigma}^2_h$. Because $z_i = \tilde{z}_i + e_i$ with $\tilde{z}_i$ constant, $S^2_{z,h} = S^2_{\tilde{z},h} + 2\,\mathrm{Cov}_h(\tilde{z}, e) + S^2_{e,h}$ and $C_h = S^2_{\tilde{z},h} + \mathrm{Cov}_h(\tilde{z}, e)$; the covariance with the constants $\tilde{z}_i$ is linear in e and has expectation zero. □

A.4 Proof of Proposition 8

(a) The term $\sum_h N_h A^2_h$ does not depend on n. For any $n_h > 0$ with $\sum_h n_h = L$, the Cauchy–Schwarz inequality gives

$$\Big(\sum_h N_h A_h\Big)^2 = \Big(\sum_h \frac{N_h A_h}{\sqrt{n_h}}\sqrt{n_h}\Big)^2 \le \Big(\sum_h \frac{N^2_h A^2_h}{n_h}\Big)\Big(\sum_h n_h\Big) = \Phi_0(n)\, L,$$

with equality if and only if $N_h A_h/\sqrt{n_h}$ is proportional to $\sqrt{n_h}$, that is $n_h \propto N_h A_h$. With the budget constraint this gives n^* and the value of $\Phi(n^*)$. Equivalently, a Lagrange multiplier μ for $\sum_h n_h = L$ gives the stationarity condition $-N_h^2 A_h^2/n_h^2 + \mu = 0$, that is $n_h = N_h A_h/\sqrt{\mu}$.

(b) Φ_0 is convex on $(0,\infty)^H$ (strictly convex in the coordinates with $A_h > 0$) and the constraints are linear, so the Karush–Kuhn–Tucker conditions are necessary and sufficient. With multipliers μ for the budget and $\alpha_h, \beta_h \geq 0$ for $n_h \geq \ell_h$ and $n_h \leq N_h$, they read $-N_h^2 A_h^2/n_h^2 + \mu - \alpha_h + \beta_h = 0$ with complementary slackness. If neither bound is active, $n_h = N_h A_h/\lambda$ with $\lambda = \sqrt{\mu}$; if the upper bound is active, $\beta_h \geq 0$ forces $N_h A_h/\lambda \geq N_h$; if the lower bound is active, $\alpha_h \geq 0$ forces $N_h A_h/\lambda \leq \ell_h$. In all cases $n_h = \min\{N_h, \max(\ell_h, N_h A_h/\lambda)\}$. The map $\lambda \mapsto \sum_h n_h(\lambda)$ is continuous and non-increasing, from $\sum_h N_h \geq L$ as $\lambda \to 0$ to $\sum_h \ell_h \leq L$ as $\lambda \to \infty$, so a solution exists.

(c) If $\tilde{n}_h \geq n_h - 1$ and $n_h \geq n_0 > 1$, then $1/\tilde{n}_h \leq 1/(n_h - 1) = (1/n_h)\, n_h/(n_h - 1) \leq (1/n_h)\, n_0/(n_0 - 1)$ for every h; multiply by $N_h^2 A_h^2$ and sum. The implementation first rounds down and then distributes the remaining labels, so $\tilde{n}_h \geq \lfloor n_h \rfloor \geq n_h - 1$.

(d) Without finite-population corrections, $V_t^{SRS} = N^2 \sigma_t^2/L$, so $w_t V_t = (N^2/L) V_t / V_t^{SRS} = (N^2/L)/\mathrm{Eff}_t$. Summing over t gives $\Phi_0(n) L/N^2 = \sum_t 1/\mathrm{Eff}_t$, and at n^*

$$\sum_t \frac{1}{\mathrm{Eff}_t} = \frac{(\sum_h N_h A_h)^2}{N^2} = \Big(\sum_h W_h A_h\Big)^2.$$

For the lower bound, the within-stratum variances satisfy $\sum_h W_h \sigma_{ht}^2 \leq \sigma_t^2$ (the between-strata part of the variance decomposition is non-negative), so by Jensen's inequality

$$\Big(\sum_h W_h A_h\Big)^2 \leq \sum_h W_h A_h^2 = \sum_t \frac{\sum_h W_h \sigma_{ht}^2}{\sigma_t^2} \leq |\mathcal{T}|. \qquad \square$$

A.5 Proof of Proposition 9

Proposition 8(a)–(c) use only that the A_h are non-negative numbers, so they apply with A_h^ψ. Proposition 10(b) applied with $A_h = A_h^\psi$ and the anticipated value $\tilde{A}_h = \tilde{A}_h^0$ gives (28) with $x_h = \tilde{A}_h^0/A_h^\psi$. By (27) and the assumption $\tilde{A}_h^\psi = A_h^\psi$, $x_h^2 = \sum_t w_t(\bar{\sigma}_{ht}^2 + S_{\tilde{z},ht}^2)/\sum_t w_t \bar{\sigma}_{ht}^2 = 1 + \tilde{\nu}_h$. The bound is Proposition 10(c) with $x_{\min} \geq 1$ and $x_{\max} \leq (1+\nu_{\max})^{1/2}$. For the plain estimator allocated with the residual rule, $A_h = A_h^0$, $\tilde{A}_h = \tilde{A}_h^\psi$, so x_h is replaced by $1/x_h \in [(1+\nu_{\max})^{-1/2}, 1]$ and $\omega_h \propto N_h A_h^0$; the ratio $x_{\max}/x_{\min}$ is the same, hence so is the Kantorovich bound $(s+1)^2/(4s)$ with $s = (1+\nu_{\max})^{1/2}$. $\square$

A.6 Proof of Corollary 1

Equation (29). Without finite-population corrections, $V_t^{SRS} = N^2 \sigma_t^2/L$ and, by Proposition 8(a) applied to the single target, $V_t(n^*) = (\sum_h N_h \sigma_{ht})^2/L$; the ratio is (29). The variance decomposition $\sigma_t^2 = \sum_h W_h \sigma_{ht}^2 + \sum_h W_h (\bar{z}_h - \bar{z})^2$ and Jensen's inequality $\sum_h W_h \sigma_{ht}^2 \geq (\sum_h W_h \sigma_{ht})^2$ give $\mathrm{Eff}_t \geq 1$, with equality if and only if both inequalities are equalities.

Refinement. If a stratum h is split into h' and h'', the same two inequalities applied within h give $W_h \sigma_h \geq W_{h'} \sigma_{h'} + W_{h''} \sigma_{h''}$, so the denominator of (29) cannot increase.

(a) By Proposition 2 the influence values of θ_{mj}^{TPR} have mean zero in each group, hence overall, and $N^{-1}\sum_i z_i^2 = N^{-1}\sum_g N_g \overline{u^2}_g / P_g^2$, where $\overline{u^2}_g = r_g - 2\delta_g \cdot \delta_g p_g + \delta_g^2 p_g = r_g - \delta_g^2 p_g$ is the group mean of u_i^2 (we used $N_g^{-1}\sum_{G_i=g} Y_i D_i = \delta_g p_g$). With $P_g = N\zeta_g p_g$, $\sigma_t^2 = N^{-2}\sum_g (r_g - \delta_g^2 p_g)/(\zeta_g p_g^2)$. In the stratum $\mathcal{D}_g$, $|D_i| = 1$, so $r_h = \pi_h = \pi_{\mathcal{D},g}$ and $c_h = c_{\mathcal{D},g} = \delta_g p_g N_g/(N_g \rho_g)$; (16) with divisor N_h gives $\sigma_{\mathcal{D}_g} = \sigma_{\mathcal{D},g}/P_g$, and $W_{\mathcal{D}_g}\sigma_{\mathcal{D}_g} = \zeta_g \rho_g \sigma_{\mathcal{D},g}/(N\zeta_g p_g) = \rho_g \sigma_{\mathcal{D},g}/(N p_g)$. In $\mathcal{A}_g$, the number of positives is $N_g(p_g - r_g)$, so its positive rate is $\pi_{\mathcal{A},g}$, and (18) gives $W_{\mathcal{A}_g}\sigma_{\mathcal{A}_g} = (1-\rho_g)|\delta_g|\{\pi_{\mathcal{A},g}(1-\pi_{\mathcal{A},g})\}^{1/2}/(Np_g)$. Inserting into (29), the factors N^{-2} cancel and (30) follows. For the error difference, $z_i = D_i(1-2Y_i)/N$ vanishes on $\mathcal{A}_g$, takes values $\pm 1/N$ on $\mathcal{D}_g$ with mean μ_g/N, so $\sigma_{\mathcal{D}_g} = (1-\mu_g^2)^{1/2}/N$, and $\sigma_t^2 = (\rho - \theta^2)/N^2$ because $z_i^2 = |D_i|/N^2$; (31) follows. The last statement of (a) is the refinement property.

(b) With $\delta_g = 0$, $c_{\mathcal{D},g} = 0$ and $\sigma_{\mathcal{D},g}^2 = \pi_{\mathcal{D},g} = r_g/\rho_g$, so $\rho_g \sigma_{\mathcal{D},g} = (\rho_g r_g)^{1/2}$ and the agreement terms vanish. With $\rho_g = \rho$, $\mathrm{Eff}^{TPR} = \sum_g r_g/(\zeta_g p_g^2)/\{\rho(\sum_g \sqrt{r_g}/p_g)^2\} = \Gamma/\rho$. By Cauchy–Schwarz, $(\sum_g \sqrt{\zeta_g}\cdot\sqrt{r_g}/(\sqrt{\zeta_g}p_g))^2 \le \sum_g \zeta_g \cdot \sum_g r_g/(\zeta_g p_g^2)$, so $\Gamma \ge 1$. Applying (29) to the two strata $\{G=0\}$, $\{G=1\}$, whose influence values have mean zero and variances $r_g/(N\zeta_g p_g)^2$, gives exactly Γ. For the error difference with $\mu_g = 0$ (hence $\theta = \sum_g \zeta_g \rho_g \mu_g = 0$), (31) equals $\rho/\rho^2 = 1/\rho$.

(c) With one group, $\zeta = 1$, $\delta = \rho\pi_{\mathcal{D}} a/p$ and $c_{\mathcal{D}} = a\pi_{\mathcal{D}}$. The numerator of (30) is $(\rho\pi_{\mathcal{D}} - \delta^2 p)/p^2 = \rho\pi_{\mathcal{D}}\{1+O(\rho)\}/p^2$. Since $\delta = O(\rho)$, $\sigma_{\mathcal{D}}^2 = \pi_{\mathcal{D}} - a^2\pi_{\mathcal{D}}^2 + O(\rho) = \pi_{\mathcal{D}}\beta^2 + O(\rho)$, and $(1-\rho)|\delta|\{\pi_{\mathcal{A}}(1-\pi_{\mathcal{A}})\}^{1/2} = \rho\sqrt{\pi_{\mathcal{D}}}\,\lambda + O(\rho^2)$. The denominator is therefore $\rho^2\pi_{\mathcal{D}}(\beta+\lambda)^2\{1+O(\rho)\}/p^2$ (assuming $\beta + \lambda > 0$, which fails only in the degenerate case $|a| = \pi_{\mathcal{D}} = 1$), and the ratio is (33). The value of $p = \rho\pi_{\mathcal{D}} + (1-\rho)\pi_{\mathcal{A}}$ changes by $O(\rho)$, which is absorbed in the $O(\rho)$ term.

(d) When $\rho_g = 1$ the strata $\mathcal{A}_g$ are empty and $\mathcal{D}_g$ is the whole group, so the design is stratification by group, and (30), (31) are continuous in ρ_g. For $\delta_g = 0$ the limit of (30) is Γ by (b). For the error difference with $\rho_g = 1$, μ_g is the group-wise error difference and the limit is $(1-\theta^2)/(\sum_g \zeta_g\sqrt{1-\mu_g^2})^2$, which equals one when $\mu_0 = \mu_1 = \theta$.

Equation (34). If $e_h = 0$ for all h, (17) gives $\sigma_{ht}^2 = \rho_h/N^2$, and $\sigma_t^2 = \rho/N^2$ with $\theta = 0$. Since $0 \le \rho_h \le 1$, $\sqrt{\rho_h} \ge \rho_h$, so $(\sum_h W_h\sqrt{\rho_h})^2 \ge \rho^2$ and $\mathrm{Eff} \le 1/\rho$; Jensen's inequality gives $(\sum_h W_h\sqrt{\rho_h})^2 \le \rho$ and $\mathrm{Eff} \ge 1$. Equality in the upper bound requires $\rho_h \in \{0,1\}$ for every h. For (35), let $\mathcal{H}_D$ be the strata with $0 < K < M$, so that $\sum_{h\in\mathcal{H}_D} W_h = \rho_{\mathrm{any}}$ and $\rho_h = 0$ outside $\mathcal{H}_D$. Cauchy–Schwarz gives

$$\Big(\sum_{h\in\mathcal{H}_D} W_h^{1/2}\cdot W_h^{1/2}\rho_h^{1/2}\Big)^2 \le \rho_{\mathrm{any}}\sum_h W_h\rho_h = \rho_{\mathrm{any}}\rho,$$

so $\mathrm{Eff} \ge \rho/(\rho_{\mathrm{any}}\rho) = 1/\rho_{\mathrm{any}}$, with equality if and only if ρ_h is constant on $\mathcal{H}_D$. (When strata are merged across $K = 0$ or $K = M$ and $0 < K < M$ because of the minimum stratum size, ρ_{any} is replaced by the total share of the strata that contain any instance with $0 < K < M$.) □

A.7 Proof of Proposition 10

(a) The proxy is a fixed function of the pool, so $\tilde{n}$ satisfies (A1) whenever the minimum per stratum is at least two; Proposition 1 applies without change.

(b) With $\tilde{n}_h = L N_h \tilde{A}_h / \sum_k N_k \tilde{A}_k$,

$$\Phi_0(\tilde{n}) = \sum_h \frac{N_h^2 A_h^2}{\tilde{n}_h} = \frac{1}{L}\Big(\sum_k N_k \tilde{A}_k\Big)\Big(\sum_h \frac{N_h A_h^2}{\tilde{A}_h}\Big),$$

and $\Phi_0(n^*) = L^{-1}(\sum_h N_h A_h)^2$ by Proposition 8(a). Writing $\tilde{A}_h = x_h A_h$,

$$\sum_h N_h \tilde{A}_h = \Big(\sum_k N_k A_k\Big)\sum_h \omega_h x_h, \qquad \sum_h \frac{N_h A_h^2}{\tilde{A}_h} = \Big(\sum_k N_k A_k\Big)\sum_h \frac{\omega_h}{x_h},$$

which gives (36). By Cauchy–Schwarz,

$$1 = \Big(\sum_h \omega_h \sqrt{x_h}/\sqrt{x_h}\Big)^2 \le \Big(\sum_h \omega_h x_h\Big)\Big(\sum_h \omega_h / x_h\Big),$$

with equality if and only if x_h is constant on the strata with $\omega_h > 0$.

(c) For $x \in [x_{\min}, x_{\max}]$, $(x - x_{\min})(x_{\max} - x) \ge 0$, that is $x + x_{\min}x_{\max}/x \le x_{\min} + x_{\max}$. Averaging with weights ω_h, $\sum \omega_h x_h + x_{\min}x_{\max}\sum \omega_h/x_h \le x_{\min} + x_{\max}$. By the arithmetic–geometric mean inequality the left side is at least $2\{x_{\min}x_{\max}(\sum \omega_h x_h)(\sum \omega_h/x_h)\}^{1/2}$; squaring gives the bound.

(d) Proportional allocation corresponds to $\tilde{A}_h \equiv 1$, i.e. $x_h = 1/A_h$, which gives the stated comparison. For the negative statement consider two strata of equal size $N/2$ with equal true values $A_1 = A_2$ and equal means, so that SRS, proportional allocation and the optimal allocation coincide. If the proxy gives $\tilde{A}_1 \approx 0$, stratum 1 receives the minimum ℓ labels and stratum 2 receives $L - \ell$, and $\Phi_0(\tilde{n})/\Phi_0(n^*) = \frac{L}{4}\{1/\ell + 1/(L-\ell)\}$, which is about $L/(4\ell)$; with $L = 1000$ and $\ell = 10$ the weighted variance is 25 times that of SRS.

(e) Since $n_h \ge \eta n_h^{\mathrm{prop}}$ and $n_h \ge (1-\eta)\tilde{n}_h$, $1/n_h \le 1/(\eta n_h^{\mathrm{prop}})$ and $1/n_h \le 1/\{(1-\eta)\tilde{n}_h\}$; multiply by $N_h^2 A_h^2$ and sum. For the comparison of proportional allocation with SRS, fix a target and write S^2, S_h^2, $\bar{z}$, $\bar{z}_h$ for its variances and means. The analysis-of-variance identity $(N-1)S^2 = \sum_h (N_h - 1)S_h^2 + \sum_h N_h(\bar{z}_h - \bar{z})^2$ gives

$$NS^2 - \sum_h N_h S_h^2 = \frac{N}{N-1}\sum_h N_h(\bar{z}_h - \bar{z})^2 - \sum_h \frac{N - N_h}{N-1} S_h^2 \ge -\sum_h S_h^2 \ge -\frac{1}{\min_h N_h}\sum_h N_h S_h^2.$$

With $f = L/N$, proportional allocation has variance $V^{\mathrm{prop}} = (1-f)(N/L)\sum_h N_h S_h^2$ and SRS has $V^{SRS} = (1-f)(N/L)NS^2$, so $V^{\mathrm{prop}}(1 - 1/\min_h N_h) \le V^{SRS}$. Summing over targets with weights w_t gives the same inequality for the weighted variance, and combining with the first part proves (e). □

A.8 Proof of Proposition 4

(a) Within stratum h, $e_i = z_i - \tilde{z}_i$, so $S_{e,h}^2 = S_{z,h}^2 - 2C_h + S_{\tilde{z},h}^2$ with $C_h = \mathrm{Cov}_h(z, \tilde{z}) = \beta_h S_{\tilde{z},h}^2 = R_h S_{z,h} S_{\tilde{z},h}$. Multiplying by c_h and summing, $V_0 - V_\psi = \sum_h c_h(2C_h - S_{\tilde{z},h}^2) = \sum_h c_h S_{\tilde{z},h}^2(2\beta_h - 1)$. The sufficient condition is $\beta_h > 1/2 \iff R_h S_{z,h}/S_{\tilde{z},h} > 1/2$.

(b) For the member λ, the residual influence value in stratum h is $z_i - \lambda_h \tilde{z}_i$ (the proxy $\lambda_h \psi_i$ is label-free, so Proposition 1 applies and the linearised variance has this form). Hence

$S^2_{z-\lambda\tilde z,h} = S^2_{z,h} - 2\lambda_h C_h + \lambda_h^2 S^2_{\tilde z,h} = S^2_{z,h} - \beta_h^2 S^2_{\tilde z,h} + (\lambda_h - \beta_h)^2 S^2_{\tilde z,h}$, and $\beta_h^2 S^2_{\tilde z,h} = C_h^2/S^2_{\tilde z,h} = R_h^2 S^2_{z,h}$, which is the displayed formula. Setting $\lambda \equiv 0$ and $\lambda \equiv 1$ gives the two decompositions. The function $\lambda \mapsto (\lambda - \beta_h)^2$ is a parabola centred at β_h, and the clipped value is the point of $[0,1]$ closest to β_h, so it is no larger than at $\lambda = 0$ or $\lambda = 1$.

(c) If $\mathrm{Cov}_h(e, \tilde z) = 0$ then $C_h = \mathrm{Cov}_h(\tilde z + e, \tilde z) = S^2_{\tilde z,h}$, so $\beta_h = 1$ and (a) gives $V_0 - V_\psi = \sum_h c_h S^2_{\tilde z,h}$. □

A.9 Proof of Proposition 5

(a) The c_h are constants (n_h is label-free), so by (20) $E_\pi V_0 - E_\pi V_\psi = \sum_h c_h\{(S^2_{\tilde z,h} + \bar\sigma^2_h) - \bar\sigma^2_h\} = \sum_h c_h S^2_{\tilde z,h}$, which vanishes iff $S^2_{\tilde z,h} = 0$ in every stratum with $c_h > 0$, that is every stratum that is not fully enumerated.

(b) $E_\pi V_\psi = \sum_h a_h$ and $E_\pi V_0 = \sum_h c_h(\bar\sigma^2_h + S^2_{\tilde z,h}) = \sum_h a_h(1+\nu_h)$, which gives the ratio and $G \geq 1$. By (20) the expected within-stratum regression slope is one and $E_\pi S^2_{z,h} = \bar\sigma^2_h(1+\nu_h)$, so the (expected) squared correlation is $S^4_{\tilde z,h}/(S^2_{\tilde z,h} S^2_{z,h}) = \nu_h/(1+\nu_h)$, hence $1 + \nu_h = 1/(1 - R_h^2)$.

(c) Consider a stratum in which $|D_i| = 1$ for all i, with δ_g, γ_g negligible, so that for TPR $z_i = s_g Y_i D_i/P_g$, $\tilde z_i = s_g \pi_i D_i/P_g$, $e_i = s_g(Y_i - \pi_i)D_i/P_g$ and $d_i = s_g D_i/P_g$. Ignoring the common factor $\kappa_h = N_h/(N_h - 1)$, which cancels in ν_h, and using $D_i^2 = 1$: $\bar\sigma^2_h = P_g^{-2}\overline{\pi(1-\pi)}$; $S^2_{\tilde z,h} = P_g^{-2}\{\overline{\pi^2} - (\overline{\pi D})^2\} = P_g^{-2}\{\overline{\pi^2} - \bar\pi^2\bar a^2\}$, using that D and π are uncorrelated within the stratum. Hence $1+\nu = \{\overline{\pi(1-\pi)} + \overline{\pi^2} - \bar\pi^2\bar a^2\}/\overline{\pi(1-\pi)} = \bar\pi(1 - \bar\pi\bar a^2)/\overline{\pi(1-\pi)}$. For FPR, (Y, π) are replaced by $(1-Y, 1-\pi)$, which leaves $\overline{\pi(1-\pi)}$ unchanged, and $\bar\pi$ by $1 - \bar\pi$. For ERR, $d_i = -2D_i/N$, $\tilde z_i = D_i(1 - 2\pi_i)/N$, so $\bar\sigma^2_h = 4N^{-2}\overline{\pi(1-\pi)}$ and $S^2_{\tilde z,h} = N^{-2}\{\overline{(1-2\pi)^2} - \bar a^2(1-2\bar\pi)^2\}$; since $(1-2\pi)^2 + 4\pi(1-\pi) = 1$, $\bar\sigma^2_h + S^2_{\tilde z,h} = N^{-2}\{1 - \bar a^2(1-2\bar\pi)^2\}$, which gives the third formula. In an agreement stratum $D_i = 0$: $e_i^{ERR} = \tilde z_i^{ERR} = 0$; for TPR $d_i = -s_g\delta_g/P_g$ is constant, $\bar\sigma^2_h = (\delta_g/P_g)^2\overline{\pi(1-\pi)}$ and $S^2_{\tilde z,h} = (\delta_g/P_g)^2 S^2_{\pi,h}$, which gives the formula; FPR is the same. The numerical examples are the case $\pi_i \equiv \bar\pi$, $\bar a = 0$: $\overline{\pi(1-\pi)} = \bar\pi(1-\bar\pi)$, so the three gains are $\bar\pi/\{\bar\pi(1-\bar\pi)\}$, $(1-\bar\pi)/\{\bar\pi(1-\bar\pi)\}$ and $1/\{4\bar\pi(1-\bar\pi)\}$. □

A.10 Proof of Proposition 6

(a) is Proposition 1 (only label-freeness of the proxy is used).

(b) Under independent labels with $P(Y_i = 1) = q_i$, write $z_i = \tilde z_i^q + (Y_i - q_i)d_i$ and $e_i = z_i - \tilde z_i^\pi = (Y_i - q_i)d_i + (\tilde z_i^q - \tilde z_i^\pi)$, using $\tilde z_i^q - \tilde z_i^\pi = (q_i - \pi_i)d_i$. The first term has mean zero and independent coordinates, the second is constant. As in the proof of Proposition 3, $E_q S^2_{z,h} = S^2_{\tilde z^q,h} + \bar\sigma^2_h(q)$ and $E_q S^2_{e,h} = \bar\sigma^2_h(q) + S^2_{\tilde z^q - \tilde z^\pi,h}$. Subtracting and multiplying by c_h gives (24). If $\tilde z_i^\pi - \bar{\tilde z}_h^\pi = \kappa(\tilde z_i^q - \bar{\tilde z}_h^q)$, then $S^2_{\tilde z^q - \tilde z^\pi,h} = (1-\kappa)^2 S^2_{\tilde z^q,h}$, which is smaller than $S^2_{\tilde z^q,h}$ iff $0 < \kappa < 2$.

(c) By Minkowski's inequality for standard deviations, $S_{e,h} \leq S_{z,h} + S_{\tilde z,h} \leq S_{z,h}(1 + \max_k S_{\tilde z,k}/S_{z,k})$, with equality in the first step iff $z_i - \bar z_h = -s_h(\tilde z_i - \bar{\tilde z}_h)$ for a constant $s_h > 0$ (perfect anti-correlation, $S_{\tilde z,h}/S_{z,h} = 1/s_h$), and in the second iff this ratio is the same in every stratum. Multiply by c_h and sum.

(d) The first statement is Proposition 4(b). With $\beta_h = 0$, (a) of that proposition gives $V_0 - V_\psi = -\sum_h c_h S^2_{\tilde z,h}$. □

A.11 Proof of Proposition 7

By (13) with coefficients $\hat{\lambda}_h$ and λ_h^∞, $\hat{T}_{\hat{\lambda}\psi} - \hat{T}_{\lambda^\infty\psi} = -\sum_h N_h(\hat{\lambda}_h - \lambda_h^\infty)(\bar{\psi}_{s_h} - \bar{\psi}_h)$, because $F_h = N_h\bar{\psi}_h$. Each $\bar{\psi}_{s_h} - \bar{\psi}_h$ is $O_p(\{(1-f_h)/n_h\}^{1/2})$ coordinatewise, so, with $W_h n_h^{-1/2} \le r$ and a fixed number of strata, $\|\hat{\tau}_{\hat{\lambda}\psi} - \hat{\tau}_{\lambda^\infty\psi}\| = o_p(r)$. The Taylor expansion (37) applies to both estimators (it needs only (A2) and the bound on the contributions), so $\hat{\theta}_{\hat{\lambda}\psi} - \hat{\theta}_{\lambda^\infty\psi} = o_p(r) = o_p(V^{1/2})$ by (A3), and Slutsky's lemma gives the same limit. By Proposition 4(b), $\lambda_h^\infty = \min\{1, \max(0, \beta_h)\}$ gives $V(\lambda^\infty) \le \min(V_0, V_\psi)$. (The sample slope $\hat{\beta}_h$ is a ratio of sample moments of bounded variables and is consistent for β_h whenever $S^2_{\tilde{z},h}$ is bounded below by a positive constant times N^{-2}; otherwise the stratum's contribution is negligible by (A4).)

For the second-order statement, write $\hat{\beta}_h - \beta_h \approx \widehat{\mathrm{Cov}}(\varepsilon, \tilde{z})/S^2_{\tilde{z},h}$ with $\varepsilon_i = z_i - \beta_h\tilde{z}_i$, a population-regression residual with variance $S^2_{\varepsilon,h} = (1-R_h^2)S^2_{z,h}$. Its variance is $\approx S^2_{\varepsilon,h}/(n_h S^2_{\tilde{z},h})$. The leading error term of the first part is the product $(\hat{\beta}_h - \beta_h)\, N_h(\bar{\psi}_{s_h} - \bar{\psi}_h)$ of two asymptotically uncorrelated factors with variances $S^2_{\varepsilon,h}/(n_h S^2_{\tilde{z},h})$ and $N_h^2(1-f_h)S^2_{\tilde{z},h}/n_h$ (after applying the gradient), whose product is $c_h S^2_{\varepsilon,h}/n_h = c_h(1-R_h^2)S^2_{z,h}/n_h$, a fraction $1/n_h$ of the stratum's optimal contribution $c_h(1-R_h^2)S^2_{z,h}$. The plug-in estimator uses the residuals $\hat{z}_i - \hat{\beta}_h\hat{\tilde{z}}_i$ in the sample, whose sample variance (divisor $n_h - 1$) is, by the usual least-squares calculation with an intercept and a slope, smaller in expectation than the variance of the fixed-β_h residuals by the factor $(n_h - 2)/(n_h - 1)$. Both statements are first-order expansions in $1/n_h$ and are meant as an explanation of the finite-sample behaviour, not as a bound. □

A.12 Justification of Remark 1

Let Z be the set of strata in which $s^2_{\hat{e},h} = 0$ and $n_h \ge 2$, $f_h < 1$. The floored estimator replaces $s^2_{\hat{e},h}$ by $\tilde{s}^2_h$ for $h \in Z$ and is unchanged otherwise; since $\tilde{s}^2_h \ge 0$, $\hat{V}_\psi^{\mathrm{floor}} \ge \hat{V}_\psi$ pointwise, and the point estimate is the same. If $d_i = 0$ for all $i \in U_h$, then $\hat{d}_i = 0$ and $\tilde{s}^2_h = 0$ (structural zero), so no floor is added. If $N^2 S^2_{e,h} \ge c > 0$, then $a^\top \Sigma_h^\xi a$ is bounded away from zero, and by Lemma 1 and the consistency of $\hat{a}$ the probability $P(s^2_{\hat{e},h} = 0) \to 0$, so $P(h \in Z) \to 0$. Let $\mathcal{H}_0$ be the strata with $N^2 S^2_{e,h} \to 0$. Their contribution to V_ψ is vanishing relative to V_ψ whenever (A4) holds in its second form, and their contribution to $\hat{V}_\psi^{\mathrm{floor}}$ is at most $\sum_{h\in\mathcal{H}_0} c_h \tilde{s}^2_h$. If this is $o(V_\psi)$, then $\hat{V}_\psi^{\mathrm{floor}}/V_\psi \to 1$ and Proposition 1(d) holds for the floored interval; if not, the floored interval is conservative, $\liminf P(\theta \in \hat{\theta}_\psi \pm z_{1-\alpha/2}\{\hat{V}_\psi^{\mathrm{floor}}\}^{1/2}) \ge 1-\alpha$, because $\hat{V}_\psi^{\mathrm{floor}} \ge \hat{V}_\psi$ and $\hat{V}_\psi/V_\psi \to 1$. The same argument applies to $\max\{s^2_{\hat{e},h}, \eta\tilde{s}^2_h\}$ if $\eta\tilde{s}^2_h < S^2_{e,h}$ in the limit for strata that carry variance, which holds with $\eta \le 1$ for a calibrated proxy because then $E_\pi S^2_{e,h} = \bar{\sigma}^2_h$ (20). □

B Ablations: full tables

This appendix gives the per-cell results behind Section 9: 29 setting–budget cells, $R = 150$ replications and $B = 999$ bootstrap resamples. All values are stored in `results/label_ablations.json`, which is produced by the script `experiments/label_ablations.py`; its `report` mode prints the tables and draws the figures. Abbreviations: Credit = Credit card default, Law = Law school, Elec. = Electricity.

Implementation details. Stratifiers: the binary stratifier crosses group with the indicator $0 < K_i < M$. The three-model pattern uses the models with the highest agreement between $\hat{y}_{mi}$ and $\mathbf{1}\{\pi_i \geq 0.5\}$ (chosen without labels; e.g. histogram GB, random forest and MLP in Adult-sex), with the eight patterns ordered by their number of ones so that merging of small strata goes to a neighbouring pattern. Proxy bins split each (G, K) cell at the quantiles of π_i within the cell. All stratifiers merge strata below 50 units within group, as in the default. When H strata at 10 labels would use more than 80% of the budget, the floor is lowered to $\max(2, \lfloor 0.8L/H \rfloor)$. This affects only the proxy-bin stratifiers at $L \leq 500$ (floors 3–9) and the grid designs with floor 20 at $L = 250$ (floors 10–18). Proxies: "single model" uses the predicted probabilities of the first logistic regression or of Gaussian naive Bayes; "noise" adds $N(0, 2^2)$ to $\text{logit}(\pi_i)$; "shuffle" permutes π within each group (fixed seed). Satterthwaite degrees of freedom are $\nu = (\sum_h a_h s^2_{z,h})^2 / \sum_h (a_h s^2_{z,h})^2/(n_h - 1)$ with $a_h = N_h^2(1 - n_h/N_h)/n_h$, computed per target. The bootstrap resamples n_h units with replacement within each stratum and takes the 2.5% and 97.5% quantiles of the plug-in estimate. The proxy variance floor is defined in Section 9. A3 subsets were drawn once per size with seed 20261104 and are the same in every setting.

Table 15: Efficiency relative to SRS (median over the 108 targets of $\text{MSE}_{\text{SRS}}/\text{MSE}$) for every cell, stratifier (A1) and proxy (A2). Pat.: default; Bin.: group × agree/disagree; J3: three-model pattern; KP2, KP3: $K\times$ 2 or 3 proxy bins; LR, NB: single-model proxies; C: constant 0.5; Sh.: shuffled; Noi.: logit noise; Rev.: $1 - \pi$; Prop.: proportional; Orc.: true-label allocation.

Setting	L	Pat.	Bin.	J3	KP2	KP3	LR	NB	C	Sh.	Noi.	Rev.	Prop.	Orc.
Adult-sex	250	3.63	1.56	1.29	3.77	4.04	3.69	3.67	3.71	3.40	3.13	3.47	2.81	3.79
Adult-sex	500	3.90	1.73	1.43	3.88	3.87	3.93	4.07	3.90	3.83	3.79	3.71	2.25	3.99
Adult-sex	1000	4.48	1.79	1.51	4.82	4.99	4.59	4.73	4.31	4.13	4.39	4.55	1.81	4.60
Adult-race	250	6.02	2.55	1.82	5.56	5.33	5.60	6.39	4.97	5.50	5.80	5.60	4.63	5.16
Adult-race	500	6.30	2.69	2.01	5.58	5.12	6.82	6.72	6.10	6.15	5.93	6.12	3.28	6.27
Adult-race	1000	8.30	3.00	2.62	7.76	6.88	6.53	7.09	7.53	7.27	7.22	7.18	2.75	7.61
Bank	250	7.58	6.26	2.42	6.84	7.17	7.21	7.39	7.27	7.59	7.12	7.80	5.66	7.53
Bank	500	8.00	5.52	2.58	8.13	7.15	7.90	7.58	8.54	8.26	8.65	8.78	3.61	7.99
Bank	1000	9.85	7.79	2.72	10.41	11.05	9.62	8.99	9.58	10.44	9.06	10.25	2.67	10.96
Credit	250	2.83	1.56	1.64	2.90	3.25	3.26	3.58	3.07	2.95	3.20	3.55	3.01	3.08
Credit	500	3.97	1.49	1.61	3.48	3.39	3.71	3.88	4.02	3.83	4.18	3.85	2.58	3.60
Credit	1000	4.20	1.61	1.71	4.12	4.11	3.85	3.95	4.32	4.06	3.84	4.46	2.12	4.00
Law	250	15.71	4.87	1.90	15.49	14.28	20.57	19.58	18.78	17.67	17.18	20.07	6.27	23.82
Law	500	36.85	4.97	1.86	39.52	21.97	31.40	27.43	34.92	36.97	38.05	31.87	5.59	38.30
Law	1000	78.85	8.57	2.15	89.20	116.24	96.94	82.83	74.60	60.27	68.27	72.81	6.22	146.50
MAGIC	250	2.56	2.08	1.08	2.76	2.58	2.43	2.54	2.41	2.48	2.29	2.42	2.19	2.46
MAGIC	500	2.53	2.10	1.24	2.66	2.65	2.97	2.83	2.84	2.68	2.70	2.72	1.70	2.79
MAGIC	1000	3.15	2.54	1.21	3.56	3.30	3.26	3.24	3.09	3.44	3.46	3.24	1.49	3.36
EEG	250	1.72	1.25	0.97	2.09	2.13	1.70	1.66	1.79	1.72	1.70	1.65	1.51	1.74
EEG	500	1.75	1.26	0.95	1.92	2.01	1.70	1.71	1.63	1.67	1.73	1.63	1.42	1.79
EEG	1000	1.95	1.45	1.20	2.46	2.40	1.91	2.04	1.96	1.94	1.90	2.18	1.47	2.09
Phoneme	250	2.48	2.29	1.23	2.15	1.79	2.44	2.58	2.50	2.37	2.46	2.48	1.55	2.31
Phoneme	500	3.59	3.10	1.27	3.34	3.02	3.48	3.60	3.55	3.58	3.38	3.79	1.46	3.45
Spambase	250	5.55	3.48	1.40	4.12	3.05	5.68	6.57	5.26	5.14	5.52	5.99	1.80	6.01
Churn	250	6.51	6.46	1.62	5.79	4.75	6.83	10.27	6.94	7.27	6.36	6.21	1.90	6.48
Churn	500	31.59	37.97	1.59	26.24	17.36	58.94	>1000	>1000	274.06	53.55	32.37	1.60	51.74
Elec.	250	2.73	2.44	0.97	2.67	2.96	2.82	2.59	2.79	2.71	2.79	2.78	2.38	2.68
Elec.	500	2.82	2.53	1.18	2.77	2.76	2.66	2.77	2.87	2.87	2.67	2.97	1.87	2.93
Elec.	1000	3.37	2.65	1.37	3.25	3.14	3.35	3.26	3.23	2.99	3.21	3.28	1.49	3.17

C Data sets and models

All data sets were obtained from OpenML through scikit-learn's `fetch_openml` interface. Adult (version 2, 48,842 records; target income above 50K; groups: sex, female versus male, and race,

Table 16: Coverage of the default design per cell (A4). H: number of strata after merging. Mean: Wald coverage averaged over targets; the remaining coverage columns give the worst target under Wald, t (Satterthwaite), stratified bootstrap and Wald with the proxy variance floor; Width: average width of floored intervals relative to Wald. Last two columns: worst-target coverage of SRS with Wald and with the floor. The Monte Carlo reference for the worst of 108 targets at exact 95% coverage is 0.90.

				Worst target, PATTERN					Worst, SRS	
Setting	L	H	Mean	Wald	t	Boot.	Floor	Width	Wald	Floor
Adult-sex	250	19	0.944	0.753	0.767	0.807	0.927	1.24	0.540	0.607
Adult-sex	500	19	0.945	0.880	0.887	0.893	0.893	1.08	0.733	0.733
Adult-sex	1000	19	0.946	0.873	0.873	0.900	0.900	1.04	0.867	0.867
Adult-race	250	17	0.945	0.873	0.887	0.893	0.893	1.13	0.733	0.733
Adult-race	500	17	0.954	0.907	0.907	0.913	0.907	1.05	0.720	0.720
Adult-race	1000	17	0.948	0.913	0.913	0.940	0.920	1.05	0.873	0.873
Bank	250	16	0.939	0.860	0.860	0.893	0.907	1.23	0.416	0.611
Bank	500	16	0.940	0.860	0.860	0.913	0.907	1.14	0.553	0.593
Bank	1000	16	0.947	0.900	0.900	0.940	0.907	1.05	0.800	0.800
Credit	250	19	0.946	0.853	0.860	0.660	0.913	1.08	0.373	0.893
Credit	500	19	0.953	0.913	0.913	0.800	0.913	1.01	0.540	0.887
Credit	1000	19	0.952	0.913	0.913	0.933	0.913	1.00	0.713	0.907
Law	250	10	0.944	0.633	0.633	0.700	0.960	1.77	0.267	0.567
Law	500	10	0.971	0.920	0.920	0.947	0.927	1.90	0.400	0.673
Law	1000	10	0.968	0.900	0.907	0.953	0.940	2.37	0.640	0.867
MAGIC	250	19	0.945	0.820	0.833	0.747	0.927	1.07	0.540	0.907
MAGIC	500	19	0.938	0.880	0.880	0.887	0.893	1.02	0.753	0.913
MAGIC	1000	19	0.945	0.887	0.887	0.933	0.893	1.01	0.900	0.900
EEG	250	18	0.948	0.907	0.913	0.900	0.913	1.16	0.893	0.893
EEG	500	18	0.947	0.907	0.907	0.920	0.913	1.12	0.907	0.907
EEG	1000	18	0.949	0.900	0.900	0.933	0.900	1.07	0.887	0.887
Phoneme	250	14	0.948	0.907	0.907	0.920	0.907	1.06	0.813	0.887
Phoneme	500	14	0.950	0.900	0.907	0.967	0.907	1.05	0.907	0.907
Spambase	250	9	0.943	0.900	0.900	0.920	0.907	1.06	0.800	0.827
Churn	250	6	0.942	0.473	0.473	0.473	0.920	1.49	0.140	0.867
Churn	500	6	0.927	0.800	0.800	0.960	0.860	1.26	0.260	0.880
Elec.	250	20	0.948	0.893	0.893	0.893	0.893	1.03	0.820	0.880
Elec.	500	20	0.949	0.860	0.860	0.873	0.913	1.02	0.873	0.913
Elec.	1000	20	0.953	0.900	0.900	0.933	0.907	1.00	0.907	0.907

Table 17: Worst-target Wald coverage over the grid of label floors (min n_h) and collapse thresholds (A4). The column $(10, 50)$ is the default.

		min $n_h = 5$			min $n_h = 10$			min $n_h = 20$		
Setting	L	25	50	100	25	50	100	25	50	100
Adult-sex	250	0.88	0.81	0.85	0.77	0.75	0.85	0.78	0.73	0.75
Adult-sex	500	0.88	0.89	0.91	0.88	0.88	0.89	0.85	0.91	0.91
Adult-sex	1000	0.90	0.90	0.89	0.90	0.87	0.91	0.91	0.91	0.90
Adult-race	250	0.85	0.89	0.89	0.85	0.87	0.91	0.81	0.83	0.91
Adult-race	500	0.90	0.91	0.91	0.88	0.91	0.89	0.91	0.89	0.89
Adult-race	1000	0.89	0.89	0.88	0.90	0.91	0.90	0.90	0.90	0.90
Bank	250	0.83	0.79	0.78	0.81	0.86	0.85	0.84	0.85	0.82
Bank	500	0.85	0.85	0.89	0.89	0.86	0.83	0.87	0.83	0.85
Bank	1000	0.90	0.90	0.89	0.90	0.90	0.87	0.87	0.91	0.87
Credit	250	0.79	0.81	0.78	0.83	0.85	0.77	0.87	0.89	0.83
Credit	500	0.88	0.91	0.91	0.90	0.91	0.91	0.90	0.89	0.89
Credit	1000	0.91	0.90	0.89	0.89	0.91	0.91	0.91	0.87	0.91
Law	250	0.80	0.68	0.77	0.79	0.63	0.73	0.85	0.62	0.87
Law	500	0.33	0.89	0.91	0.28	0.92	0.87	0.26	0.89	0.90
Law	1000	0.67	0.93	0.91	0.65	0.90	0.91	0.67	0.89	0.84
MAGIC	250	0.85	0.87	0.83	0.83	0.82	0.83	0.85	0.84	0.86
MAGIC	500	0.90	0.90	0.90	0.89	0.88	0.91	0.92	0.91	0.91
MAGIC	1000	0.93	0.91	0.91	0.91	0.89	0.89	0.89	0.91	0.90
EEG	250	0.89	0.90	0.89	0.89	0.91	0.89	0.90	0.89	0.88
EEG	500	0.89	0.89	0.90	0.91	0.91	0.91	0.90	0.90	0.90
EEG	1000	0.90	0.91	0.91	0.90	0.90	0.92	0.89	0.91	0.89
Phoneme	250	0.89	0.89	0.89	0.88	0.91	0.91	0.88	0.88	0.89
Phoneme	500	0.89	0.87	0.90	0.89	0.90	0.89	0.90	0.89	0.88
Spambase	250	0.89	0.89	0.87	0.90	0.90	0.87	0.86	0.85	0.83
Churn	250	0.55	0.49	0.47	0.57	0.47	0.43	0.53	0.53	0.51
Churn	500	0.69	0.75	0.85	0.59	0.80	0.83	0.69	0.72	0.87
Elec.	250	0.89	0.89	0.91	0.89	0.89	0.90	0.90	0.89	0.89
Elec.	500	0.91	0.91	0.91	0.91	0.86	0.93	0.89	0.89	0.91
Elec.	1000	0.91	0.91	0.93	0.91	0.90	0.91	0.90	0.90	0.90

non-White versus White); Bank Marketing (version 1, 45,211 records; target subscription; group: age below 30); default of credit card clients (version 1, 30,000 records; target default; group: sex); law school admission (binarised version, 20,800 records; group: race, non-White versus White); MAGIC gamma telescope (19,020 records); EEG eye state (14,980 records); Phoneme (5,404 records); Spambase (4,601 records); Churn (5,000 records); Electricity (45,312 records, subsampled to 30,000). For the last six the group is the indicator that the first numeric feature exceeds its median, and the feature is removed from the inputs. In all data sets the minority class is the positive class; for Adult, Bank Marketing and credit-card default this coincides with the conventional positive class (income above 50K, subscription, default).

Each data set was split once, stratified by the label, into a training part (60%) and a pool (40%), with seed 0. Numeric features were standardised and categorical features one-hot encoded inside each model's pipeline. The nine models are: logistic regression (default, and with inverse regularisation strength 0.01), decision trees of depth 3 and 12, a random forest with 200 trees and minimum leaf size 3, histogram gradient boosting with default settings, 25-nearest neighbours, Gaussian naive Bayes, and a multilayer perceptron with hidden layers of 64 and 32 units and early stopping. Pool predictions, predicted probabilities, group indicators and labels are cached by `experiments/preference_validation.py` and used by all experiments.

Table 18: Equalised-odds gaps per cell (A5). Margin: true EO gap between the best and the next distinct model (in Law school six models tie at $EO_m = 0$, so identifying a best model is trivial). P(best): probability that the estimated smallest-gap model has the smallest true gap. τ: mean Kendall τ. Eff.: median over models of $\mathrm{MSE}_{\mathrm{SRS}}/\mathrm{MSE}$ of $\widehat{EO}_m$ for PATTERN and for the gap-targeted allocation.

			P(best)					τ		Eff.	
Setting	L	Margin	SRS	GROUP	PAT.	Gap	Orc.	SRS	PAT.	PAT.	Gap
Adult-sex	250	0.002	0.19	0.23	0.29	0.31	0.26	0.06	0.24	1.98	2.05
Adult-sex	500	0.002	0.27	0.25	0.33	0.45	0.43	0.20	0.37	2.12	2.83
Adult-sex	1000	0.002	0.32	0.45	0.48	0.49	0.55	0.30	0.56	1.95	2.67
Adult-race	250	0.025	0.13	0.13	0.27	0.31	0.29	0.03	0.19	5.70	5.43
Adult-race	500	0.025	0.18	0.21	0.37	0.33	0.40	0.09	0.29	4.97	5.52
Adult-race	1000	0.025	0.22	0.29	0.57	0.53	0.56	0.14	0.46	3.05	4.26
Bank	250	0.002	0.11	0.19	0.20	0.19	0.21	0.01	0.13	2.98	3.88
Bank	500	0.002	0.17	0.16	0.27	0.24	0.29	0.10	0.13	2.73	5.05
Bank	1000	0.002	0.12	0.19	0.40	0.34	0.39	0.11	0.29	3.00	5.84
Credit	250	0.000	0.15	0.16	0.11	0.16	0.17	0.05	0.23	0.86	1.36
Credit	500	0.000	0.16	0.16	0.13	0.15	0.18	0.07	0.22	1.36	1.54
Credit	1000	0.000	0.10	0.15	0.13	0.10	0.10	0.09	0.23	1.93	2.38
Law	250	0.002	1.00	1.00	1.00	1.00	1.00	0.87	0.99	37.64	7.74
Law	500	0.002	1.00	1.00	1.00	1.00	1.00	0.90	1.00	88.43	10.19
Law	1000	0.002	1.00	1.00	1.00	1.00	1.00	0.94	1.00	208.35	9.61
MAGIC	250	0.008	0.17	0.17	0.39	0.34	0.33	0.40	0.48	0.92	1.38
MAGIC	500	0.008	0.29	0.35	0.56	0.55	0.65	0.49	0.58	0.95	1.46
MAGIC	1000	0.008	0.57	0.43	0.76	0.69	0.75	0.60	0.70	0.81	1.61
EEG	250	0.039	1.00	1.00	0.99	0.99	0.97	0.64	0.68	1.21	1.69
EEG	500	0.039	0.98	1.00	1.00	1.00	1.00	0.74	0.77	1.47	1.56
EEG	1000	0.039	1.00	1.00	1.00	1.00	1.00	0.80	0.85	1.33	1.25
Phoneme	250	0.027	0.28	0.39	0.46	0.34	0.44	0.45	0.50	1.42	1.45
Phoneme	500	0.027	0.47	0.45	0.51	0.59	0.55	0.53	0.60	0.99	2.16
Spambase	250	0.027	0.31	0.34	0.76	0.67	0.71	0.35	0.73	1.83	2.23
Churn	250	0.002	0.17	0.15	0.29	0.23	0.28	0.04	0.06	3.08	2.95
Churn	500	0.002	0.17	0.22	0.45	0.51	0.39	0.06	0.19	2.71	3.30
Elec.	250	0.007	0.35	0.33	0.35	0.41	0.35	0.62	0.69	0.70	1.12
Elec.	500	0.007	0.46	0.47	0.49	0.47	0.54	0.77	0.77	0.53	1.13
Elec.	1000	0.007	0.57	0.57	0.59	0.60	0.65	0.83	0.89	0.51	1.48

Table 19: Model subsets (A3) at $L = 250$: for each of the nine random subsets (three of each size M), the share of the pool on which the subset's models do not all agree and the median-over-targets efficiency of PATTERN relative to SRS, written as share/efficiency. Model indices: 0 logistic regression, 1 strongly regularised logistic regression, 2 tree of depth 3, 3 tree of depth 12, 4 random forest, 5 histogram gradient boosting, 6 25-NN, 7 Gaussian naive Bayes, 8 MLP; –: all models agree everywhere (dropped); 10^4: clipped.

	$M = 2$			$M = 3$			$M = 5$		
Setting	{6,8}	{7,8}	{2,7}	{0,5,7}	{2,6,7}	{3,4,5}	{0,2,5,7,8}	{3,5,6,7,8}	{0,2,3,4,5}
Adult-sex	0.06/10.8	0.40/2.7	0.46/3.4	0.46/3.3	0.48/3.0	0.08/11.8	0.49/3.2	0.49/3.1	0.12/6.2
Adult-race	0.06/34.5	0.41/3.6	0.46/5.1	0.45/4.4	0.48/4.1	0.08/15.0	0.49/5.8	0.49/4.5	0.12/10.8
Bank	0.07/7.0	0.12/10.9	0.12/5.9	0.13/9.4	0.14/6.1	0.07/14.7	0.16/8.4	0.17/7.4	0.09/11.1
Credit	0.04/23.3	0.47/1.3	0.48/1.5	0.52/3.0	0.50/3.5	0.09/10.2	0.53/3.7	0.54/3.0	0.12/7.0
Law	0.06/343	0.17/45.8	0.17/35.8	$0.17/10^4$	0.19/13.0	–	$0.17/10^4$	0.19/30.9	–
MAGIC	0.10/6.7	0.22/4.1	0.24/2.6	0.29/2.9	0.28/3.3	0.11/9.4	0.36/2.7	0.31/3.0	0.29/3.0
EEG	0.09/16.6	0.44/1.7	0.47/1.5	0.55/2.2	0.61/2.1	0.18/5.8	0.69/1.7	0.56/2.6	0.56/1.9
Phoneme	0.07/23.7	0.13/12.2	0.15/9.1	0.25/4.1	0.23/4.4	0.15/10.8	0.31/3.2	0.31/3.1	0.34/3.2
Spambase	0.09/11.9	0.17/6.1	0.22/7.0	0.20/11.8	0.28/6.7	0.09/12.8	0.26/7.7	0.28/7.1	0.15/6.5
Churn	0.10/7.8	0.11/12.9	0.07/114	0.15/9.7	0.08/54.7	0.06/1432	0.17/8.4	0.18/7.5	0.16/9.3
Elec.	0.10/8.5	0.15/4.5	0.12/4.7	0.21/4.4	0.21/3.5	0.16/6.2	0.25/3.5	0.30/2.6	0.25/3.4